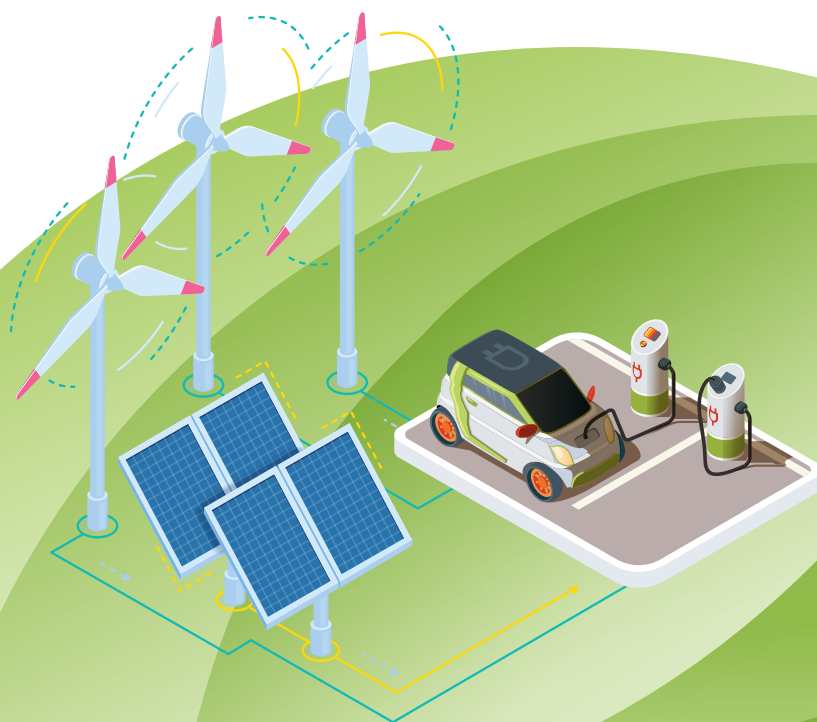




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THE NATIONAL ENERGY COUNCIL

INDONESIA ENERGY OUTLOOK 2 0 2 2





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BUREAU OF ENERGY POLICY AND ASSEMBLY FACILITATION
SECRETARIATE GENERAL
OF THE NATIONAL ENERGY COUNCIL
2022

REMARK BY CHAIRMAN OF THE NATIONAL ENERGY COUNCIL



I would like to appreciate the effort taken by the Secretariate General of the National Energy Council for publishing the book of Indonesia Energy Outlook 2022. This book does not only give the overview of the current energy condition, but also shows Indonesia energy projection in the future as well as the overview of energy condition per region that would be very beneficial as an input in making the policy and planning in Indonesia's energy sector.

The economic growth strongly influences the energy supply-demand including the restriction on mobility globally due to the Covid-19 pandemic which is still happening until today. Furthermore, the government's policies on the utilization of Solar Power Plant and the acceleration of electric vehicle should be used as a consideration in evaluating and formulating medium and long-term strategy.

In this opportunity, allow me to convey my gratitude to Members of the National Energy Council, Secretariate General of the National Energy Council, especially for the Editorial Board for the hard work. I hope the book of Indonesia Energy Outlook 2022 would be beneficial for all the stakeholders and energy policy makers.

Jakarta, December 2022

Minister of Energy and Mineral Resources as
the Chairman of National Energy Council

Arifin Tasrif

FOREWORD BY MEMBER OF THE NATIONAL ENERGY COUNCIL



Let us praise to God the Almighty for His blessing that the National Energy Council has finished the book of Indonesia Energy Outlook 2022.

Indonesia Energy Outlook portrays the energy in Indonesia including the current energy condition and energy projection in the future.

Indonesia Energy Outlook 2022 is jointly formulated by Members of the National Energy Council from Stakeholders and Editorial Board at the Secretariate General of the National Energy Council with the assistance from the experts. The formulation of Indonesia Energy Outlook also involves units in the Ministry of Energy and Mineral Resources namely the Data and Information Technology Center, Directorate General of Electricity, Directorate General of Oil and Gas, Directorate General of New Renewable Energy and Energy Conservation, as well as Directorate General of Mineral and Coal especially on the collection of current data and planning data.

In Energy Outlook 2022, the energy supply and demand projection are carried out for the next 10 years in two (2) scenarios namely Business as Usual (BaU) which uses the existing policy and Optimistic (OPT) scenario which uses assumptions toward developed country 2045 and Net Zero Emission (NZE) 2060. Next, the projection from the two scenarios will be compared with the projection that has been set in National Energy General Planning (RUEN).

In this 2022 publication, Indonesia Energy Outlook presents the result of projection based on 5 regions namely Sumatra, Java-Bali, Kalimantan, Sulawesi and Nusa Tenggara, Maluku and Papua (NUSMAPA).

Jakarta, December 2022
Member of National Energy Council from
Industrial Stakeholder

Herman Darnel Ibrahim

FOREWORD BY SECRETARY GENERAL OF THE NATIONAL ENERGY COUNCIL



Indonesia Energy Outlook 2022 is an annual publication by the Secretariate General of the National Energy Council since 2014.

The energy supply and demand projection are conducted based on LEAP (Low Emissions Analysis Platform) modelling using 2021 as the baseline year taken from the Data and Information Center MEMR and BPS as well as data of planning from related ministries.

In the effort to improve the quality of energy supply and demand projection data, Secretariate General of the National Energy Council would enhance the relationship with related parties to publish the book of Indonesia Energy Outlook as a reliable and trusted reference.

Jakarta, December 2022

Secretary General of the National Energy Council

Djoko Siswanto

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- Members of the National Energy Council from stakeholders;
- Directorate General of Oil and Gas, Directorate General of Electricity, Directorate General of Mineral and Coal, Directorate General of New Renewable Energy and Energy Conservation, Data and Information Center MEMR and PT PLN (Persero);
- Danish Energy Agency;
- Energy experts for their contribution in formulating Indonesia Energy Outlook 2022.

DISCLOSURE

Indonesia Energy Outlook (IEO) 2022 is an analysis on the long-term national energy demand and supply projection (2022–2032) with particular assumptions which are developed for the purpose of future energy scenario planning. The assumption and projection are based on energy technology development including fossil and renewable based on current data and condition. The data in Indonesia Energy Outlook is taken from official publication as well as temporary data or data that are continuously updated by the sources.

EXECUTIVE SUMMARY

Indonesia Energy Outlook (IEO) 2022 presents national energy demand and supply projection in 2022-2032 based on assumptions of social, economy and technology development in the future by using baseline data year 2021.

The energy demand and supply analysis are conducted based on LEAP (Low Emissions Analysis Platform) calculation model. LEAP is an energy planning modelling application to take an integrated energy demand and supply analysis.

Population growth and economic growth are the basic assumptions which are developed to obtain an illustration on energy demand up to the year 2032, biofuel mandatory, electric vehicle (EV) development, and power capacity.

In 2032, final energy consumption in BaU scenario will increase with the average growth of 4.8% per year into 207 million TOE. The biggest final energy consumption shares in 2032 come from transportation and industry with 42.6% and 37.9% respectively. The final energy consumption in OPT scenario will increase 6.6% per year into 248 million TOE. Unlike BaU scenario, the biggest final energy consumption in OPT scenario comes from industry with 49.2%, followed by transportation with 36.4%.

Based on region, the biggest final energy consumption in 2032 occurs in Java-Bali (45.7%) in BaU scenario, while the smallest final energy consumption occurs in Nusmapa (10.4%).

Electricity demand in 2032 in OPT scenario is projected to reach 60 million TOE (702 TWh) or almost twice of BaU scenario that is influenced by fuel to electric vehicle substitution program.

Meanwhile, electricity production in 2032 is projected to reach 791 TWh in OPT scenario and 489 TWh in BaU scenario. Power production in both scenarios in 2032 is still dominated by coal of 75.1% in BaU and 48.9% in OPT.

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1

INTRODUCTION

1 INTRODUCTION

1.1 BACKGROUND

Indonesia Energy Outlook (IEO) which is published annually is a result of an analysis to give an overview on national energy condition especially on energy supply and demand projection until 2032. The 2022-2032 projection uses two scenarios namely Business as Usual (BaU) and Optimistic scenario (OPT). The BaU scenario uses the consumptions based on the current condition, while OPT scenario uses the assumptions based the condition toward developed country 2045 and NZE 2060.

The main data source in IEO 2022 is Handbook of Energy and Economic Statistics of Indonesia (HEESI) 2021 publication by Data and Information Center MEMR, RUPTL PLN 2021-2030, Statistics Indonesia 2022 – BPS , and Industrial Statistics of BPS 2019.

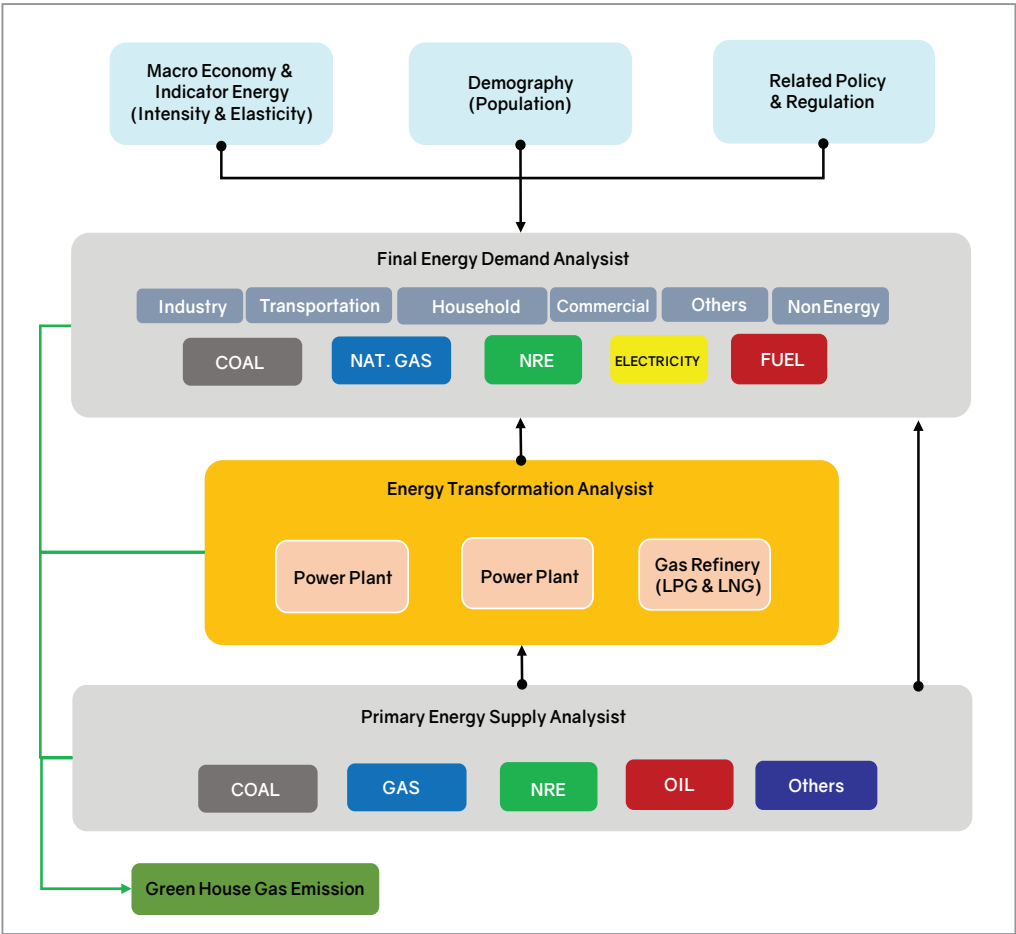
1.2 METHODOLOGY

1.2.1 Modeling Analysis Framework

Modeling analysis is divided into three stages namely final energy demand analysis, energy transformation, and primary energy supply. Final energy demand analysis is done by using GDP growth assumption and population growth. It also takes into account the policy, strategic planning and roadmap on current energy development. The primary energy supply analysis also takes into account the use of various energy sources and energy resources potential including the current policies as current energy technology development. Meanwhile, energy transformation analysis is done by considering RUPTL. The modeling analysis framework is shown in Picture 1.1.

The energy supply and demand analysis are conducted based on calculation of LEAP model as a simulation model in energy planning which can conduct an integrated energy supply and demand analysis. In LEAP model, energy demand projection is calculated based on the multiplication of energy consumption activity and energy consumption intensity. Energy activity is described by economic growth, population and production. Meanwhile, energy intensity is the level of energy consumption per GDP or per population or per production in particular period. Energy intensity can be considered as fixed during simulation period or declining to show energy efficiency increase.

Based on the modelling analysis framework in Picture 1.1, the parameters in final energy demand projection are social economic data including population and economic growth, historical data on energy consumption to find out energy intensity and energy consumption pattern as the impact of people's lifestyle that is influenced by the projection of GDP increase or a more efficient technology.



Picture 1.1 Modeling Analysis Framework

In 2022, the energy supply demand projection per region will also be carried out to obtain the overview of energy demand and supply in each region. The projection is done in five regions namely Sumatra, Java-Bali, Kalimantan, Sulawesi, and Nusmapa (Nusa Tenggara, Maluku and Papua). The consideration in determining the five regions is the geographical condition of several provinces located close to each other.

In the modelling, energy consumption uses the data on each province. It is then calculated based on per region. The data of energy consumption per province is obtained from units in MEMR and energy Business Entities. Data on fuel and LPG consumption/sales per province is specifically derived from Directorate General of Oil and Gas. Meanwhile, the main data of electricity sales is derived from PLN sales data with additional identification of non-PLN consumption data and electricity from smelter industry. Data on coal consumption per province is obtained from Directorate General of Mineral and Coal. However, the sales of electricity per sub sector refers to data from Industrial Statistics BPS. Data on gas consumption is based on data on sales at HEESI, while consumption per industrial sub sector uses Industrial Statistics BPS data. The data on household consumption per province uses data from Indonesia Residential End Use Survey that is the result of a CLASP study with the Directorate General of EBTKE.

1.2.2 Energy Projection Scenario

1.2.2.1 BAU Scenario

In the book of Statistics Indonesia 2022 published by BPS, Indonesian economic growth in 2020 is -2.07% and 3.7% in 2021 or experiences a decline from the previous year of 5% in 2019 due to the Covid-19 pandemic. Meanwhile the economic growth in 2022 refers to Law No. 6 of 2021 on State Budget and Expenditure 2022. It is predicted that Indonesia economic growth will reach 5.2% in 2022 and 5.3%-5.9% in 2023.

BAU scenario refers to current condition and the future projection which is assumed based on historical data in the last several years such as additional city gas, electric stove, electric vehicle, and others. The power plant development refers to RUPTL 2021-2030 with assumption that the project completion is delayed for two years.

1.2.2.2 OPT Scenario

OPT scenario uses the assumptions that lead to developed country 2045 and NZE 2060. The population growth assumption is the same with the previous scenario, but the economic growth is higher. The use of electric vehicle and power plant additional capacity from NRE especially Solar PP and co-firing Coal Fired Power Plant is quite different compared to BaU scenario. The details of assumption in the two scenarios above can be seen in Table 1.1

Table 1.1 Scenario Assumption

INDICATOR	Unit	BaU	OPT
DEMOGRAPHY & ECONOMY			
Population growth	%	0.9	
Economic growth (Average)	%	• 2023-2025 of 5.6 • 2026-2032 of 6-6.2	
DEMAND			
Electric Car 2032			
- Java	%	10	20
- Outside Java	%	7	20
Electric Motorcycle 2032			
- Java	%	5	25
- Outside Java	%	3	25
Electric Bus 2032			
- Java	%	5	10
- Outside Java	%	3	10
Biofuel Mandatory 2032			
- Bioethanol	%	1	5
- Biodiesel	%	30	40
ELECTRICITY			
Power Plant Capacity		RUPTL 2021-2030 with COD delayed 2 years	RUPTL 2021-2030 Optimization of Solar PP and Co-firing Coal Fired PP

1.2.3 Energy-Related Policy Assumption

In order to support energy demand projection, several energy-related policies are being considered, including:

1. National Energy Policy

National Energy Policy (KEN) mandates renewable energy mix target in primary energy mix of at least 23% in 2025 and minimize the use of oil to be less than 25% in 2025. Furthermore, energy efficiency is also targeted to decline 1% per year to promote energy saving in all sectors. Several targets in KEN also becomes the consideration in energy demand projection such as optimization of gas for domestic and coal downstreaming.

2. National Energy General Plan (RUEN)

National Energy General Plan (RUEN) is a mandate of Law No. 30 of 2007 on Energy. Based on Article 17 paragraph (1) of Law No. 30, the government formulates the draft of RUEN based on KEN. The targets in RUEN that are taken into consideration in energy demand projection are LPG to city gas substitution, DME utilization, biofuel utilization, and EV utilization.

3. MEMR Strategic Planning

Several strategic planning programs in MEMR that are taken into consideration in calculating the energy demand projection are the development of city gas, gas for public vehicle and biofuel utilization.

4. RUPTL 2021–2030

The data on capacity of planned power plant refers to RUPTL 2021–2030. In BaU scenario, the power plant development refers to RUPTL with the project completion is delayed for two years and the OPT scenario follows RUPTL with additional optimization of Solar PP and co-firing Coal Fired Power Plant.

5. Biofuel Roadmap

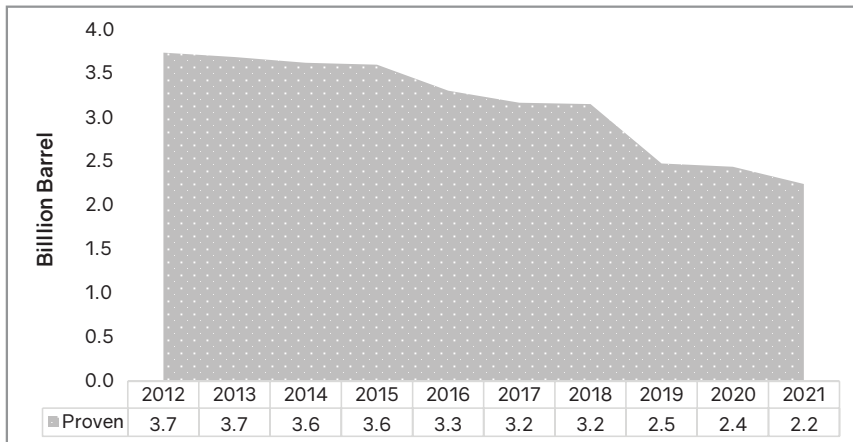
Biofuel mandatory is taken into consideration as one of the assumptions in final energy demand projection in transportation, industry, commercial and power plant sector.

1.3 CURRENT ENERGY CONDITION

In 2021, total primary energy production consisting of oil, gas, coal and renewable energy reaches 481 million TOE. Around 58.2% or 280 million TOE from the total production is exported, especially coal and LNG. However, Indonesia is also importing energy especially crude oil and fuel products amounting to 49 million TOE and small amount of high rank coal to meet the demand of steel industry.

1.3.1 Oil

Based on BP Statistic Review 2021 data, the number of Indonesia oil reserves is only 0.1% from the world reserves. Indonesia oil reserves experience a decline from 7.4 billion barrels (3.7 billion barrels for potential reserves and 3.7 billion barrels for proven reserves) in 2012 to 3.9 billion barrels (1.7 billion barrels for potential reserves and 2.2 billion barrels for proven reserves) in 2021 (Picture 1.2).



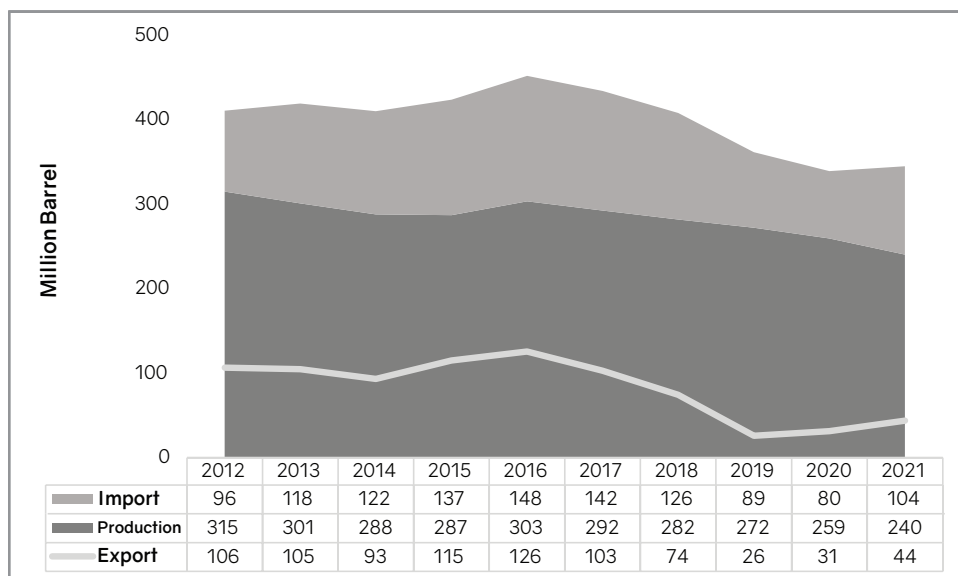
Source: HEESI, 2021

Picture 1.2 Oil Reserves (Proven) 2012–2021

Oil production in the last 10 years shows the declining trend, from 315 million barrel (862 thousand bph) in 2012 into 240 million barrel (659 thousand bph) in 2021. The declining production occurs since oil producing wells are mainly mature wells, while new producing wells are relatively limited.

The declining production and the increasing refinery input demand cause Indonesia to import oil especially from Middle East. The import tends to increase from 96 million barrels in 2012 into 104 million barrels in 2021 that is influenced by the fluctuating crude oil demand for refinery input. The crude oil import for refinery input in 2012–2018 is around 30–50%, but it declines into 26–35% in 2019–2021.

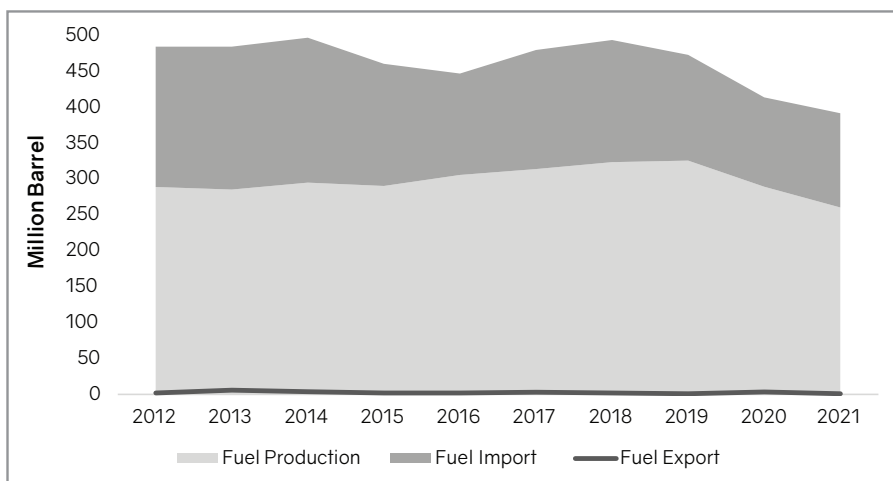
On the other hand, Indonesia oil export continues to decline from 106 million barrel in 2012 into 44 million barrels in 2021. This is based on the Minister of Energy and Mineral Resources Regulation No. 42 of 2018 on the Priority of Crude Oil Utilization for Domestic Need. It means that domestic oil production is used maximally for the domestic need. The trend of oil production, export and import since 2012 to 2021 can be seen in Picture 1.3.



Source: HEESI, 2021

Picture 1.3 Crude Oil Production, Export and Import 2012–2021

Within 10 years, the average fuel production from domestic is 297 million barrels per year and the average fuel import is around 164 million barrels. The trend of fuel production, export and import in the last 10 years can be seen in Picture 1.4.

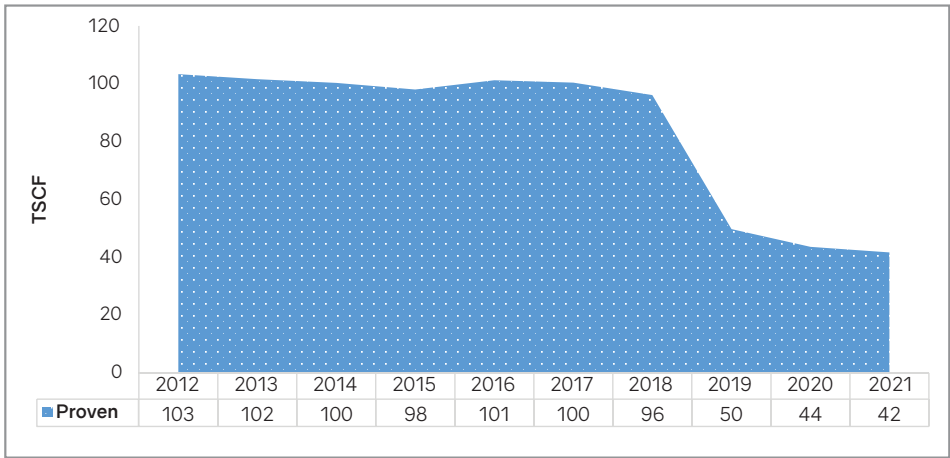


Source: HEESI, 2021

Picture 1.4 Fuel Production, Import and Export 2012–2021

1.3.2 Gas

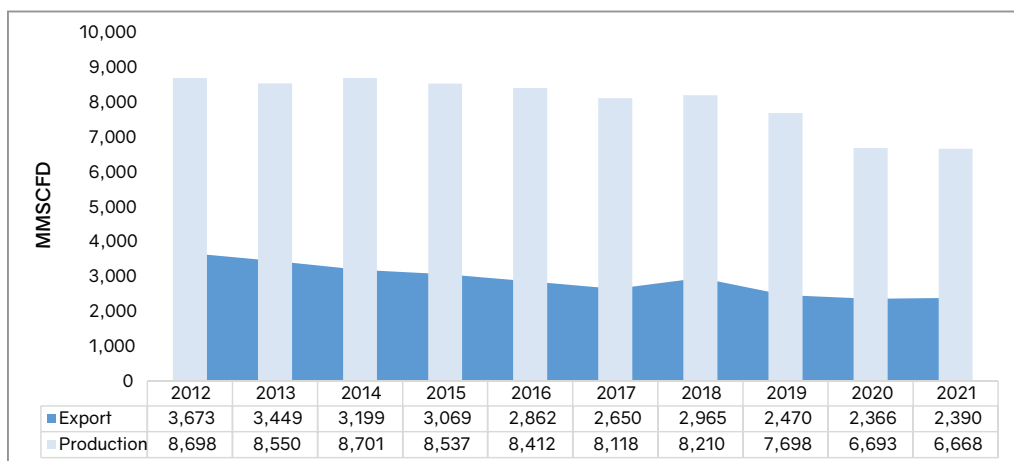
Based on HEESI, Indonesia's total proven gas reserves in 2021 is 42 TSCF. It declines from 103 TSCF in 2012 as shown in Picture 1.5.



Source: HEESI, 2021

Picture 1.5 Gas Reserves 2012-2021

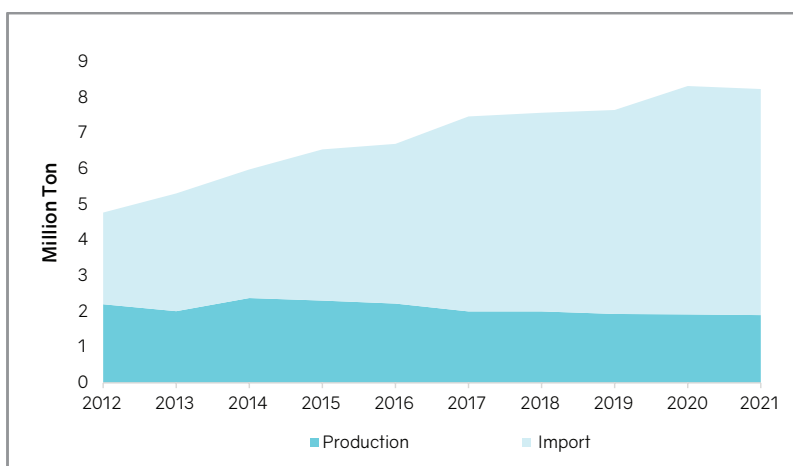
In 2021 Indonesia's total natural gas production is around 6,668 MMSCFD that is used mainly to meet domestic consumption in the industrial sector either as energy or feed stock (fertilizer industry), power plants and city gas (household and commercial) as well as gas for transportation sector. Gas production is declining from around 8,698 MMSCFD in 2012. One of them is influenced by the limited reserves in Arun. Furthermore, the activity of gas production in Tangguh field (Train 3) is not yet started. In addition, gas has become the export commodity in the form of LNG and piped gas of around 2,390 MMSCFD in 2021 but the gas export is declining due to the program in prioritizing gas for domestic need that shows an increase from 57.8% in 2012 into 64.2%. The export percentage through pipeline and LNG toward the total gas production is declining from 42.2% in 2012 into 35.5% in 2021 (Picture 1.6).



Source: HEESI, 2021

Picture 1.6 Gas Production and Export 2012–2021

Gas primary energy including LPG is derived from LPG refinery production and LPG import. LPG production from 2012 to 2021 is around 2 million tons and LPG imports continue to increase. The trend of LPG production and import is shown in Picture 1.7.

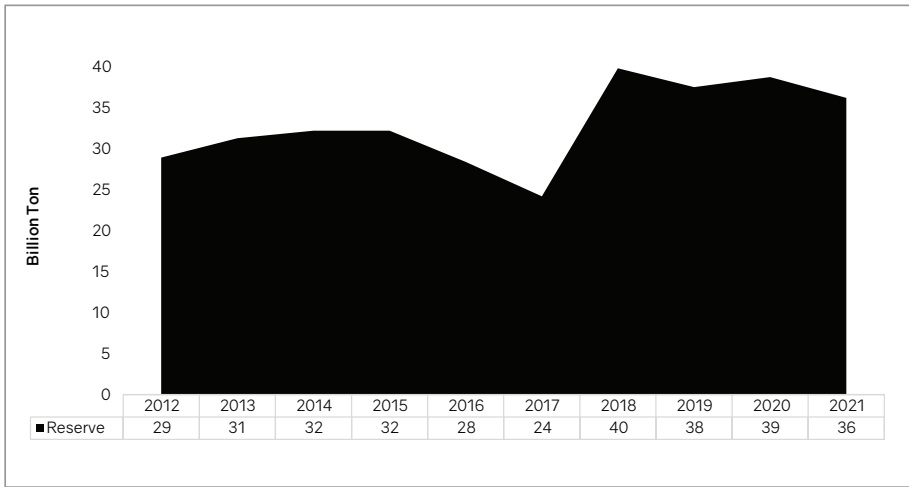


Source: HEESI, 2021

Picture 1.7 LPG Production and Import 2012–2021

1.3.3 Coal

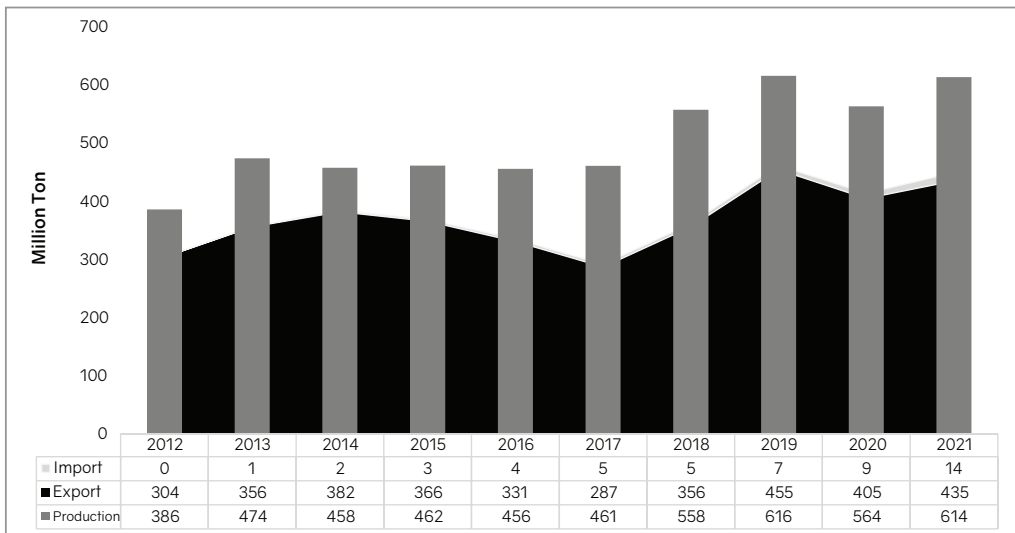
Indonesia's total coal reserves in 2021 are 36 billion tons and it continues to increase since 2012 despite of the increasing production (Picture 1.8). It is due to the increasing exploration in the last 10 years.



Source: HEESI, 2021

Picture 1.8 Coal Reserves 2012–2021

Coal production in 2012–2021 increased significantly from 386 million tons in 2012 into 614 million tons in 2021. Coal export is around 70.9% (435 million tons) from the total production. It is mostly exported to China and India. The high percentage of Indonesia coal export has made Indonesia one of the biggest coal exporters in the world besides Australia. The coal production, export and import in the last 10 years is shown in Picture 1.9.



Source: HEESI, 2021

Picture 1.9 Coal Production, Export and Import 2012–2021

1.3.4 Renewable Energy

The declining fossil energy production especially oil and gas as well as the global commitment in reducing green house gas emission have encouraged the government to continuously enhance the role of new and renewable energy to maintain energy security and independence. Based on Government Regulation No. 79 of 2014 on National Energy Policy, the new and renewable energy mix target is at least 23% by 2025 and 31% by 2050. Indonesia has quite abundant new renewable energy potential to meet the primary energy mix target. In 2021, the updating data on NRE potential has been conducted as seen in Table 1.2 below.

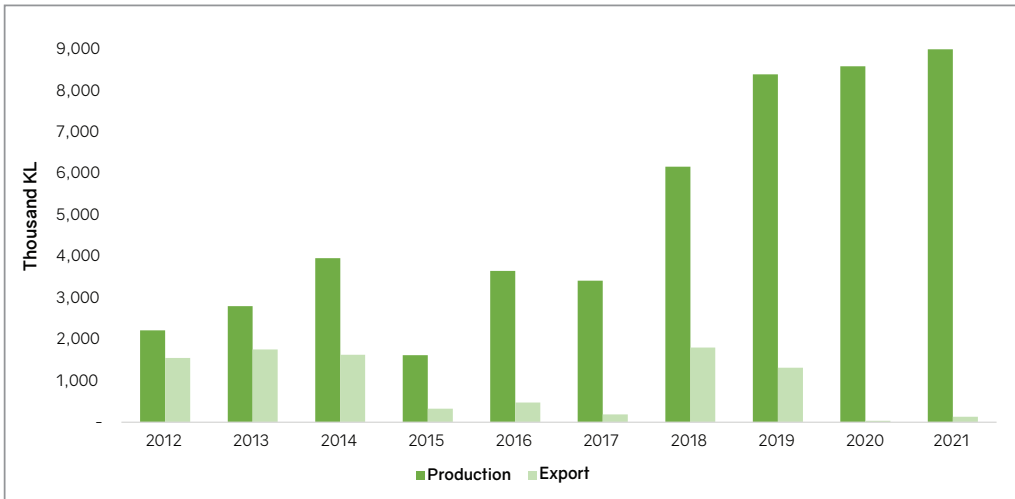
Table 1.2 NRE Potential and Power Plant Capacity 2021

NRE Commodity	Total Potential 2021 (GW)	Power Plant Capacity (GW)	% Utilization
Tidal	17.9	-	-
Geothermal	23.9	2.3	9.6%
Bioenergy	56.9	2.3	4.0%
Wind	154.9	0.2	0.1%
Hydro	95.0	6.6	6.9%
Solar	3,294.4	0.2	0.01%
Total	3,643.0	11.6	0.3%

The total renewable energy potential for power plant is 3,643 GW, but only 0.3% or 11.6 GW is utilized.

The low NRE utilization for electricity generation is due to the relatively high cost of NRE-based power plant. Thus, it is difficult to compete with fossil power plant especially coal. In addition, the lack of support from domestic industry in NRE component and the issue in low interest funding have contributed to the slow development of renewable energy.

The trend of biodiesel production and export is shown in Picture 1.10.



Source: HEESI, 2021

Picture 1.10 Biodiesel Production and Export 2012 - 2021

1.3.5 New Energy

The definition of new energy is based on the general provision of Law No. 30 of 2007 that energy from new energy sources may be derived from new technology, both from renewable energy sources or non-renewable energy sources. Several energy types that are classified into new energy are nuclear, hydrogen, coal bed methane, liquified coal, and gasified coal but until now these energy types have not been developed maximally. In RUEN, only gasified coal that has the capacity development target, while other types of energy have not determined the development target. RUEN sets the target to develop gasified coal power plant with the capacity of 44 MW. The issues on price and technology have become the challenges in developing new energy.

a. Nuclear

Based on data published by BRIN, Indonesia has the potential of 89,498 ton of uranium and 143,234 ton of thorium. The heat of 1 gram fission of U^{235} is equivalent to the heat of 2-3 tons of coal. The average abundance of U^{235} is 0.7% as long as no new nuclear raw material reserves are found. Therefore, according to the uranium potential data, the total potential only from Indonesian uranium is equivalent to 1.88 billion tons of coal.

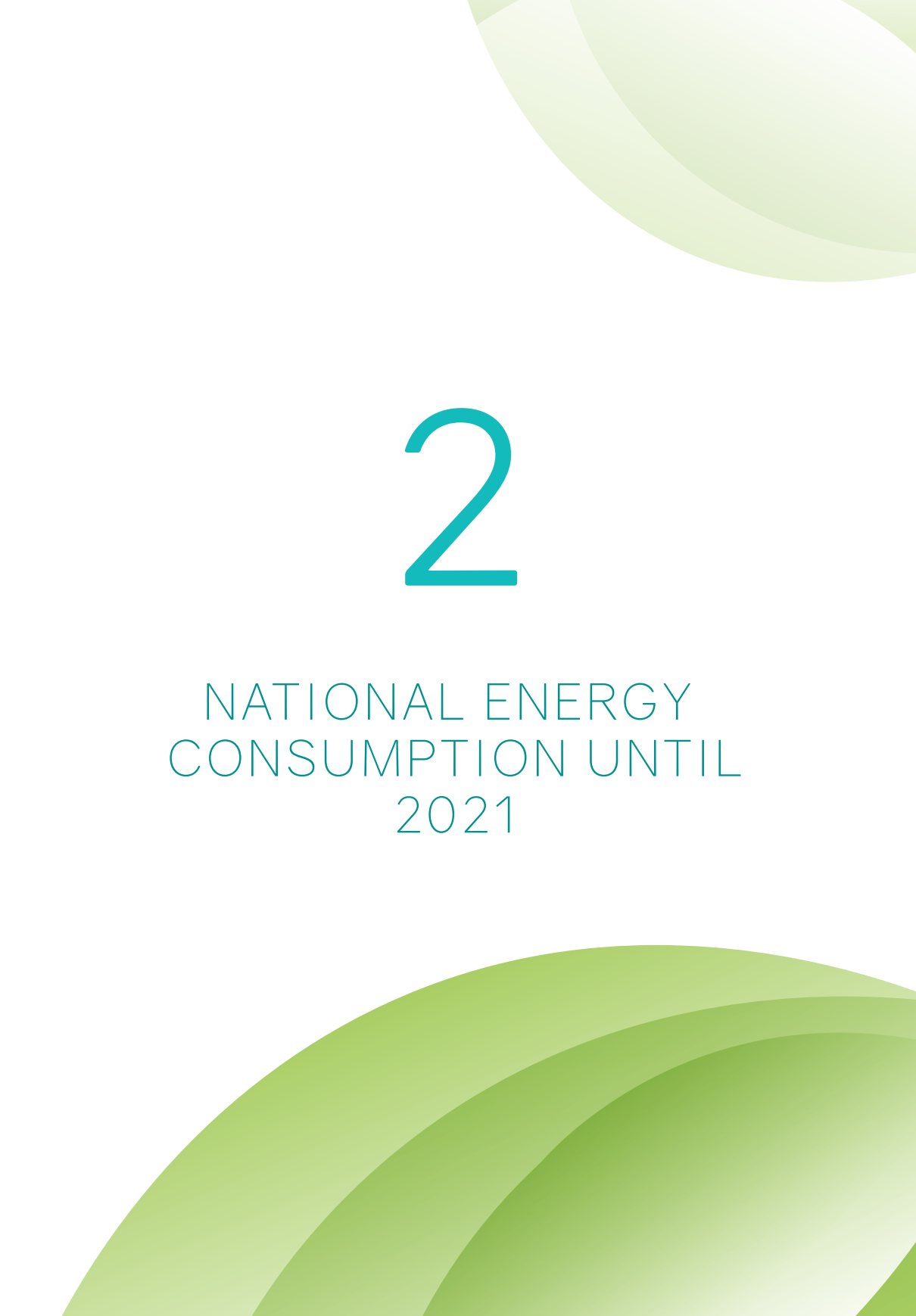
b. Coal Bed Methane

The abundant coal reserves and resources in Indonesia contain another energy source potential including coal bed methane (CBM). CBM that is formed together with coal seams underground can be used as an alternative gas energy source. Based on Research and Development Agency MEMR data, Indonesia CBM resources are 453 trillion cubic feet (TCF) spread in 11 coal and oil and gas basins. One of them is South Sumatra Basin with 180 TCF resources but until today, the potential is not yet used massively.

c. Coal Gasification

The government through the Ministry of Energy and Mineral Resources planned to develop coal gasification in the form of dimethyl ether (DME) as an alternative substitution to LPG for household. DME is planned to be developed from the abundant resources of domestic coal. The Presidential Regulation on RUEN mandates the development of 1 million ton DME in 2025. Furthermore, Indonesia Energy Grand Strategy has targeted DME development from coal to stop LPG import by 2030 from State Owned Enterprise IUP and extension of PKP2B. There are three business entities that have been planning to develop DME. They are PT Bukit Asam with production plan of 1.4 million tons per year, PT KPC with production plan of 1.2 million tons per year and PT Arutmin with production plan of 2 million tons per year. They are going to start the production in 2025.

In addition, three coal business entities located in East Kalimantan and North Kalimantan plan to develop underground coal gasification but it is still in pilot project stage. These business entities are PT Kideco Jaya Agung in East Kalimantan, PT Indominco in East Kalimantan, and PT Medco Energi Mining International (MEMI) and Phoenix Energi Ltd in North Kalimantan.



2

NATIONAL ENERGY
CONSUMPTION UNTIL
2021

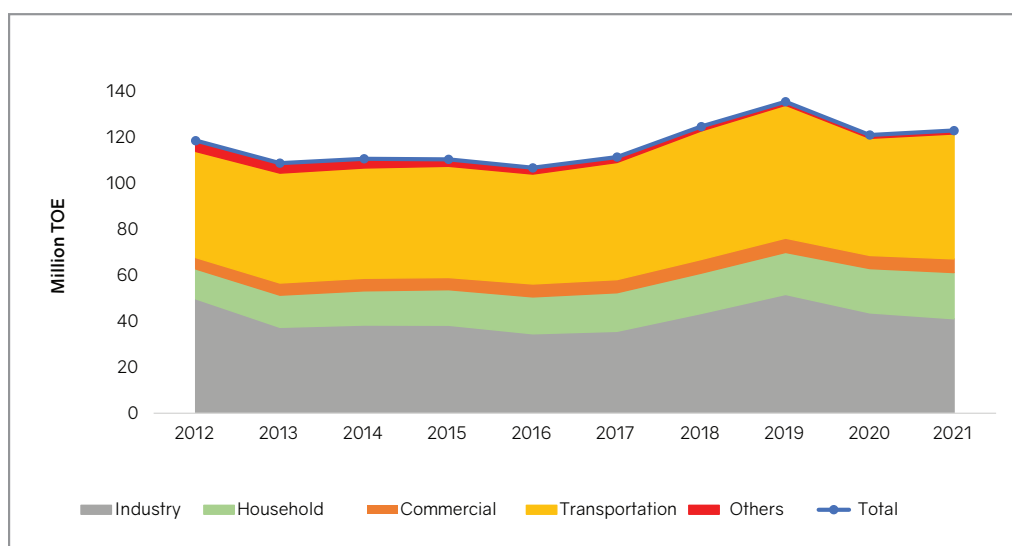
2 NATIONAL ENERGY CONSUMPTION UNTIL 2021

2.1 FINAL ENERGY CONSUMPTION

2.1.1 Final Energy Consumption per Sector

Final energy consumption in 2021 increased after a sharp decline in 2020 due to Covid-19 pandemic. In 2021, final energy consumption reached 123 million TOE or increased around 1.6% due to economic recovery along the year 2021.

Based on the consuming sector, transportation still dominates the biggest portion of 44.2% from the total final energy consumption as shown in Picture 2.1. Compared to the previous year, energy consumption in transportation increased 6.7%. The increase occurred along with the increasing public mobility as the social restriction and the number of Covid-19 pandemic cases are declining.



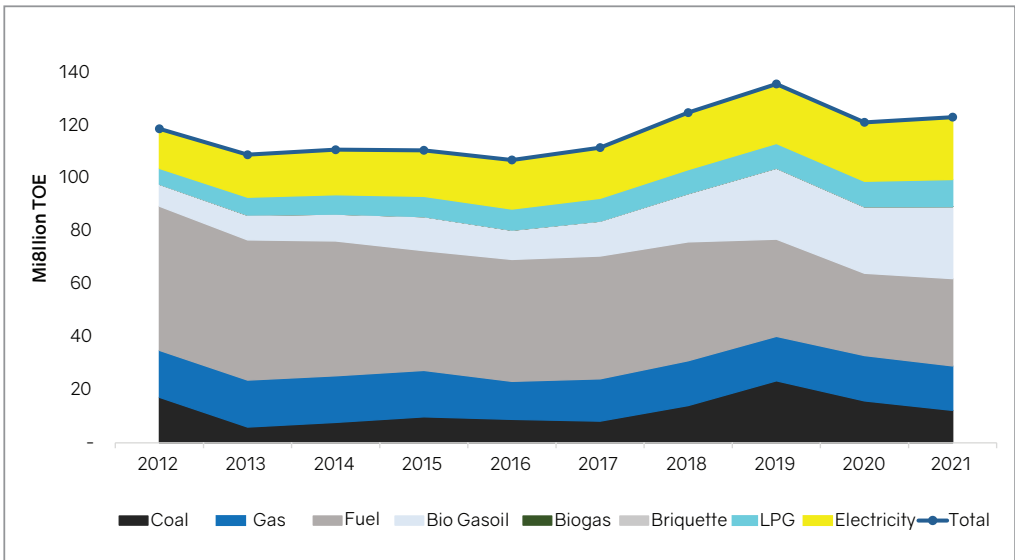
Source: HEESI, 2021

Picture 2.1 Final Energy Consumption per Sector 2012 – 2021

Industry is the second biggest consumer which contributes 33.5% to the final energy consumption. However, compared to the previous year, energy use in industry declined 5.8%. It shows that the industry has not been fully recovering. Next, household becomes the third biggest energy consumer with the percentage of 16.3% from total final energy.

2.1.2 Final Energy Consumption per Energy Type

Based on type of energy, energy consumption by the public is still dominated by fuel, reaching 33 million TOE (26.8%) as shown in Picture 2.2. It is due to the increasing public mobility and number of vehicles. In addition, fuel to gas and electric vehicle substitution has not been able to be carried out optimally since the number of gas filling station is still limited and the regulation on electric vehicle has just passed in August 2019.



Source: HEESI, 2021

Picture 2.2 Final Energy Consumption per Energy Type 2012 – 2021

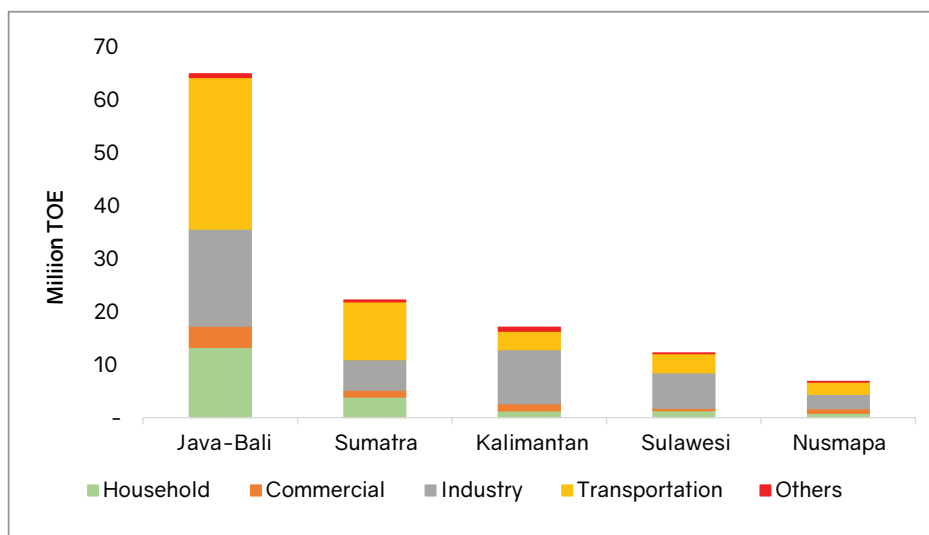
Biofuel (biogasoil and biogas) is in the second place with the consumption of around 27 million TOE (22.1%). The achievement is part of the government's effort to promote the utilization of biofuel through B30 mandatory. Meanwhile, electricity and LPG consumption by the public in 2021 reached 23.6% million TOE and 10.2 million TOE respectively. The consumption of these two energy products increased from the previous year.

2.1.3 Final Energy Consumption per Region

Based on the region, final energy consumption in 2021 is centered in Java-Bali region with the share of 52.7% from the total Indonesia final energy consumption. The next biggest share is in Sumatra with 22 million TOE (18.0%), Kalimantan with 17 million TOE (13.9%), Sulawesi with 12 million TOE (9.9%), and Nusmapa with 7 million TOE (5.5%) in the last position of

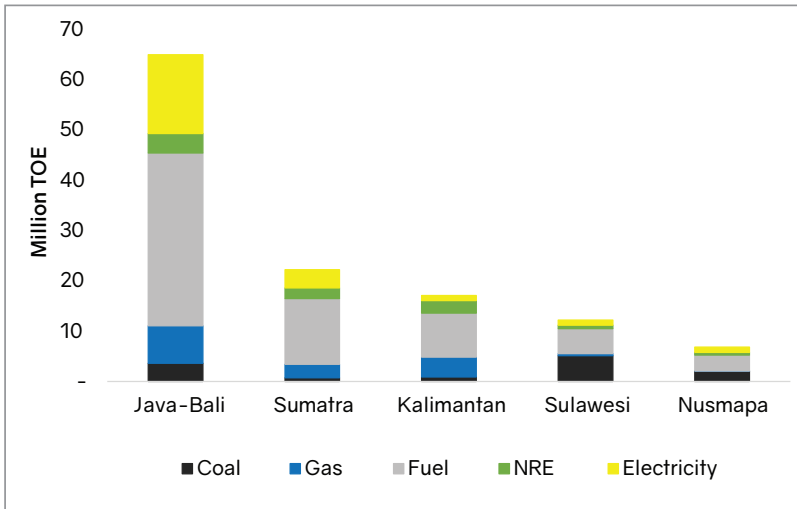
final energy consumption. This number is in line with the distribution of Indonesia population about 58% in Java-Bali region, and 22% in Sumatra region.

The use of energy in Java-Bali and Sumatra is dominated by transportation, while in Kalimantan and Sulawesi it is dominated by industry. The details of final energy consumption per region is shown in Picture 2.3.



Picture 2.3 Final Energy Consumption per Region in 2021 based on Consuming Sector

From the type of energy, all regions still depend on oil, especially in the form of fuel to meet the demand from transportation. Specifically for Sulawesi, coal consumption (5 million TOE) is not far different from oil consumption (4.9 million TOE). It is supported by the growing smelter industry since 2018 along with the issuance of policy on mineral export restriction for company which has not established domestic processing facility (Mining Law Article 102 and 103). The imbalance occurs in electricity commodity which is mostly used in Java-Bali reaching 69.9% from total national. It shows that electricity network infrastructure outside Java-Bali is still far below in Java-Bali. Meanwhile, for NRE, in total, utilization is mostly found in Java-Bali of about 4 million TOE. However, the highest percentage in the mix is in Kalimantan region reaching 14.6% (or equivalent to 3 million TOE). Complete picture in Figure 2.4.

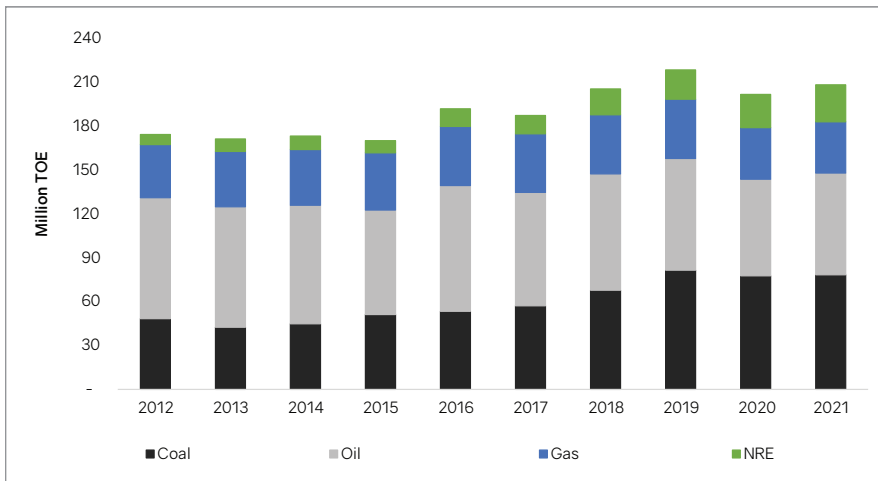


Picture 2.4 Final Energy Consumption per Region 2021 based on Energy Type

2.2 PRIMARY ENERGY SUPPLY

2.2.1 Primary Energy Supply per Energy Type

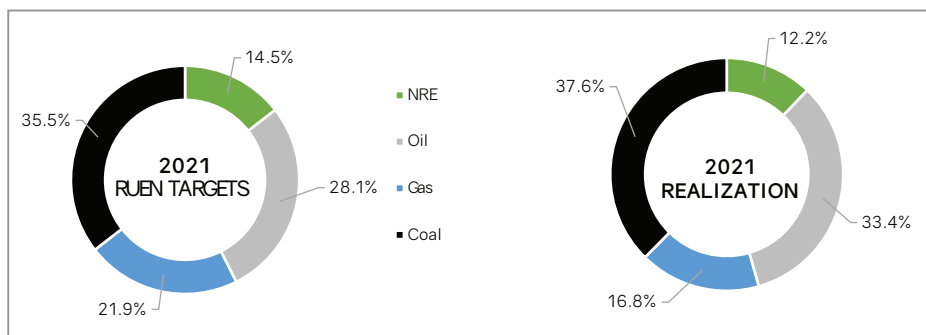
Total primary energy supply in 2021 increased 3.3% from the previous year as shown in Picture 2.5. Based on energy type, coal utilization still dominated around 37.6% followed by oil of 33.4% and gas of 16.8%.



Source: HEESI, 2021

Picture 2.5 Primary Energy Supply 2012 – 2021

In 2021, NRE supply reached 25 million TOE, or 12.2% of the total primary energy supply. This figure increased from the previous year's realization of only 11.3%. However, this progress is considered insufficient because the condition of NRE achievement is still below the RUEN projection as shown in Picture 2.6. There is still hope to increase the NRE energy mix in the next 4 years despite that the target of 23% NRE in 2025 may be difficult to achieve.

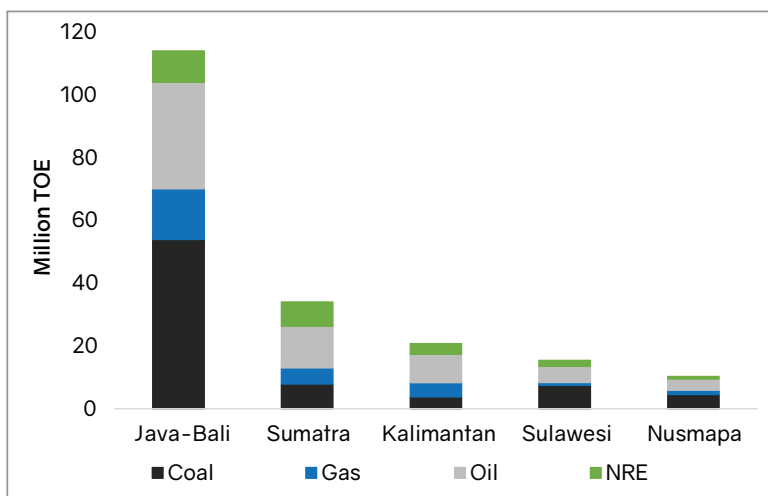


Picture 2.6 Primary Energy Mix 2021

2.2.2 Primary Energy Supply per Region

In line with final energy consumption, 58.6% of the total national primary energy in 2021 is concentrated in Java-Bali region, which is 114 million TOE. In Java-Bali region, coal is still the mainstay commodity, reaching 47.3%. Meanwhile, the utilization of NRE has only reached a mix of 8.8%.

In Sumatra region, primary energy supply reached 17.5% of the total national. Energy utilization in Sumatra region is dominated by oil with the share of 39.1%. However, the utilization of NRE in Sumatra region is the largest compared to other regions, which is 22.7%. Meanwhile, the utilization of NRE reached only 16.5% in Kalimantan, 12.6% in Sulawesi and 7.6% in Nusmapa. The details of primary energy consumption in 2021 in each region can be seen in Picture 2.7.



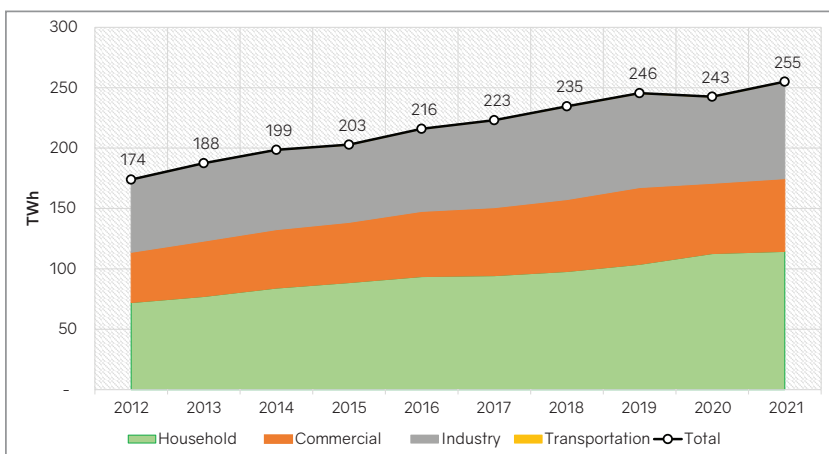
Picture 2.7 Primary Energy Supply per Region 2021

2.3 ELECTRICITY CONSUMPTION

2.3.1 Electricity Consumption per Sector

The growth of electricity consumption in the last 10 years shows an increase from 174 TWh in 2012 to 255 TWh in 2021. It leads electricity to experience the highest growth compared to other types of energy after experiencing a decline in electricity consumption in 2020 due to the Covid-19 pandemic. The electricity consumption is electricity that is included in the PLN network (on-grid), but in total electricity consumption in Indonesia will be higher because it does not include electricity consumed directly from power plants in the industrial sector.

Households are the largest sector that consumes electricity in 2021, followed by the industrial sector, commercial sector and transportation sector. The demand for electricity in the household sector is 114 TWh or 44.9% of the total national electricity demand. This is influenced by the number of households reaching around 69 million in 2021. The demand for electricity in the industrial sector is 80 TWh (31.4%) and the commercial sector is 60 TWh (23.6%). The rest of the demand is from the transportation sector of 0.3 TWh (0.1%). Currently, the transportation sector's electricity consumption is only used for the Jabodetabek and Java electric trains with a consumption of 317 GWh. The demand for electricity by sector is shown in Picture 2.8.



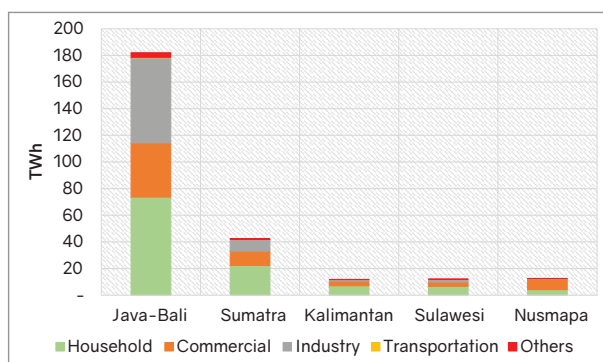
Source: HEESI, 2021

Picture 2.8 Electricity Consumption per Sector 2012 – 2021

2.3.2 Electricity Consumption per Region

In general, the largest total electricity consumption is in Java-Bali region with 69.9% of the total national followed by Sumatra with only 16.3%, Sulawesi and Kalimantan with 4.6% and 4.5% respectively, and Nusmapa with 2.5%.

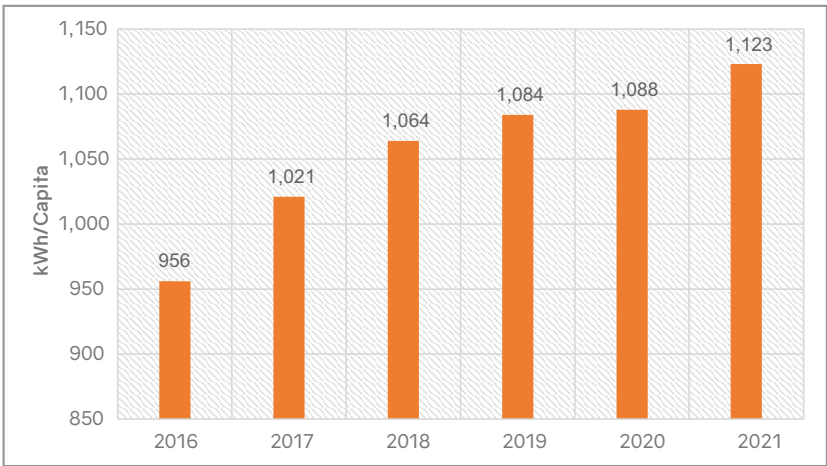
The condition of electricity consumption per region shows that the progress of infrastructure development outside Java-Bali is still very far from the conditions in Java-Bali region. The conditions in eastern Indonesia especially Papua and Maluku, contribute no more than 3%. In addition, the level of population density is also one of the main factors causing disparity. Total electricity customers in 2021 reached 79 million, around 61% are Java household. This fact further strengthens the need for accelerated development and equity outside Java, as shown in Picture 2.9



Picture 2.9 Electricity Consumption per Region 2021

2.3.3 Electricity Consumption per Capita

Electricity consumption per capita over the last 5 years grew by 2.9%. The trend of electricity consumption per capita in 2016-2021 from the Directorate General of Electricity, MEMR can be seen in Picture 2.10.



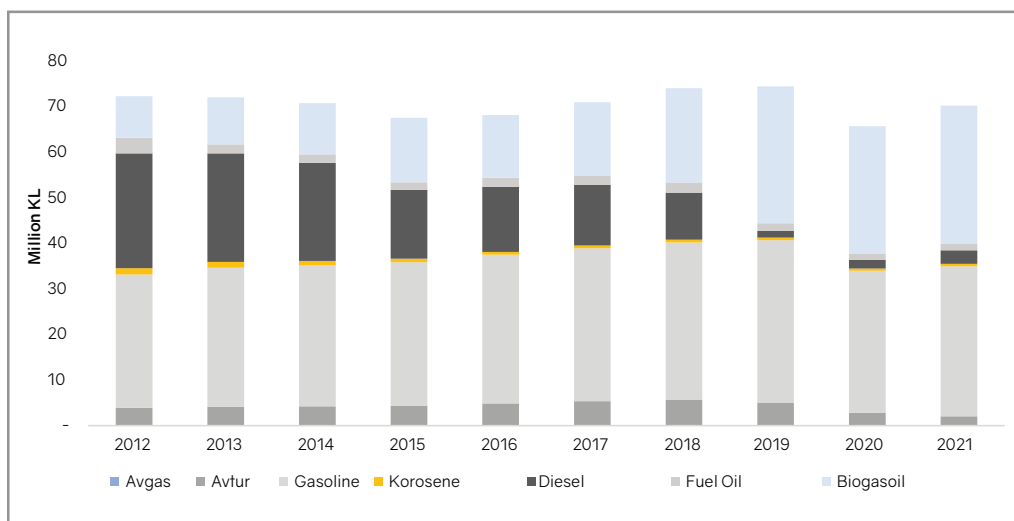
Source: Directorate General of Electricity MEMR, processed by
Secretariate General of National Energy Council

Picture 2.10 Electricity Consumption per Capita 2016-2021

2.4 FUEL CONSUMPTION

2.4.1 Fuel Consumption per Sector

In 2021, fuel consumption will increase by 5.7% compared to 2020 to 70 million KL in line with the increase in community mobility, especially the increase in activities on land, river-sea, and air transportation, due to the relaxation in social restrictions and the decrease in the number of Covid-19 cases. Under these conditions, there was an increase in the use of diesel oil by 59.9%, bio gasoil by 8.2%, fuel oil by 5.9% and gasoline by 5.6% compared to in 2020. The description of fuel consumption per type of fuel can be seen in Picture 2.11 below.

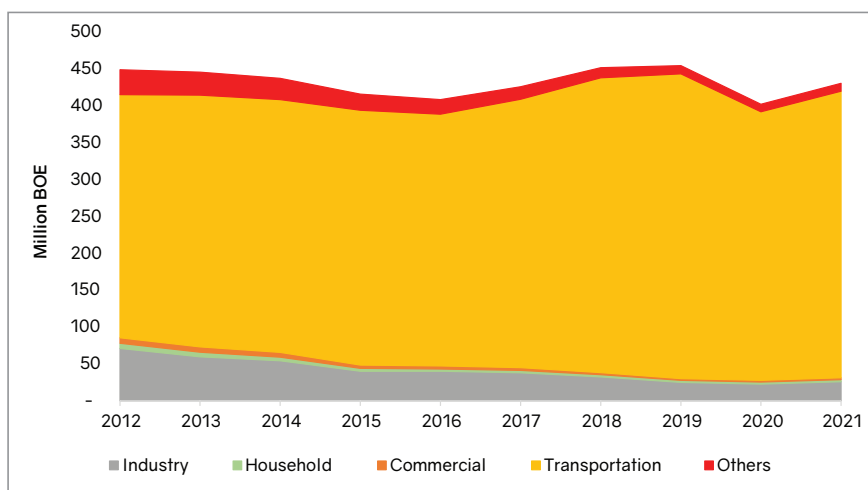


Source: HEESI, 2021

Picture 2.11 Fuel Consumption 2012–2021

Biogasoil consumption increased from 9 million KL in 2012 to 30 million KL in 2021 which is influenced by the B30 program. On the other hand, the consumption of diesel oil decreased from 25 million KL in 2012 to 3 million KL in 2021.

The largest use of fuel, when viewed by consuming sector, is dominated by the transportation sector followed by the industrial sector and other sectors as shown in Picture 2.12 below.



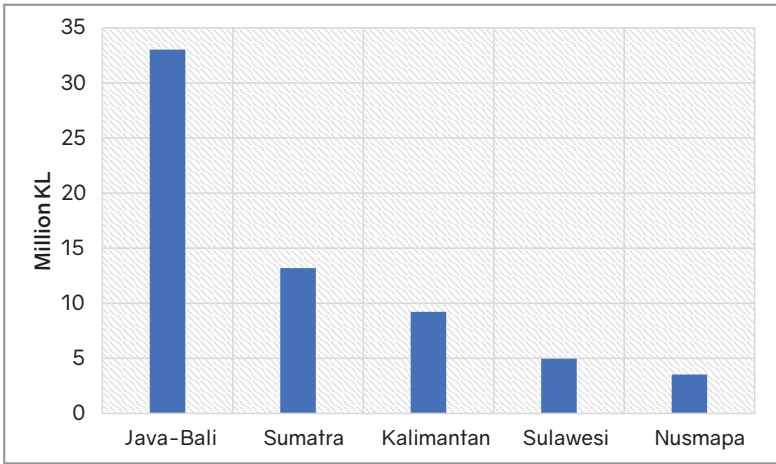
Source: HEESI, 2021

Picture 2.12 Fuel Consumption per Sector 2012 – 2021

The share of fuel consumption in transportation sector reached 73.5% in 2012 and continues to increase to 90.3% in 2021. This condition occurs since the substitution program from fuel to CNG has not been implemented due to the limited number of Gas Filling Station. In addition, the electric vehicle program that was launched a few years ago has not yet been implemented because regulations related to electric cars were only enacted in August 2019. On the other hand, the growth of road vehicles which includes cars, buses, trucks and motorcycles continues to increase. The largest vehicle growth is motorcycles, which currently reach 121.2 million units, followed by passenger cars with 16.9 million units. Thus, the number of vehicles is very influential on the increasing volume of fuel, especially gasoline.

2.4.2 Fuel Consumption per Region

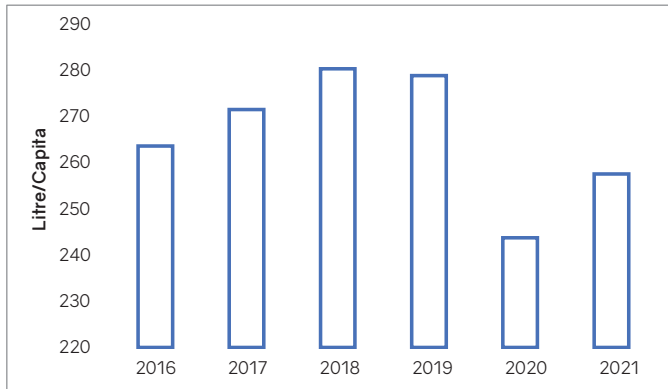
By region, most fuel consumption is in Java-Bali region at around 45.9%, followed by Sumatra at 22.3% and Kalimantan at 17.6% as shown in Picture 2.13. By province, the largest fuel consumption is in East Java, followed by West Java and Central Java. The high fuel consumption in Java-Bali is indeed in line with the high mobility and number of vehicles in the region. Based on BPS data, the number of Indonesian motorized vehicles in 2021 is 142.9 million units. Meanwhile, the number of motorcycles in Java and Bali is 75.2 million units (62.1% of the total motorcycle in Indonesia) and the number of passenger cars is 11.9 million units (70.5%).



Picture 2.13 Fuel Consumption per Region

2.4.3 Fuel Consumption per Capita

In 2021, fuel consumption per capita increased by 5.7 compared to 2016 as shown in Picture 2.14 .



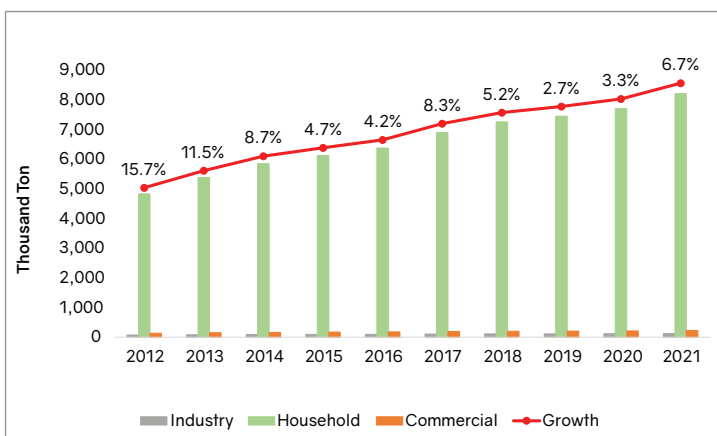
Picture 2.14 Fuel Consumption per Capita

2.5 LPG CONSUMPTION

2.5.1 LPG Consumption per Sector

Around 96% of LPG in Indonesia is consumed by the household sector as the impact of kerosene to LPG conversion program since 2007. In addition, starting in 2016 free LPG converter kits have been distributed to fishermen and farmers with the aim of saving fishermen's operational costs when fishing. LPG consumption in 2012 reached 5 million tonnes and continues to increase with an average growth rate for 2012–2021 of 6.1%.

The increase in LPG consumption, especially 3 kg LPG that is still subsidized, needs to be anticipated by the Government, given the large number of uses of 3 kg LPG which are not on target. The development of LPG consumption is shown in Picture 2.15.

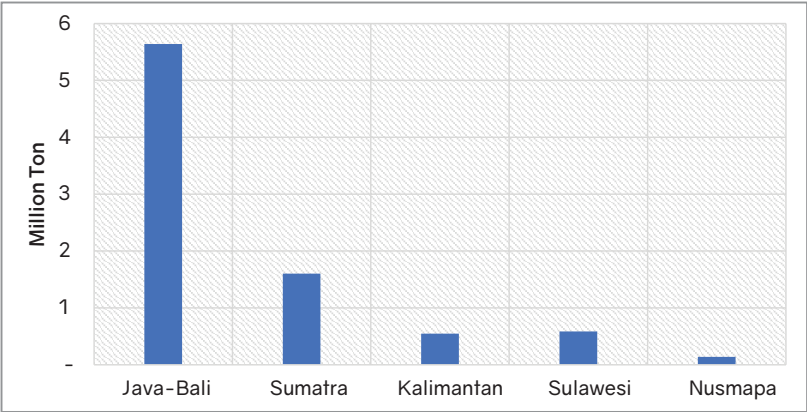


Source: HEESI, 2021

Picture 2.15 LPG Consumption 2012–2021

2.5.2 LPG Consumption per Region

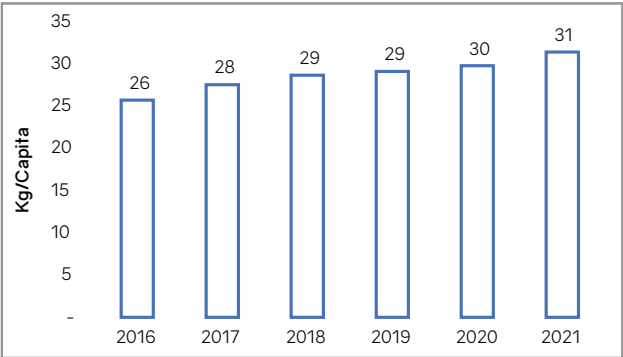
By region, the largest share of LPG consumption is in Java and Bali at 66.2%, followed by Sumatra at 18.8% and Sulawesi at 6.9% as shown in Picture 2.16. Considering that most of LPG is used for households, the LPG consumption per region is influenced by the number of households in each region. Based on BPS data, the number of households in Indonesia is 70 million, consisting of 42.4 million households in Java-Bali, 14.5 million households in Sumatra, 4.2 million households in Kalimantan, 4.7 million households in Sulawesi and 4.3 million households in Nusmapa.



Picture 2.16 LPG Consumption per Region

2.5.3 LPG Consumption per Capita

In contrast to electricity and fuel consumption, LPG consumption per capita always grows every year starting from 2016 as shown in Picture 2.17. The average growth for the last 5 years is 4.4%. The LPG consumption growth is indeed driven by the household sector with the allocation of 95.9% of the total LPG consumption.



Picture 2.17 LPG Consumption per Capita

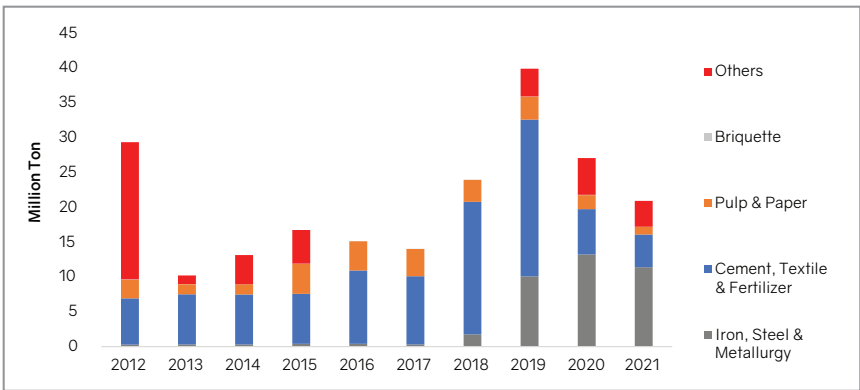
2.6 COAL CONSUMPTION

2.6.1 Coal Consumption in Industrial Sector

Coal sales in industrial sector increased from 29.3 million tons in 2012 to 20.9 million tons in 2021. Compared to sales in 2020, consumption in industrial sector in 2021 experienced a sharp decline of about 22.7%.

In 2021, coal consumption in industrial sector was dominated by the use of the iron, steel and metallurgical sectors of 11.4 million tons or around 54.5% of the total industrial sector consumption. It is then followed by the cement, textile and fertilizer industries of 4.7 million tons, other industries of 3.7 million tons, and the pulp and paper industry of 1.1 million tons.

Data on coal consumption in industrial sector can be seen in Picture 2.18.

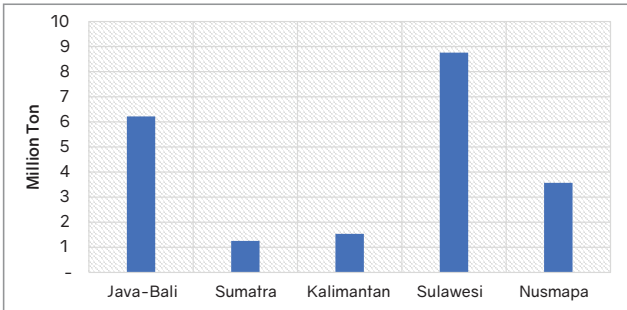


Source: HEESI, 2021

Picture 2.18 Indonesia Coal Consumption in Industry 2012–2021

2.6.2 Coal Consumption in Industry per Region

In 2021, Sulawesi region became the largest consumer of coal, reaching 41% of the total national coal consumption. It was caused by the large number of smelter factories as well as nickel, iron and steel industries that use coal as an energy source. The value of coal consumption in industrial sector per region can be seen in Picture 2.19.

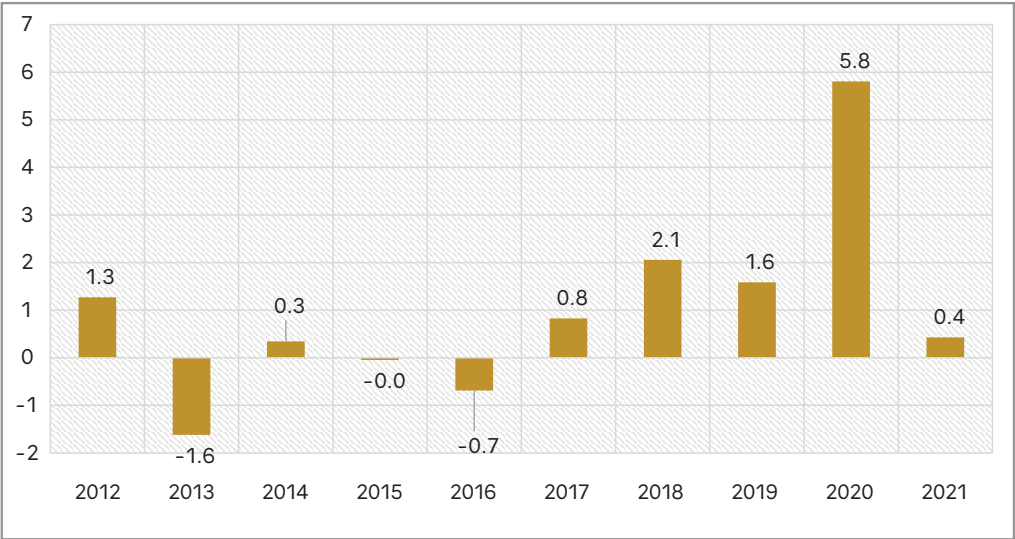


Picture 2.19 Coal Consumption in Industry per Region

2.7 ENERGY ELASTICITY AND INTENSITY

Energy elasticity is the ratio of final energy consumption growth to GDP growth in the same time period. The low or below one energy elasticity value indicates efficient use of energy because to increase 1% GDP growth needs energy demand growth below 1%.

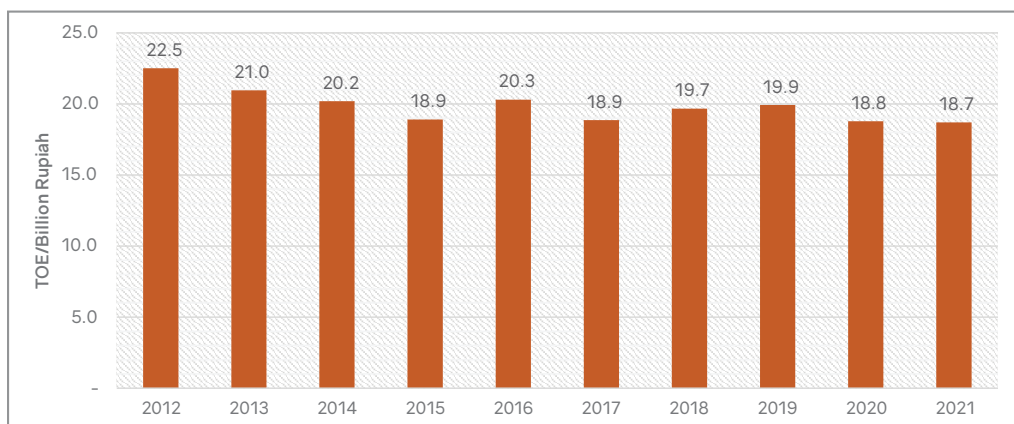
In accordance with the target of KEN, energy elasticity must be below one starting in 2025. The energy elasticity will gradually decrease from year to year, and by 2050 the energy elasticity is targeted to reach 0.46 which indicates that national energy use will be more efficient. Indonesia's energy elasticity from 2012–2021 fluctuated (Picture 2.20).



Source: HEESI, 2021

Picture 2.20 Indonesia Energy Elasticity 2012–2021

Meanwhile, primary energy intensity is the ratio of the amount of primary energy supply to the total gross domestic product (TOE/Billion Rupiah). Based on RUEN, the projection of primary energy intensity should decrease by 1% every year, but the realization of Indonesia's primary energy intensity from 2012–2021 fluctuated and tends to decline from 22.5 TOE/Billion Rupiah in 2012 to 18.7 TOE/Billion Rupiah in 2021 as shown in Picture 2.21.

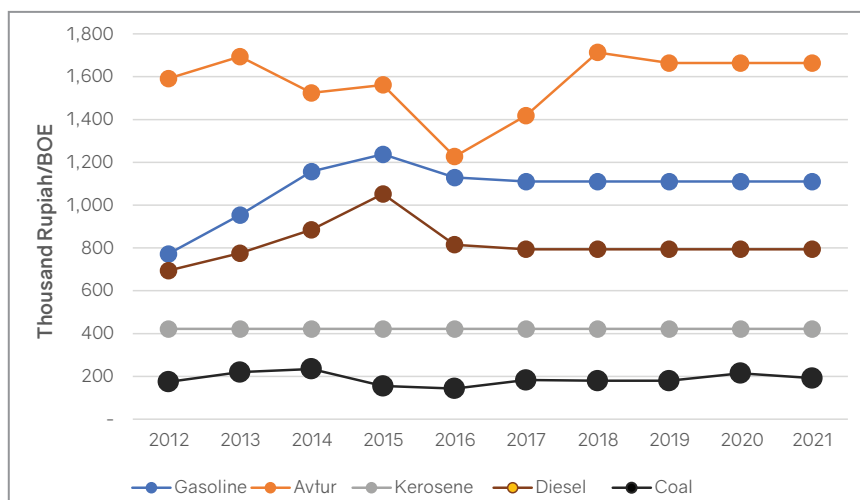


Source: HEESI, 2021

Picture 2.21 Indonesia Primary Energy Intensity 2012–2021

2.8 ENERGY PRICE

Indonesia needs energy with good quality at affordable energy prices to support the vision of becoming a developed country through the development of industry and other energy consuming sectors. If energy prices are compared in the same unit (thousand rupiah/BOE), then coal will be the cheapest energy source as shown in Picture 2.22 below.



Source: HEESI, 2021

Picture 2.22 Energy Price (Thousand Rupiah/BOE)

2.8.1 Electricity Price

Based on PLN's 2021 statistical book, the average selling price of electricity per kWh in 2021 is IDR 1,083.3. There is an increase of IDR 1,071.4 from the previous year. The price of electricity for the household sector is still subsidized while other sectors use economic prices. The determination of electricity tariffs by the Government is regulated in the Minister of Energy and Mineral Resources Regulation No. 3 of 2020 on the Fourth Amendment to the Minister of Energy and Mineral Resources Regulation No. 28 of 2016 on Electricity Tariffs provided by PT PLN (Persero). The Minister of Energy and Mineral Resources Regulation also stipulates that PT PLN (Persero) is obliged to announce the electricity tariff adjustment at the latest one month prior to the implementation of the new tariff adjustment. This is intended to increase transparency in domestic electricity tariffs. The development of electricity tariffs per sector in the last ten years is shown in Table 2.1.

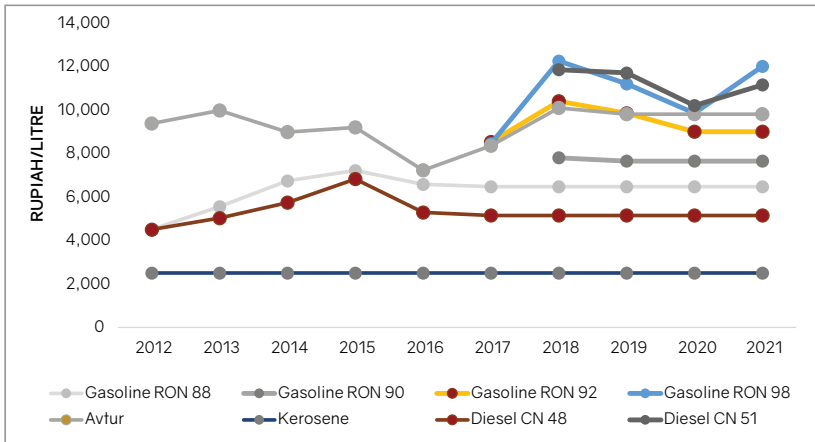
Table 2.1 Average Electricity Tariff per Costumer Group (Rupiah/kWh)

Year	Household	Industry	Business	Social	Government Office	Public Street Lighting	Average
2012	631.7	709.9	965.2	677.5	968.6	802.8	728.3
2013	692.1	796.4	1,116.6	756.9	1,092.2	911.0	818.4
2014	758.2	977.8	1,265.9	810.0	1,256.2	1,097.0	939.7
2015	837.0	1,142.7	1,284.2	812.4	1,324.5	1,459.1	1,034.5
2016	843.7	1,051.8	1,201.2	816.0	1,234.7	1,415.3	991.4
2017	1,056.0	1,088.8	1,245.6	821.3	1,278.5	1,461.5	1,105.1
2018	1,102.4	1,085.3	1,244.0	823.2	1,280.1	1,461.6	1,123.0
2019	1,098.8	1,100.7	1,258.3	830.7	1,291.5	1,465.1	1,129.6
2020	991.9	1,090.9	1,239.3	804.1	1,299.2	1,459.9	1,071.4
2021	1,024.0	1,086.2	1,234.7	806.4	1,292.9	1,447.1	1,083.3

Source: PLN Statistics, 2021

2.8.2 Fuel Price

Price of fuel that is still subsidized is diesel for goods transportation and kerosene for household. Meanwhile, since 2014, fuel subsidy has been removed and the price follows the economical price. The development of average fuel price in the last ten years is shown in Picture 2.23.

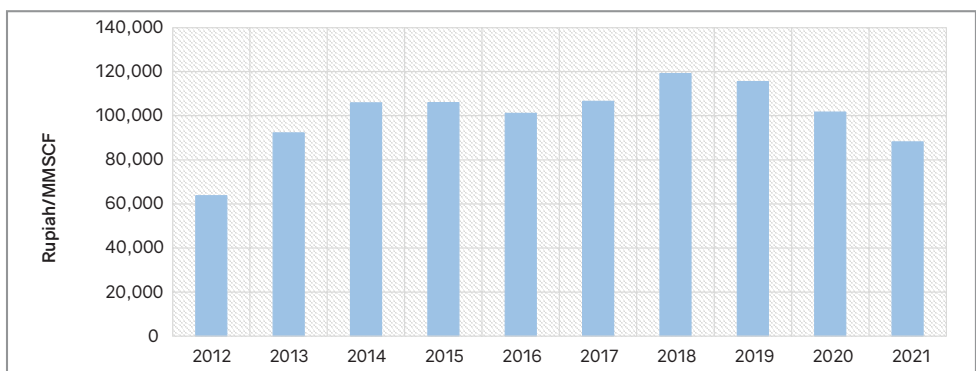


Source: HEESI and Buku Saku KESDM, 2021

Picture 2.23 Fuel Price Development 2012 – 2021

2.8.3 Natural Gas Price

Gas price is stipulated based on the agreement between seller or supplier and buyer (consumer). However, since 2015, the government has issued regulation on gas price for industry due to the high gas price in industry. In 2016, the government issued policy on gas price at 6 US dollar per MMBTU as stipulated in Presidential Regulation No. 40 of 2016 on Gas Price Stipulation. The government then made a derivative to this Presidential Regulation through Minister of Energy and Mineral Resources Regulation No. 8 of 2020 on the Procedure in Stipulating Certain User and Gas Price in Industry. The technical arrangements are stipulated in Minister of Energy and Mineral Resources Decision No. 89K/10/MEM/2020 on Certain Gas User and Price in Industry. In this Ministerial Decision, seven industries will receive gas at the price of 6 US dollar per MMBTU namely the industry of fertilizer, petrochemical, oleochemical, steel, ceramics, glass, and rubber gloves. Based on this regulation, the price scheme is valid from 2020 to 2024. Gas price in the last ten years is shown in Picture 2.24.

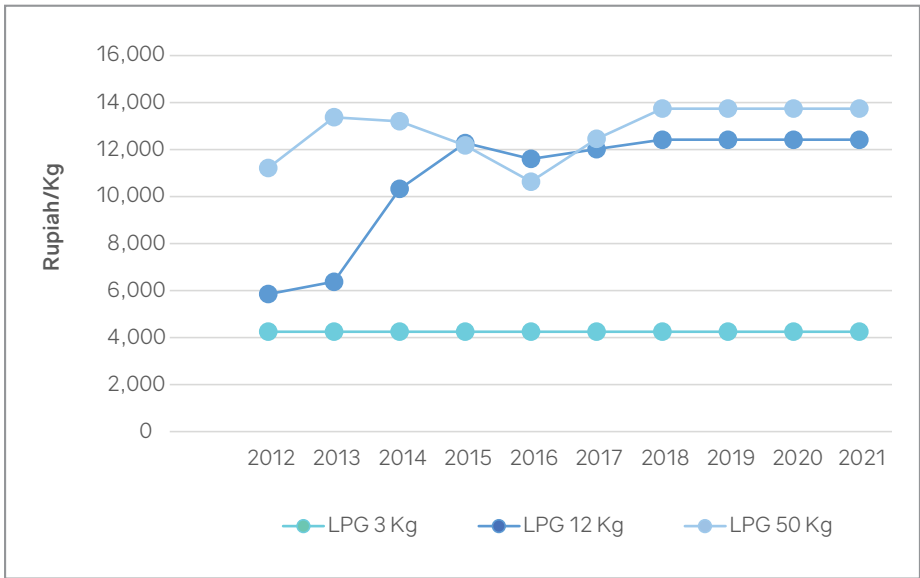


Source: PLN Statistics, 2021

Picture 2.24 Gas Price Development

2.8.4 LPG Price

Total LPG subsidy in 2021 is 67.6 trillion rupiah. LPG receives the biggest subsidy from the State Budget. The subsidized LPG is the 3 kg cylinder LPG for the poor or low-income household. In reality, the 3 kg LPG is used by high income household and commercial sector. The subsidy then does not meet the target. The price of subsidized LPG is far below normal LPG. In reality, 85% of household uses subsidized LPG. The comparison between subsidized and non subsidized LPG is seen in Picture 2.25.

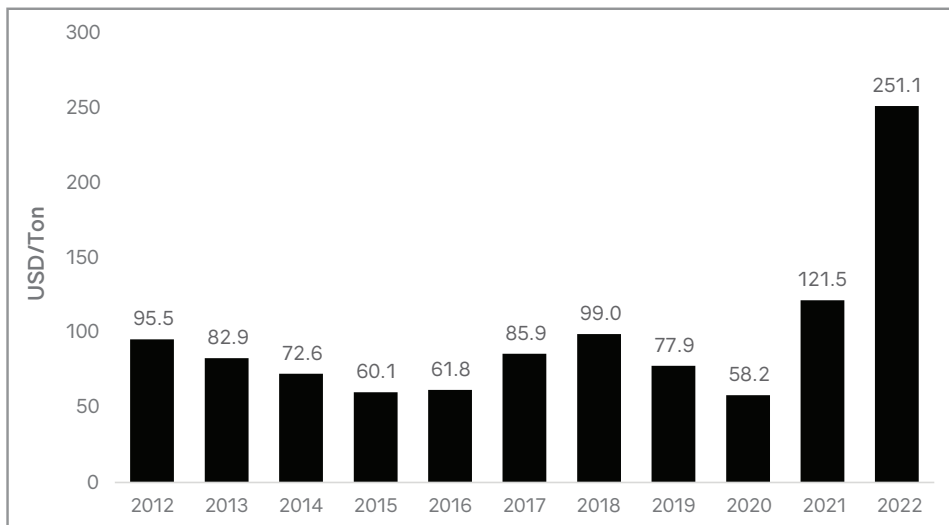


Source: HEESI, 2021

Picture 2.25 LPG Price Development

2.8.5 Coal Price

The Directorate General of Mineral and Coal issues Coal Reference Price monthly as the benchmark price for domestic coal. The average coal reference price in 2021 reached \$121 USD, which is the highest average coal reference price in the last 10 years (highest HBA is \$215 USD in November 2021). Indonesia as one of coal producing countries has had a good impact from the increase in world coal prices, but it needs to be a concern considering that domestic power plants are still dominated by Coal Fired power plant. The development of the average coal reference price from 2012 to 2022 can be seen in Picture 2.26.

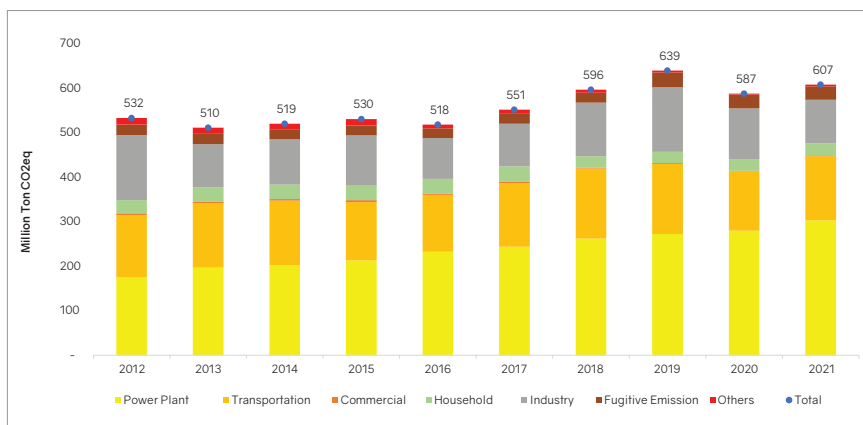


Source: DG Mineral and Coal, MEMR

Picture 2.26 Development of the Average Coal Reference Price

2.9 EMISSION

Emission is the result of fuel combustion from energy consuming sectors and power plant activity. Total emission in 2021 is 607 million ton CO₂eq. The biggest contributor is power plant sector with 49.8%, followed by transportation and industry with 23.7% and 16.1% respectively. The development of CO₂ emission in 2012-2021 is shown in Picture 2.27.



Source: Pusdatin MEMR, 2021

Picture 2.27 CO₂ Emission from Fuel Combustion

2.10 ENERGY INFRASTRUCTURE

2.10.1 Oil Refinery

At present, Indonesia's oil refinery capacity is 1,151.1 million BPH. The details of oil refineries and their capacity is resumed in Table 2.2.

Table 2.2 Indonesia Oil Refinery Capacity 2021

Refinery	Capacity (Million BPH)
Dumai	177.0
Musi	127.3
Cilacap	348.0
Balikpapan	260.0
Balongan	125.0
Cepu	3.8
Kasim	10.0
Tuban (TPPI)	100.0
Total Kapasitas	1,151.1

Source: HEESI, 2021

In the last 10 years, oil refinery capacity in Indonesia does not change a lot. The latest additional capacity occurred in 2015 with the additional 12 million BPH to Tri Wahana Universal (TWU) refinery Unit II. TWU refinery is the oil refinery located in Bojonegoro, East Java which produces main products such as straight run gasoline, marine diesel oil, and diesel. TWU refinery uses crude oil from Banyu Urip Field and Cepu Block. However, this refinery only operated for 2 years and it stopped operating in early 2018 due to lack of refinery raw material supply.

Crude oil for refinery input in 2021 is 300 million barrels from domestic production and import. Besides crude oil, oil refinery input also needs gas and intermedia. After being processed, oil refinery will produce gasoline, diesel oil, fuel oil, kerosene, avtur, avgas and other refinery products (non fuel) such as LPG, lubricant, naptha, and others. Oil refinery production in 2021 reached 255 million barrels and non fuel refinery production reached 41 million barrels.

2.10.2 LPG Refinery

Currently, Indonesia's LPG refinery capacity is 4,740.3 million to per annum but only 3,878.5 million ton per annum that is still operating until 2021. The details of LPG refinery and its capacity are resumed in Table 2.3.

Table 2.3 Indonesia LPG Refinery Capacity 2021

LPG Refinery	Capacity (Million Ton per Annum)
LPG Refinery (Oil refinery)	1,331.0
PT Pertamina (Dumai)	68.0
PT Pertamina (Plaju)	131.0
PT Pertamina (Cilacap)	318.0
PT Pertamina (Balikpapan)	91.0
PT Pertamina (Balongan)	548.0
PT TPPI	175.0
Upstream LPG Refinery	2,342.0
PT Badak NGL	1,000.0
PT Chevron*	90.0
PT Petrogas	14.0
PT Petrochina	600.0
PT Conoco Philips*	525.0
PT Saka Indonesia	113.0
Downstream LPG Refinery	1,067.3
PT Pertamina (P.Brandan)*	44.0
PT Maruta Bumi Prima*	17.0
PT Medco LPG Kaji*	73.0
PT Pertamina (Mundu)	37.0
PT Titis Sampurna	73.0
PT Sumber Daya Kelola (Tugu Barat)*	7.0
PT Bina Bangun Wibawa Mukti	55.0
PT Surya Esa Perkasa	82.0
PT Yudhistira Haka Perkasa*	44.0
PT Wahana Insannugraha	37.0
PT Media Karya Sentosa phase I*	58.0
PT Media Karya Sentosa phase II	84.0
PT Yudistira Energi	58.0
PT Gasuma Federal Indonesia	26.0
PT Pertasamtan Gas	259.0
PT Sumber Daya Kelola (Losarang)*	3.8
PT Arsynergy Resources	109.5
Total Capacity	4,740.3
Total Operating Capacity	3,878.5

Source: Oil and Gas Statistics Semester I 2021

Note: (*) Stop operating

2.10.3 LNG Refinery

At present, Indonesia's LNG refinery capacity is 31.2 million ton per annum. The details of LNG refineries and their capacity is summarized in Table 2.4.

Table 2.4 Indonesia LNG Refinery Capacity 2021

LNG Refinery	Capacity (Million Ton per Annum)
PT Arun LNG*	12.9
PT Badak	21.6
PT BP	7.6
PT Donggi Senoro	2.0
Total Capacity	44.1
Total Operating Capacity	31.2

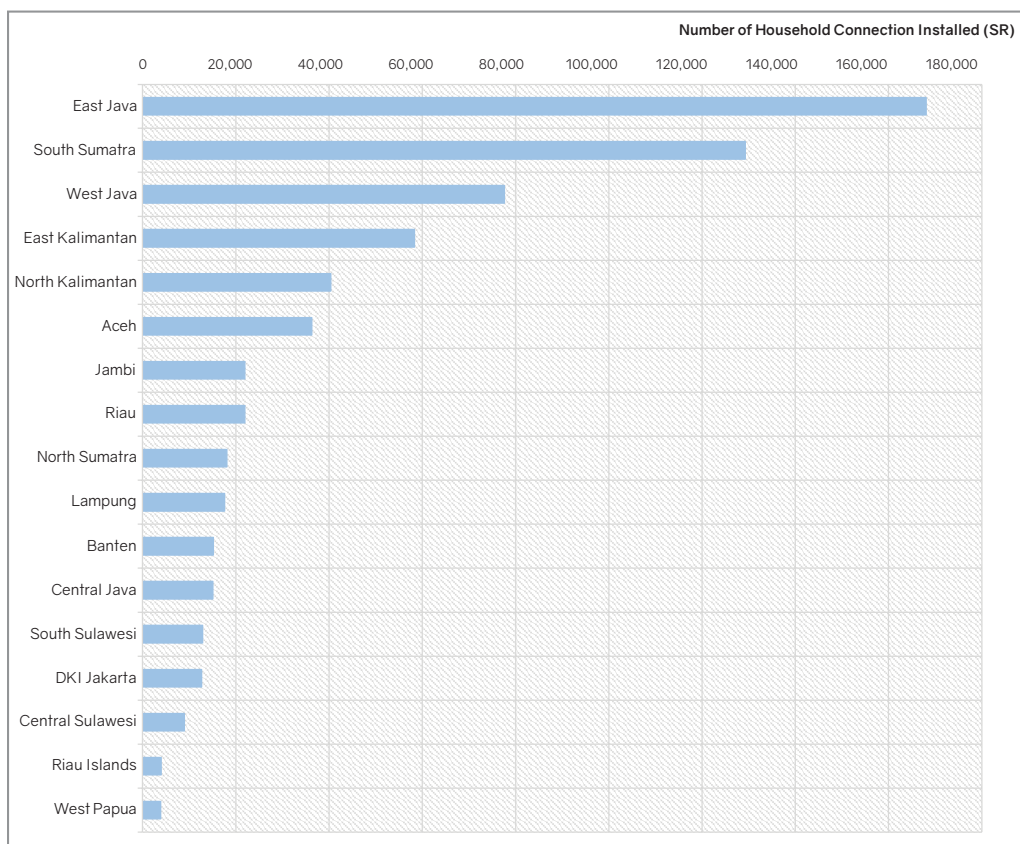
Source: Oil and Gas Statistics Semester I 2021

Note: (*) Stop operating

In October 2014, PT Arun's LNG refinery stopped operating due to the unavailable gas supply from Arun gas field. This refinery, located in Lhokseumawe, Aceh is then transformed into regasification terminal by Pertamina through PT Perta Arun Gas to meet the gas supply for industry and electricity. Total LNG production in 2021 from these three fields is 14,712.2 thousand metric ton. Most LNG is exported especially to China, Korea, Japan and Taiwan based on long-term contract reaching 416,691 thousand MMBTU in 2021 or only 23% of LNG is used for domestic need, especially to meet the demand from power sector.

2.10.4 City Gas Network

From 2009 to 2021, a natural gas distribution network for households (city gas) to connect 799 thousand households (SR) has been built from the State Budget and Non-State Budget. Specifically in 2021, 127 thousand household connections have been installed in 21 regencies/cities throughout Indonesia. As shown in Picture 2.28, the development of natural gas network infrastructure in 10 years has reached 17 provinces in all regions of Indonesia, with the most networks being built in East Java (25.4%) and South Sumatra (19.5%), in accordance with the availability of natural gas sources.

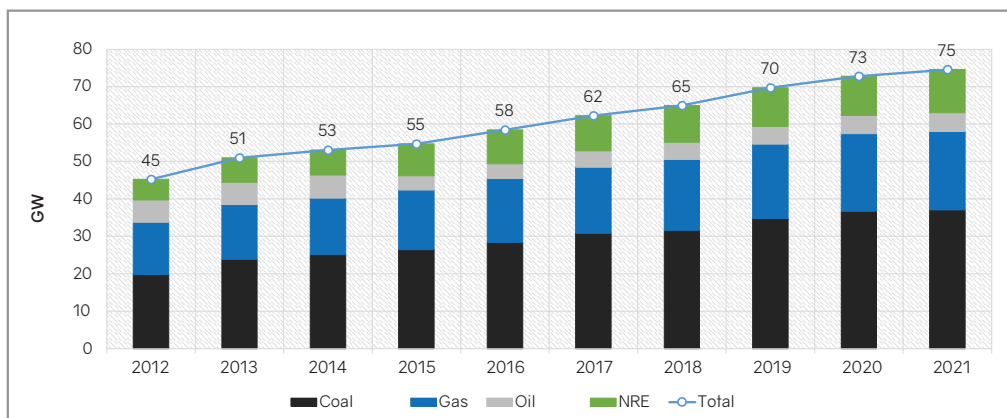


Source: Oil and Gas Statistics Semester I 2021 and presentation by Directorate General of Oil and Gas

Picture 2.28 Gas Network for Household Infrastructure Built by The Government per Province

2.10.5 Power Plant

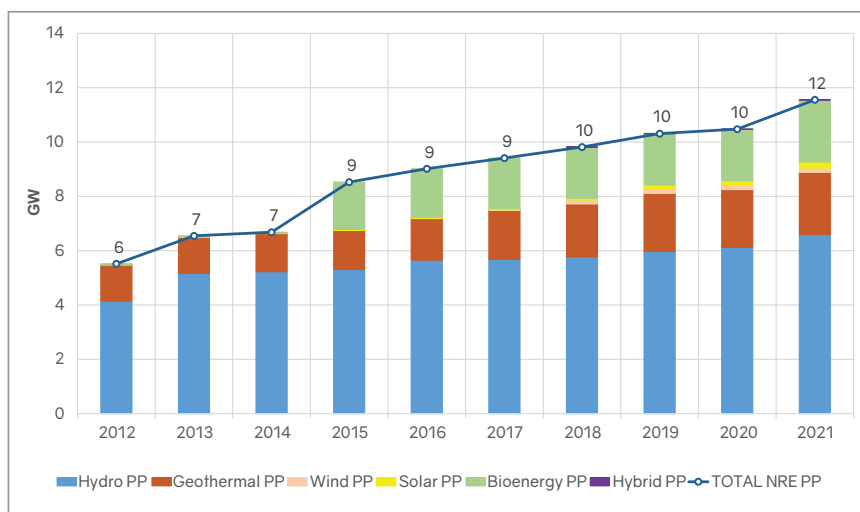
Until the end of 2021, Indonesia's total power plant capacity is 75 GW, consisting of 71 GW on grid power plant and 3 GW off grid power plant. This capacity shows that additional power plant capacity is almost doubled in the last 10 years as shown in Picture 2.29. Power plant is still dominated by coal which contributes almost half of the total national capacity. Gas follows with the contribution of 28%. Meanwhile, NRE based-power plant is only 15%. It gives additional 6 GW in the last 10 years.



Source: HEESI, 2021

Picture 2.29 Power Plant Installed Capacity per Energy Type 2012–2021

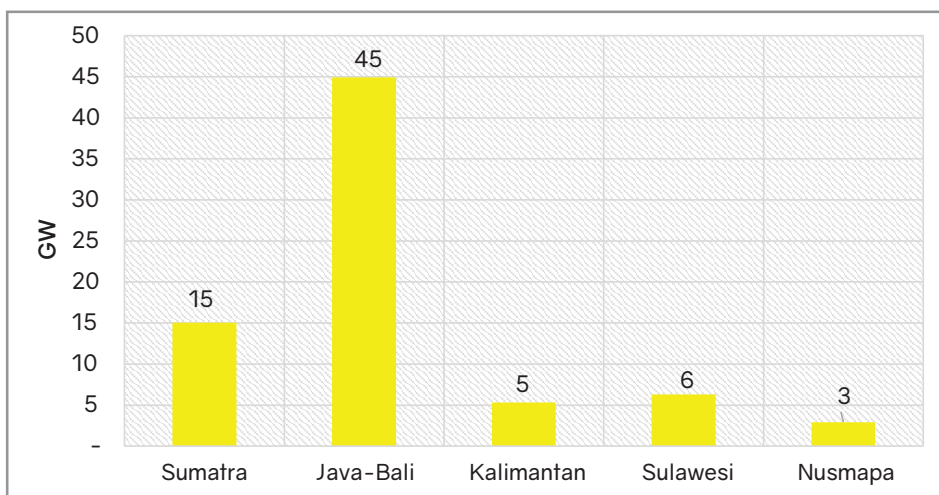
The use of NRE in electricity generation is dominated by hydropower (57%), geothermal (20%), and biomass (18%), as shown in Picture 2.30. Meanwhile, the utilization of solar power, both on grid and off grid, has only reached 225 MW.



Source: HEESI, 2021

Picture 2.30 NRE-Based Power Plant Installed Capacity 2012–2021

Power plant infrastructure is spread in all areas in Indonesia, but only 63% of power plant capacity is built in Java-Bali region (Picture 2.31). Meanwhile, power plant capacity is only 20.2% in Sumatra, 7.1% in Kalimantan, 8.5% in Sulawesi, and 3.9% in Nusmapa region from the total national installed capacity. This statistic shows that power plant infrastructure has not been developed evenly.



Picture 2.31 Power Plant Capacity per Region 2021

2.11 ENERGY POLICY

2.11.1 Oil and Gas

State oil and gas exploitation is regulated in Law of the Republic of Indonesia number 22 of 2001 on Oil and Gas, or often referred to as the Oil and Gas Law. In its implementation, there are several derivatives of the Oil and Gas namely Government Regulation No. 34 of 2005 on Oil and Gas Upstream Business Activities, Government Regulation No. 30 of 2009 on Oil and Gas Downstream Business Activities, and Presidential Regulation No. 36 of 2018 on the Implementation of Oil and Gas Upstream Business Activities Management.

Currently, the Oil and Gas Law is being discussed by the House of Representatives for revision through the formation of the Oil and Gas Law Draft. The drafting of this Law is intended to amend Law No. 22 of 2001 which was annulled by the Constitutional Court 10 years ago. One of the main issues in the Oil and Gas Law is the establishment of a definitive institution to replace the Special Task Force for Upstream Oil and Gas Business Activities (SKK Migas).

Another issue is the amendment in the 10% participating interest for Regional Owned Enterprises (ROE). The amendment is the obligation of ROE to seek financing capital or find partners to establish companies in the management of the oil and gas business. In addition, the issue to include the cost recovery system in the Oil and Gas Law Draft was also discussed. It is an effort not to regulate cost recovery in derivative regulations so that the change of government and ministers will not change policies.

2.11.2 Coal

Indonesia's mineral and coal mining activities are regulated in Law of the Republic of Indonesia Number 3 of 2020. This Law is an amendment to the previous legal basis, namely Law of the Republic of Indonesia No. 4 of 2009 on Mineral and Coal Mining. Some of the issues that become points of amendments in the revision of the Law include the following:

1. **Intersectoral Problem Solving**
Several sectors have authority that overlaps with the Ministry of Energy and Mineral Resources, such as smelter licensing and mining designations in forest areas.
2. **Conception of Mining Legal Area**
The new regulatory revisions have accommodated mining investigation and research activities in all Indonesian jurisdictions.
3. **Adjustment of Licensing Nomenclature and Withdrawal of Authority**
In the previous regulations, the form of mining business only includes: IUP, IPR, and IUPK. Meanwhile, the new regulation includes more detailed provisions in the implementation of mining businesses based on a Business Licensing from the Central Government. The types of licenses include: IUP, IUPK, IUPK as Continuation of Contract/Agreement Operation, IPR, SIPB, Assignment License, Transportation and Sales License, Mining Services Business License, and Mining Business License for Sales. In addition, it also regulates the licensing authority that can be delegated to the Regional Government.
4. **Increasing Mineral Value Added and Development and/or Utilization of Coal**
In the new regulation, there are additional obligations for 'processing and/or refining' in mineral and rock mining, as well as 'development and/or utilization' in coal mining. In addition, there are also additional detailed provisions on cooperation partners in processing and refining activities.
5. **Strengthening the Implementation of Reclamation and Post-mining**
The new regulation has accommodated the provisions on the obligation to carry out reclamation and post-mining as well as the criminal sanctions.
6. **Licensing Period for Integrated IUP or IUPK**
The new regulation has added provision on licensing period for IUP/IUPK that have been integrated with processing and/or refining facilities.
7. **Accommodating the Constitutional Court's Decision on Law No. 4 of 2009.**
In the new regulation, there are changes/adjustments to the clause regarding the mining area set by the central government after being determined by the provincial government, as well as the abolition of the minimum area of Exploration WIUP.
8. **Continuation of KK/PPK2B Operations**
In the new regulation, there are additional provisions regarding contract extension guarantees, efforts to increase state revenue, utilization of goods obtained during the PPK2B implementation period that have been designated as state assets, as well as the obligation of Production Operation IUPK holders to carry out domestic coal development/ utilization activities.

2.11.3 Renewable Energy

The government continues to create a conducive ecosystem for the development of NRE in Indonesia, including preparing and improving the regulatory framework. This year, on 13 September 2022, President Joko Widodo has just issued Presidential Regulation No.112 of 2022 on the Acceleration of Renewable Energy Development for the Power Supply. In general, this Presidential Regulation mandates the acceleration of the construction of low-emission and environmentally friendly power plants as well as the prohibition of the construction of new Coal Fired Power Plants (Coal Fired PP), without disturbing existing plants. Through this Presidential Regulation, it is hoped that it can help increase investment, accelerate the achievement of renewable energy targets, as well as reduce greenhouse gas emissions.

Based on Presidential Regulation No.112 of 2022, the construction of power plants will be carried out selectively and the construction of NRE-based power plants is targeted to be carried out hand in hand. The development of new Coal Fired PP is prohibited except for Coal Fired PP which has been stipulated in the Electricity Supply Business Plan (RUPTL) prior to the enactment of this Presidential Regulation or for Coal Fired PP that meets the following requirements:

1. Integrated with industries built to increase natural resources value added or industries included in National Strategic Projects that have major contributions to job creation and/or national economic growth.
2. Committed to reduce greenhouse gas emissions at least 35% (thirty five percent) within a period of 10 (ten) years since the operation of the Coal Fired PP compared to the average Coal Fired PP emissions in Indonesia in 2021 through technology development, carbon offsets, and/or Renewable Energy mix.
3. Operating no later than 2050.

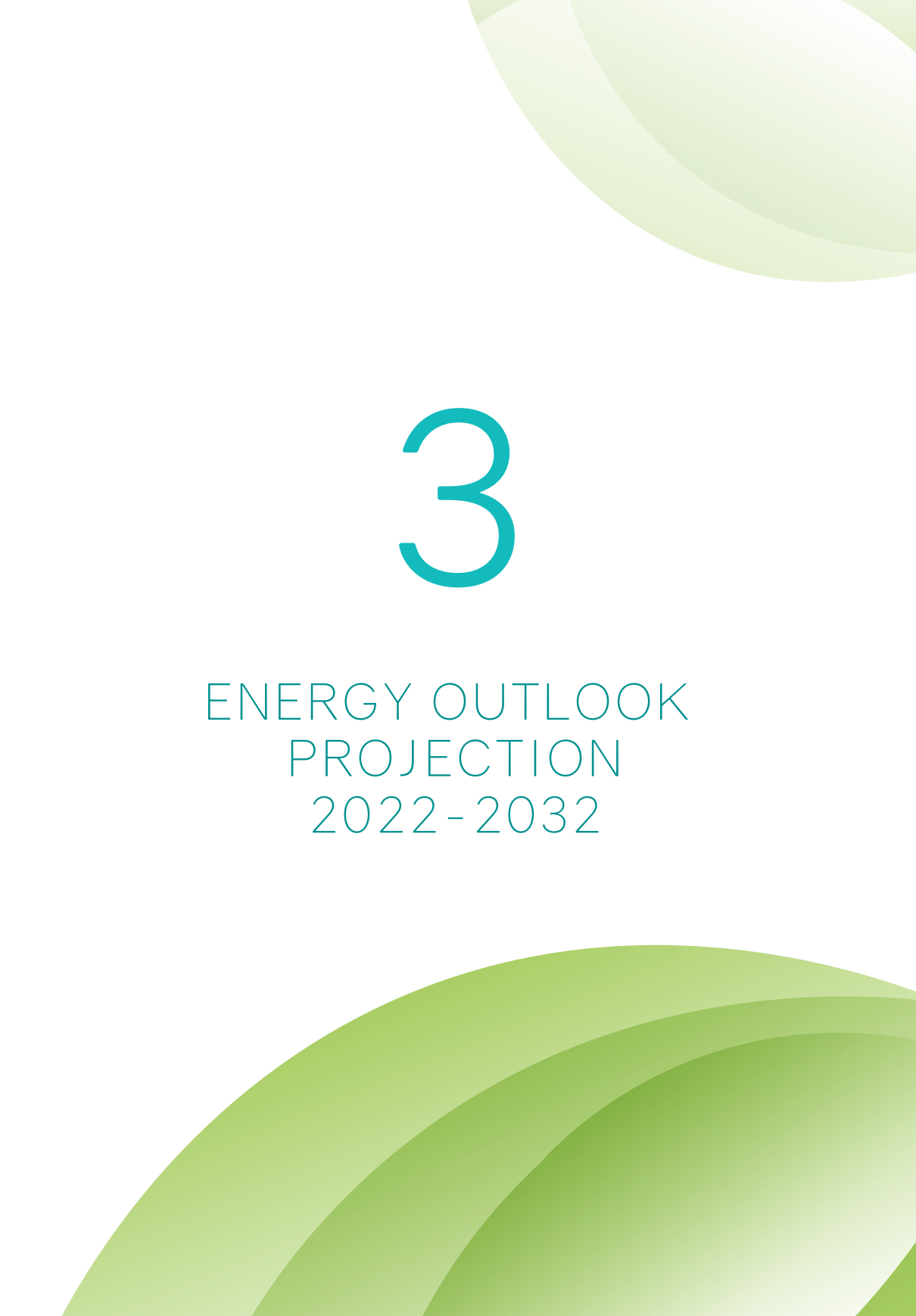
Meanwhile, the Presidential Regulation No.112 of 2022 regulates several things regarding the mechanism for the use of renewable energy in the electricity sector, such as: the purchase price of electricity, the implementation of the purchase of electricity, the Electricity Sales and Purchase Agreement, and the role of the government related to new and renewable energy such as support, fostering, and supervision. In addition to the electricity purchase tariff, the implementation of the purchase of electricity is also regulated, which can be done in two ways, namely through direct appointment or direct election. This Presidential Regulation also stipulates provisions related to incentives, both fiscal and non-fiscal. The fiscal incentives can be in the form of income tax facilities, import facilities, land tax facilities, geothermal development facilities, and/or support for financing and/or guarantee facilities. Then, non-fiscal incentives can be in the form of incentives provided by both the central government and/or regional governments.

In addition to the Presidential Regulation, the Government is currently finalizing the draft of the New Energy and Renewable Energy Law, or often abbreviated as the EBET Law. In

the future, the EBET Law is expected to become a legal umbrella for renewable energy development that can provide legal certainty, harmonize relevant laws and regulations, strengthen institutions and governance of renewable energy development, create a conducive investment climate for renewable energy investors, and optimize renewable energy resources to support industrial development and national economy

Meanwhile, technical rules regarding its utilization are still regulated in separate legal products according to each type of energy. Some of these regulations include:

1. Law No. 21 of 2014 on Geothermal;
2. Minister of Energy and Mineral Resources Regulation No. 32 of 2008 on Provision, Utilization and Trading of Biofuel as Other Fuels;
3. Minister of Energy and Mineral Resources Regulation No. 44 of 2015 on the Electricity Purchase from Municipal Waste PP by PLN;
4. Presidential Regulation No. 47 of 2017 on the Provision of Energy-Saving Solar Lamps (LTSHE) for Unelectrified Community;
5. Minister of Energy and Mineral Resources Regulation No. 50 of 2017 on the Utilization of NRE Sources for Power Supply;
6. Minister of Energy and Mineral Resources Regulation No. 41 of 2018 on the Provision and Utilization of Biofuel (Biodiesel Type);
7. Minister of Energy and Mineral Resources Regulation No. 4 of 2020 on Utilization of Renewable Energy Sources for Power Supply;
8. Minister of Energy and Mineral Resources Regulation No. 26 of 2021 on Rooftop Solar Power Plants Connected to the Power Network of Power Supply for Public Interest License Holder.



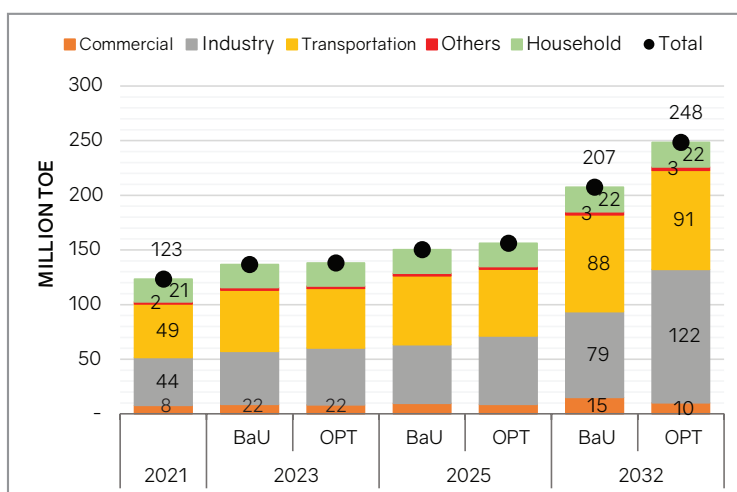
3

ENERGY OUTLOOK PROJECTION 2022-2032

3 ENERGY OUTLOOK PROJECTION 2022-2032

3.1 FINAL ENERGY CONSUMPTION

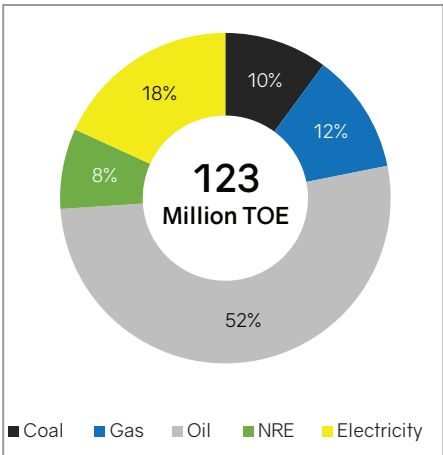
Energy demand in each scenario is influenced by economic growth, population growth and policies taken during the projection period. In the next ten years, final energy demand in the BaU scenario is projected to grow by an average of 4.8%, while the final energy demand in the OPT scenario grows at a higher average of 6.6% per year. The difference between the two scenarios is influenced by the program of electric vehicles and the use of biofuels in the transportation sector which is higher in the OPT scenario. Meanwhile, in the household sector, the assumption of LPG stoves to city gas and electricity substitution in the OPT scenario is greater. Energy demand in each scenario in 2032 is described in Picture 3.1.



Picture 3.1 Final Energy Demand per Sector until 2032

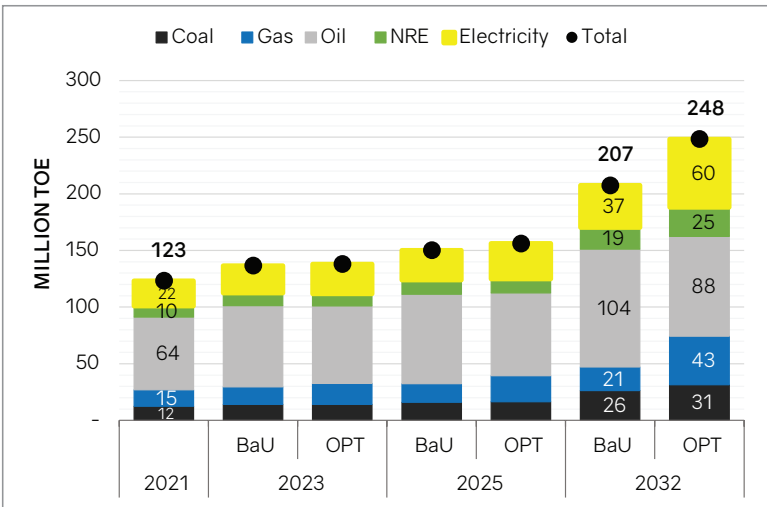
In 2021 the largest share of final energy consumption was from the transportation sector, but in 2032 in the OPT scenario, the largest share will shift to the industrial sector (49.2%). This is influenced by higher growth in the industrial sector in the OPT scenario, to pursue the vision of Indonesia Onward in 2045.

Based on type of energy, fuel becomes the biggest energy source in 2021 with 52.1% followed by electricity with 18.2%, gas with 11.8% and coal with 10.1%. Meanwhile, NRE share in final energy consumption is only 7.9% as seen in Picture 3.2.



Picture 3.2 Final Energy Share per Energy Type 2021

There is no significant change in energy use substitution in BaU scenario. Thus, in 2032 the biggest energy demand is still from fuel (50.1%). In OPT scenario, the role of fuel declines into 35.6% due to substitution policy from fuel to gas, electricity and biofuel. Meanwhile, NRE share increases into 10.2% due to the use of biogasoline and the increasing use of biodiesel. The complete overview of final energy demand share per type of energy in 2032 is presented in Picture 3.3



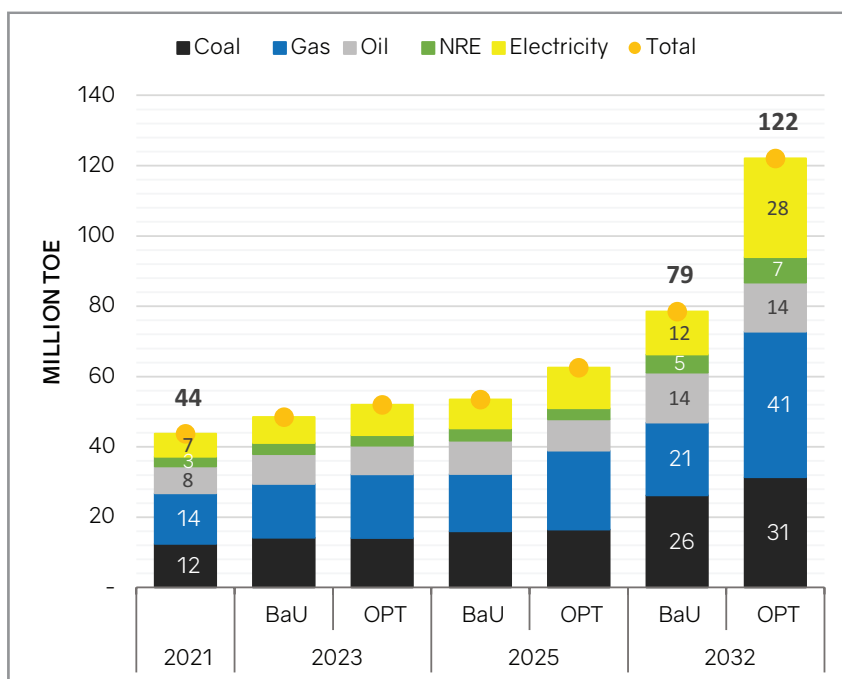
Picture 3.3 Final Energy Demand per Energy Type until 2032

3.1.1 Final Energy Consumption per Sector

3.1.1.1 Industrial Sector

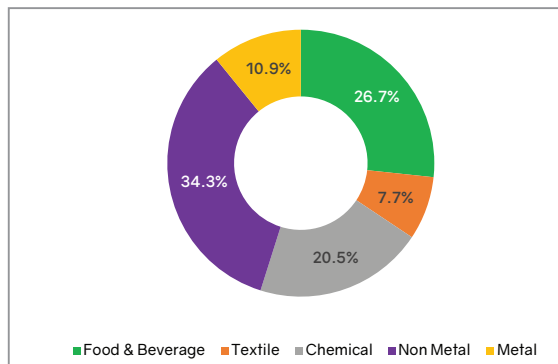
Coal, gas and electricity become the biggest energy sources consumed in industrial sector. Meanwhile, NRE (biomass) is mostly used for food as well as pulp and paper industry. Several food industries still use biomass (charcoal and wood) as fuel, while pulp and paper industry use renewable energy such as oil palm shells, rice straw, biogas and black liquor to substitute coal. Fuel is only used for power supply back up.

Industrial sector has not developed innovation plan on energy substitution, except biodiesel program that has been implemented in all sectors and biomass to coal substitution in several pulp & paper industry. Energy demand projection in industry based on two scenarios is seen in Picture 3.4.



Picture 3.4 Final Energy Demand in Industry per Energy Type

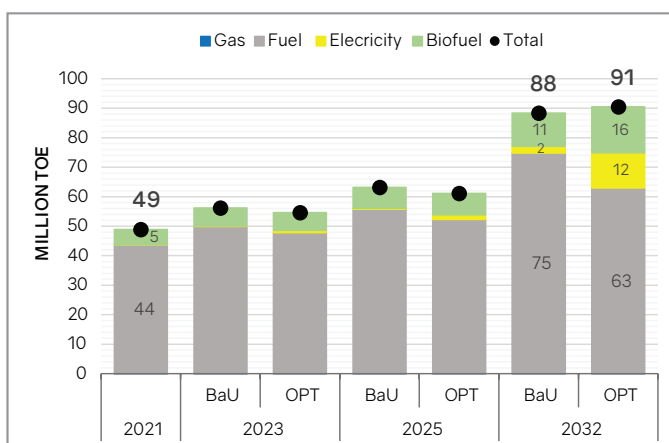
In 2032 the largest energy consumption in the industrial sector will be gas and coal. Gas is mainly used in the petrochemical industry including fertilizers (as feedstock), ceramics and metal industries. While coal is mainly used in the cement industry, pulp and paper industry as well as food industry. Based on the type of industry, there are 6 categories of industries that consume the most energy can be found in Picture 3.5.



Picture 3.5 Energy Demand Share of the Six Biggest Industries in BaU Scenario 2032

3.1.1.2 Transportation Sector

In 2021, energy consumption in transportation reached 49 million TOE and became the biggest consuming sector compared to the other sectors. Around 89.4% of final energy consumption in transportation is from fuel, while the rest 10.5% is from biofuel (biodiesel) and only 0.1% is from gas and electricity. Currently fuel consumption still highly depends on import. If this condition is not anticipated, it will burden the state and influence trade balance. Therefore, the government is reducing fuel import especially gasoline through substitution to biofuel (biodiesel and biogasoline), CNG and battery based-electric vehicle. The use of EV is higher in OPT scenario than in BaU scenario. Thus, fuel consumption is lower since it is replaced by electricity, CNG and biofuel. Energy consumption projection in transportation sector per type of energy is presented in Picture 3.6.



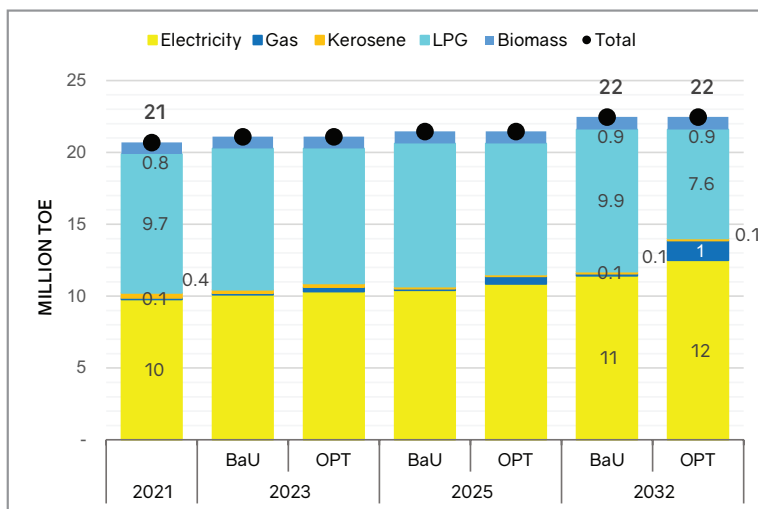
Picture 3.6 Energy Demand in Transportation in Both Scenario

In 2032, the use of biofuels in transportation, especially in OPT scenario, will reach 17.3% with the increase in the use of biodiesel to B40 and biogasoline to E5. Meanwhile, electricity

consumption in transportation is projected to increase to be around 12 million TOE (13.2 %) in the OPT scenario and only 2 million TOE (2.6%) in the BaU scenario. Thus, the use of fuel is only 69.5% in OPT scenario and 84.5% in BaU scenario of the total final energy demand in the transportation sector.

3.1.1.3 Household Sector

Energy demand in household sector is also influenced by the increasing number of households. In 2032, energy use in household sector in BaU scenario is dominated by electricity (50.8%), LPG (44.0%), biomass (3.9%), and the rest is natural gas and kerosene. Meanwhile, in OPT scenario, electricity demand in 2032 will reach 55.6% of total energy demand in the household sector, while LPG demand will be 33.7% and natural gas will be 6.1%. Efforts to reduce LPG imports with LPG to electricity and gas substitution program will reduce LPG demand from 9.9 million TOE in 2021 to 7.6 million TOE in 2032 . The overview of final energy demand in household sector is shown in Picture 3.7.



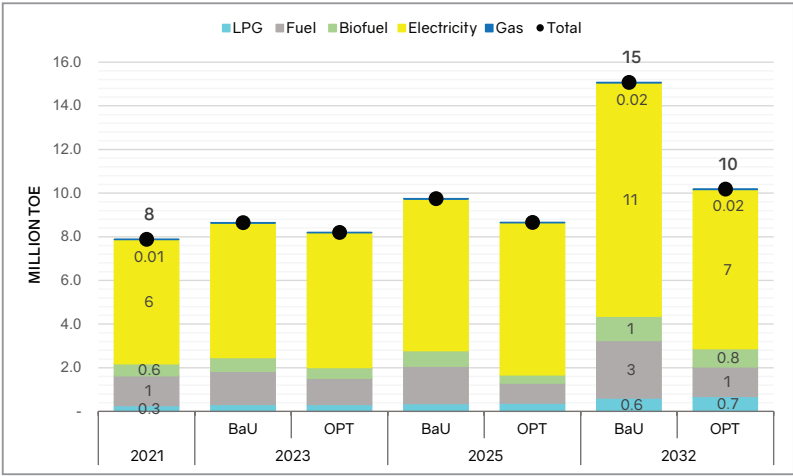
Picture 3.7 Energy Demand in Household Sector per Energy Type

The largest energy consumption growth in household sector in the OPT scenario is natural gas at 27.3% per year along with the increase in the city gas program and followed by electricity at 2.3% per year due to the electric stove substitution program. On the other hand, LPG growth experiences a decrease with the growth of -2.3% per year.

3.1.1.4 Commercial Sector

In 2021, total energy consumption in commercial sector from hotel, mall, hospital and office reached 8 million TOE consisting of 72.2% of electricity, 7% of biodiesel, 3.4% of LPG, 0.1% of gas and 17.3% of fuel (diesel). Until 2032, energy demand in commercial sector is

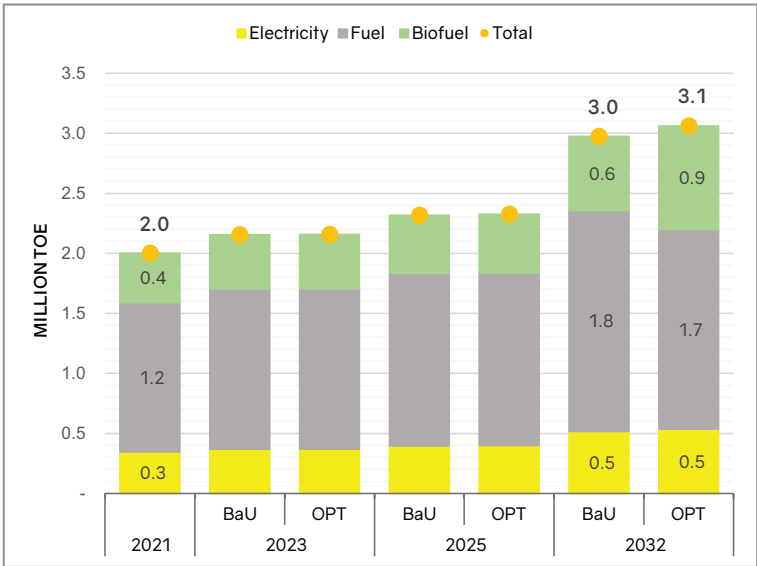
projected to increase from 8 million TOE in 2021 into 15 million TOE (BaU) and 10 million TOE (OPT). Energy demand in OPT scenario is lower as it is assumed that there is energy saving in equipment and building, including hotel and mall (Picture 3.8).



Picture 3.8 Energy Demand in Commercial Sector per Energy Type

3.1.1.5 Other Sectors

Other sector consists of three sub-sectors, namely agriculture, mining and construction. Energy consumption in other sectors in the two scenarios is not much different, around 3 million TOE in 2032 with the projected demand for each type of energy as shown in Picture 3.9.

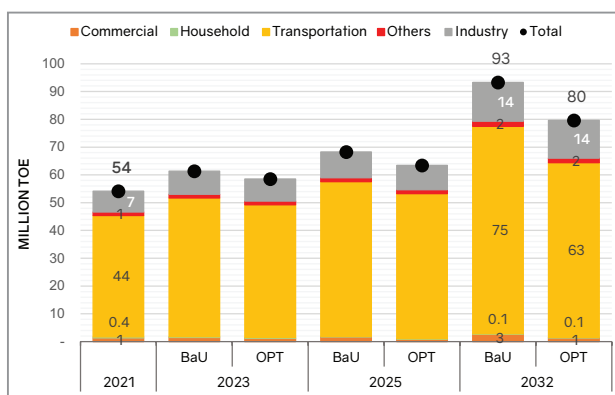


Picture 3.9 Energy Demand in Other Sectors per Energy Type

3.1.2 Final Energy Consumption per Energy Type

3.1.2.1 Fuel

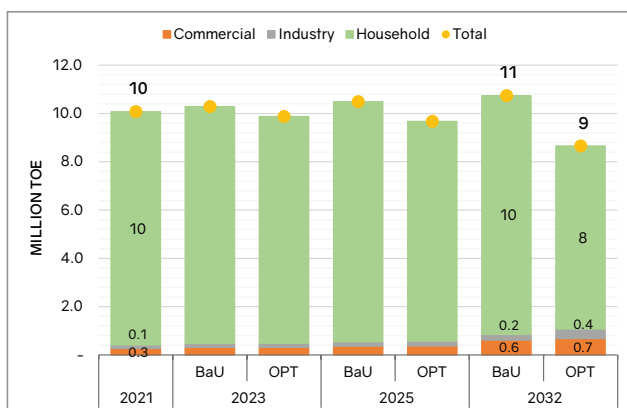
Total fuel consumption in BaU scenario grows by an average of 5.1% per year, so that fuel consumption in 2032 will reach 93 million TOE. Meanwhile in OPT scenario, fuel demand grows slower by 3% so that it will increase to 80 million TOE in 2032, in line with the more optimistic substitution of fuel vehicles to electricity and biofuels. The demand for all types of fuel increases except for the use of kerosene in the household sector that is projected to decrease to 0.015 million TOE at the end of the projection year as shown in Picture 3.10.



Picture 3.10 Fuel Consumption per Sector in BaU and OPT Scenario

3.1.2.2 LPG

LPG demand in 2032 will grow by 0.6% per year (BaU) but decreased by 1.41% per year in OPT scenario. At the end of the projection, LPG demand will reach 11 million TOE (BaU) and 9 million TOE (OPT) as shown in Picture 3.11.



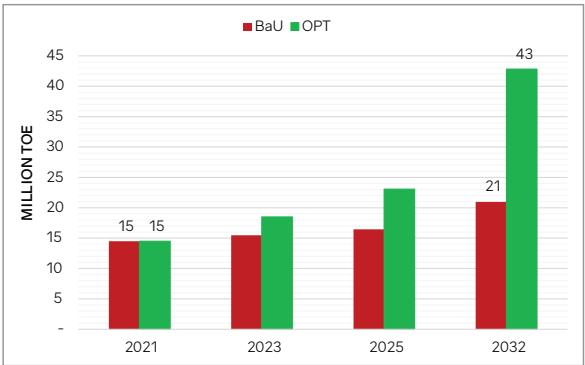
Picture 3.11 LPG Consumption per Sector

Around 95.6%, LPG consumption in 2021 will be used for household sector. By 2032, LPG share is projected to decrease to 92.1% (BaU) and 87.6% (OPT).

3.1.2.3 Gas

The industrial sector is the biggest natural gas user, while only 1% of natural gas is used in the household, transportation and commercial sectors.

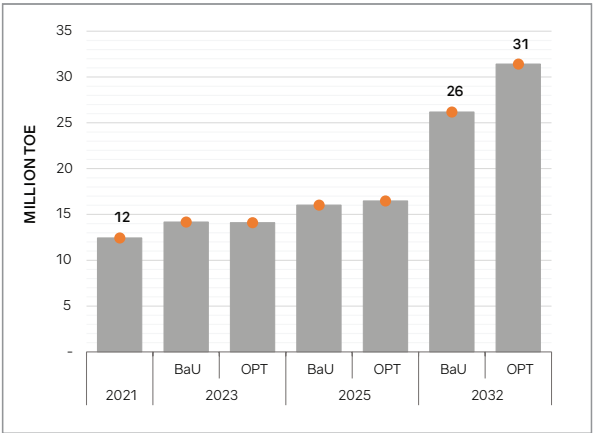
In OPT scenario, natural gas demand is projected to increase by an average of 10.3% per year to reach 43 million TOE, while in BaU scenario it will grow even lower at 3.4% and reach 21 million TOE in 2032 as shown in Picture 3.12.



Picture 3.12 Gas Consumption per Scenario

3.1.2.4 Coal

Coal demand in industrial sector grows by an average 7.0% per year in BaU scenario, reaching 26 million TOE in 2032. Meanwhile, in OPT scenario which is assumed to have high growth in industrial sector, coal demand will increase to 31 million TOE or growing at an average of 8.8% per year as shown in Picture 3.13.

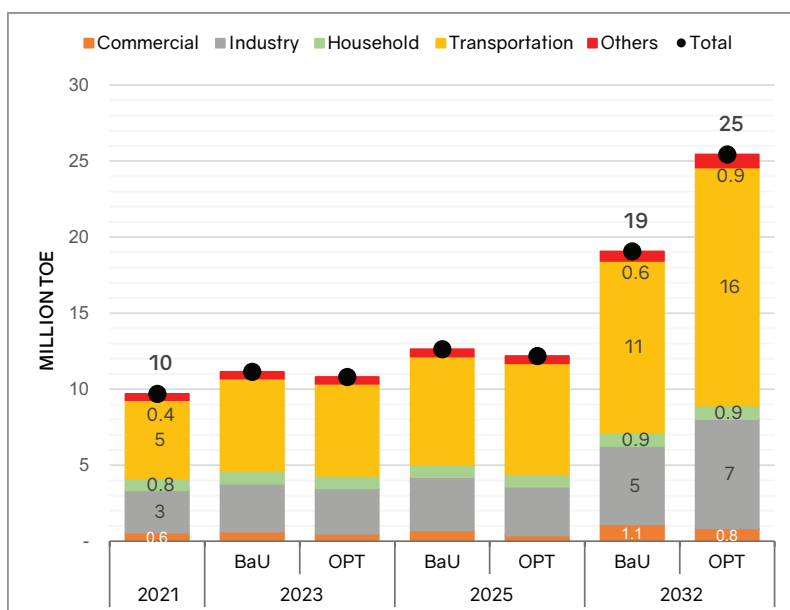


Picture 3.13 Coal Consumption in Industrial Sector

3.1.2.5 Bioenergy

Final energy consumption of bioenergy includes biodiesel, biogasoline, biogas and biomass. In 2032 in BaU scenario, bioenergy demand grows by an average of 6.3% per year into 19 million TOE. Meanwhile, in OPT scenario, bioenergy demand grows higher with an average of 9.2% per year into 25 million TOE. In OPT scenario, the higher growth in bioenergy demand is influenced by B40 and E5 programs starting in 2030.

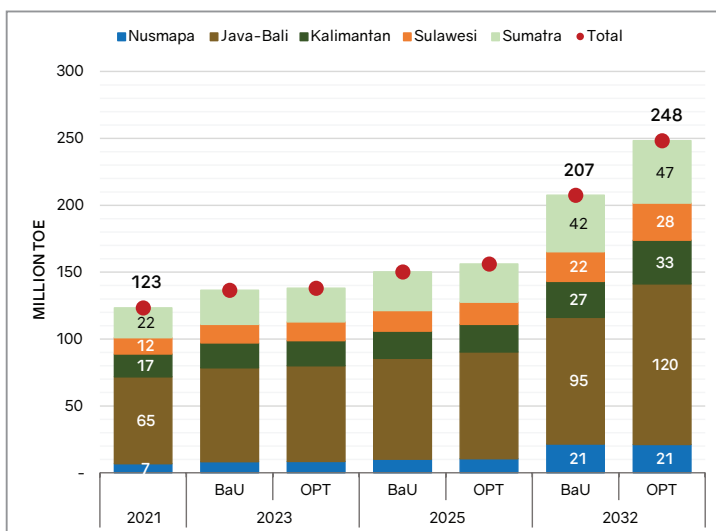
In 2032, the largest share of bioenergy demand will come from the transportation sector, namely 59.3% (11 million TOE) in BaU scenario and 61.6% (16 million TOE) in OPT scenario. The overview of fuel consumption by sector can be seen in Picture 3.14.



Picture 3.14 Bioenergy Consumption per Sector

3.1.3 Final Energy Consumption per Region

In 2021, the largest share of final energy consumption was in Java and Bali (52.7%) and Sumatra (18.0%). However, in 2032 the share of final energy in Java-Bali begins to decline to 45.7% (BaU) and 48.3% (OPT). On the other hand, the share of final energy in Sulawesi and Nusmapa in 2021 was 9.9% and 5.5%, respectively. In 2032, final energy demand is projected to increase, so that its share will increase to 10.7% (BaU) and 11.2% (OPT) for Sulawesi region. Meanwhile, Nusmapa's share will increase to 10.4% (BaU) and 8.5% (OPT) due to the relatively high industrial growth in eastern Indonesia. The overview of energy consumption by region can be seen in Picture 3.15.



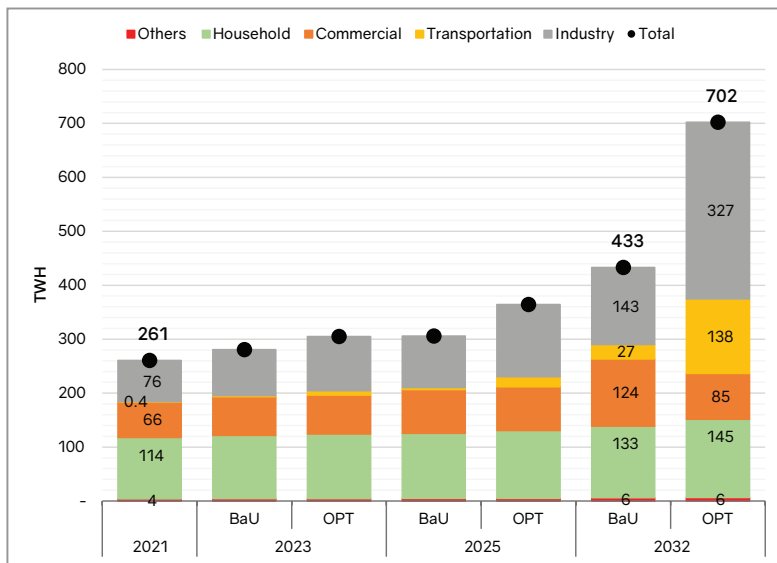
Picture 3.15 Final Energy Consumption per Region

3.2 ELECTRICITY CONSUMPTION

3.2.1 National Electricity Consumption per Sector

The total of electricity demand including electricity consumption for several smelter industries in 2032, is projected to increase to 433 TWh in the BaU scenario, and 702 TWh in the OPT scenario.

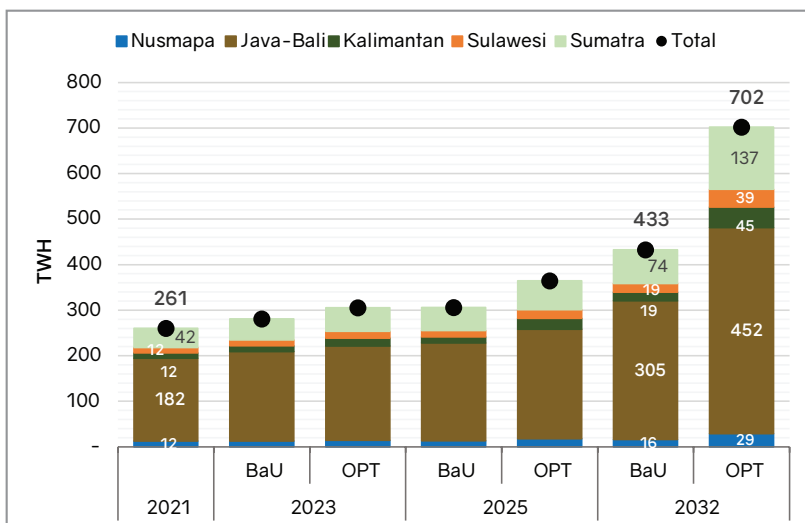
The largest share of electricity demand in 2021 comes from household sector around 43.6%, followed by industry 29.3%, commercial, 25.4%, other sectors 1.5%, and transportation around 0.2%. The share of electricity consumption in 2032 specifically in the OPT scenario will change, so the largest share will shift to the industrial sector around 46.6%, followed by transportation 19.7%, household 20.7%, commercial 12.1%, and others sector 0.9%. There is an increase in the share of electricity consumption in industrial and transportation sectors along with the planned new industrial areas, especially outside Java as well as the use of electric vehicles which has been echoed since 2022 through an appeal to the use of electric vehicles as official vehicles for the central and regional governments as well as the increasing use of electric vehicles in public transportation such as online transportation, buses, and taxis. The projection of electricity demand by sector for each scenario is shown in Picture 3.16.



Picture 3.16 National Electricity Demand per Sector

3.2.2 Electricity Consumption per Region

The largest total electricity demand in 2021 was in Java and Bali at around 69.9%. It is projected that the share will remain the same until 2032 in BaU scenario, while in OPT scenario the share will decrease to 64.4%. Meanwhile, the lowest electricity consumption was in the Nusmapa region at 4.8% in 2021 and 4% (2032). A complete overview of electricity consumption per region is shown in Picture 3.17.

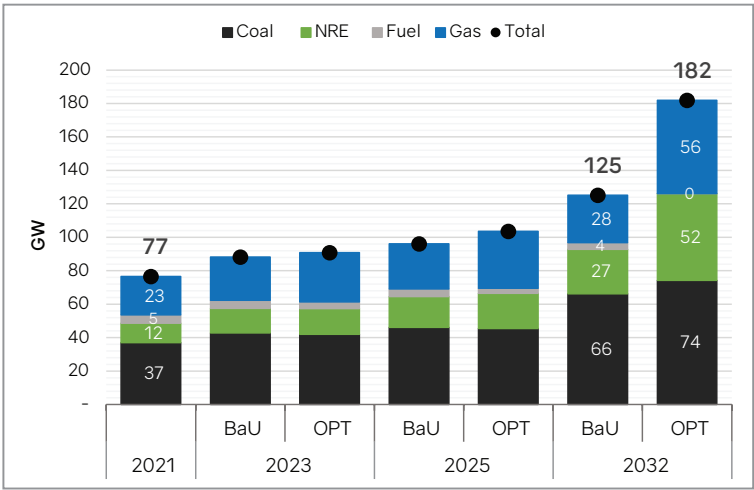


Picture 3.17 Electricity Consumption per Region

3.3 ELECTRICITY

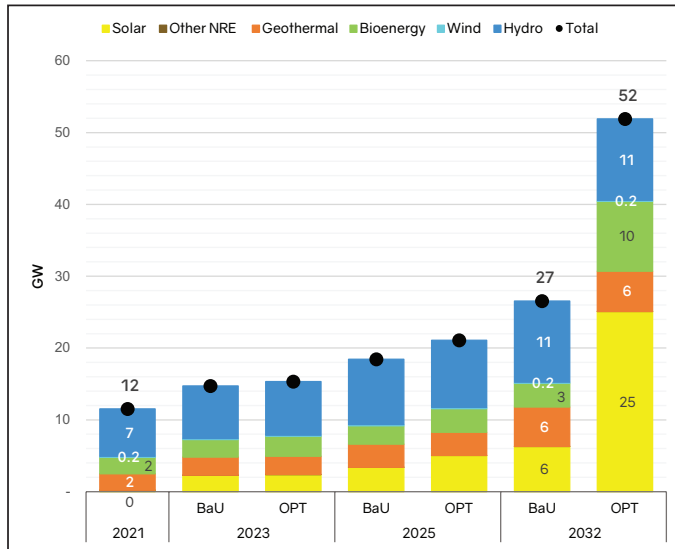
3.3.1 Power Plant Capacity

Power plant capacity in 2032 will increase by 4.6%, and 8.2% respectively (average per year) reaching 125 GW (BaU) and 182 GW (OPT). In the same year for BaU scenario, the largest power plant comes from coal (53%), followed by gas (22.6%), NRE (21.2%), and fuel. On the other hand, in OPT scenario, the utilization of NRE-based power plants will reach around 28.5%, despite that the share of coal is still around 40.8%, and the share of gas is 30.7%. This is a step to maintain electricity supply by using clean energy (gas) as shown in Picture 3.18.



Picture 3.18 Power Plant Capacity per Scenario

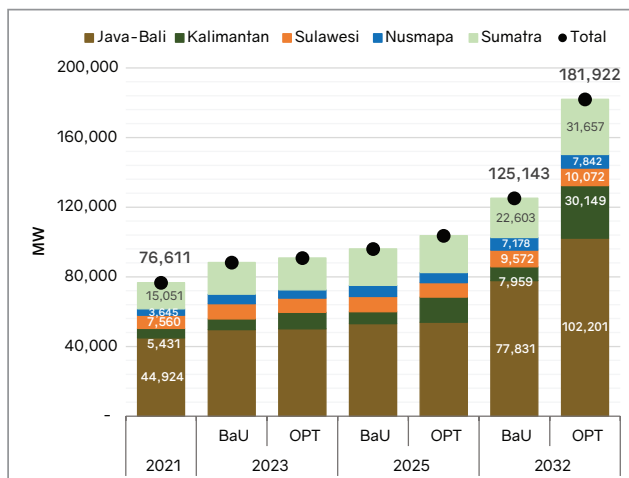
NRE power plant capacity in BaU scenario in 2032 will increase with an average annual growth of 7.9% into 27 GW, while in OPT scenario it will increase with an average annual growth of 14.7% into 52 GW. In OPT scenario, Solar PP capacity will experience the highest growth compared to other power plants. The capacity of Solar PP in OPT scenario in 2032 will reach 25 GW or 48.3% of the total NRE power plant capacity. Meanwhile, the share of Hydro PP (including Micro PP and MHP) will be around 21.8% (11 GW), Bioenergy PP will be around 18.7% (10 GW), and Geothermal PP will be around 10.8% (6 GW). The development of power plant capacity in BaU and OPT scenarios can be seen in Picture 3.19.



Picture 3.19 NRE Power Plant per Scenario

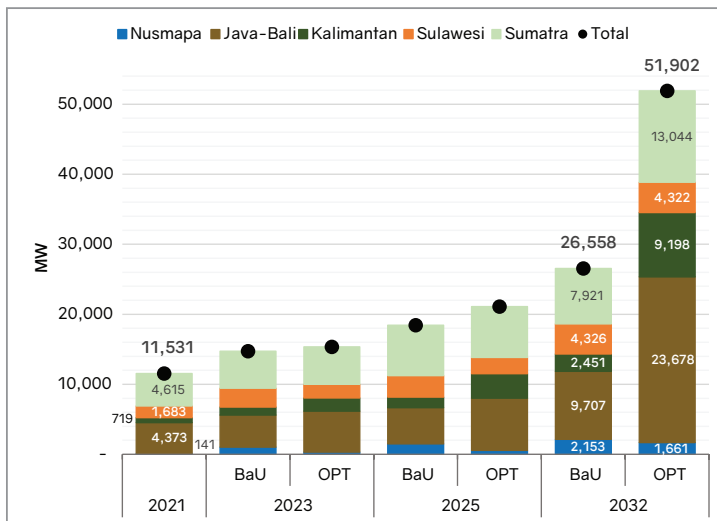
3.3.2 Power Plant Capacity per Region

In 2021, the largest power plant capacity was in Java-Bali region of 44,924 MW (58.6%), and Sumatra 15,051 MW (19.6%) while the remaining 16,636 MW was in Kalimantan, Sulawesi and Nusmapa regions. In 2032 the power plant capacity in Java-Bali region will increase to 77,831 MW (BaU) and 102,201 MW (OPT). The share of power plant capacity in Sumatra region will increase to 22,603 MW (BaU) and 31,657 MW (OPT). Meanwhile, the share of power plant capacity in Nusmapa region will only reach 7,178 MW (BaU) and 7,842 MW (OPT). A complete overview of power plant capacity per region is shown in Picture 3.20.



Picture 3.20 Power Plant Capacity per Region

The total capacity of NRE power plants in 2021 reached 11,531 MW, with the largest share in two regions namely Java-Bali of 4,615 MW (40%) and Sumatra of 4,373 MW (37.9%). Meanwhile, the remaining 2,543 MW was in Sulawesi of 1,683 MW (14.6%), Kalimantan of 719 MW (6.2%), and Nusmapa of 141 MW (1.2%). In 2032, the capacity of NRE power plants is projected to grow by 7.9% in BaU scenario and 14.7% in OPT scenario. They will reach 26,558 MW and 51,902 MW, respectively. The capacity of NRE power plant in Nusmapa region will grow higher than the other regions. In 2032 it will be around 1,661 MW. A complete overview of NRE power plant capacity per region can be seen in Picture 3.21.



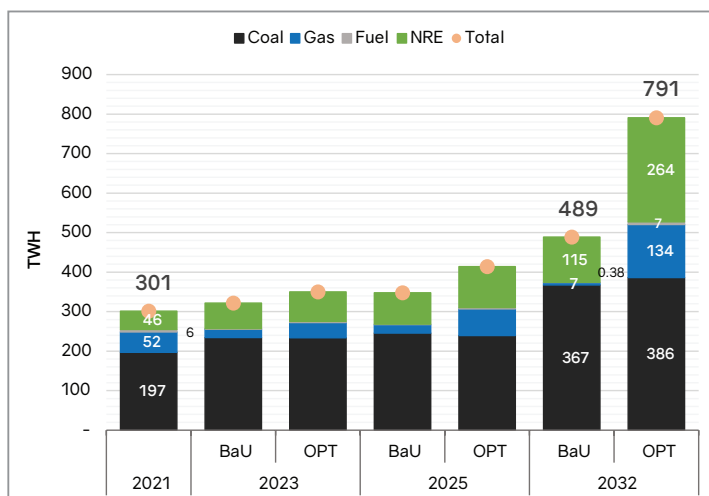
Picture 3.21 NRE Power Plant Capacity per Region

3.3.3 Electricity Production

Electricity production in 2032 will increase to around 489 TWh (BaU) and 791 TWh (OPT), taking into account the 10% loss in transmission and distribution.

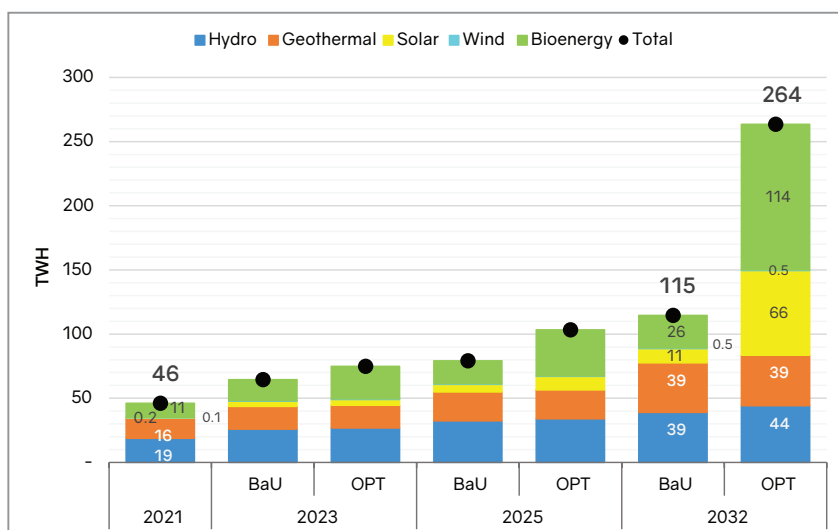
Based on the type of power plant, the share of coal-fired power plant production in 2021 is around 65.5%, and it will increase to around 75.1% in 2032 for BaU scenario. However, in OPT scenario, the share of electricity production from coal will drop quite significantly to around 48.9% in 2032.

In OPT scenario, the development of power plants is directed at optimizing NRE power plants, especially increasing Solar PP and optimizing coal cofiring by around 10% starting in 2030. In addition, future power plant development is also directed at substituting fuel to NRE or gas, so that the share of fuel in 2032 will be only about 1%. The diesel PP still exists, but it is prioritized for 3T area (Outermost, Remote and Disadvantaged Areas) that are not reached by other energy. The electricity production per type of energy for both scenarios can be seen in Picture 3.22.



Picture 3.22 Electricity Production per Energy Type

Electricity production from NRE plants in 2032 will be 115 TWh (BaU) with the largest share coming from Hydro PP, Geothermal PP, Bioenergy PP, and Solar PP. Meanwhile, in the same year, electricity production in OPT scenario will be around 264 TWh with the largest production share from Bioenergy PP at around 114 TWh (43.1%), and Solar PP production will be around 66 TWh (25%). The cheaper price of electrical components from Solar PP and the solar-rooftop program in commercial buildings and luxury homes as well as Energy Saving Solar Lamps (LTSHE) allow for faster Solar PP penetration. The projected electricity production from NRE power plants can be seen in Picture 3.23.

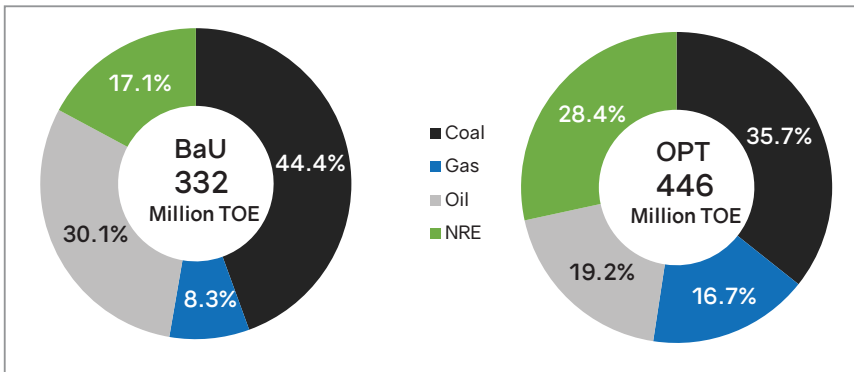


Picture 3.23 Electricity Production from NRE Power Plant

3.4 PRIMARY ENERGY SUPPLY

3.4.1 National Primary Energy Supply

Primary energy supply grows on average of 5% in BaU scenario and 7.8% in OPT scenario that it will reach 332 million TOE (BaU) and 446 million TOE (OPT). At the end of the projection year, the share of coal still dominates at 44.4% (BaU) and 35.7% (OPT). Meanwhile, the share of NRE in OPT scenario is projected to increase to 28.4% as a result of the use of biomass (NRE) to reduce coal use and increase the use of biofuels. The projection of the primary energy mix in 2032 in both scenarios can be seen in Picture 3.24.

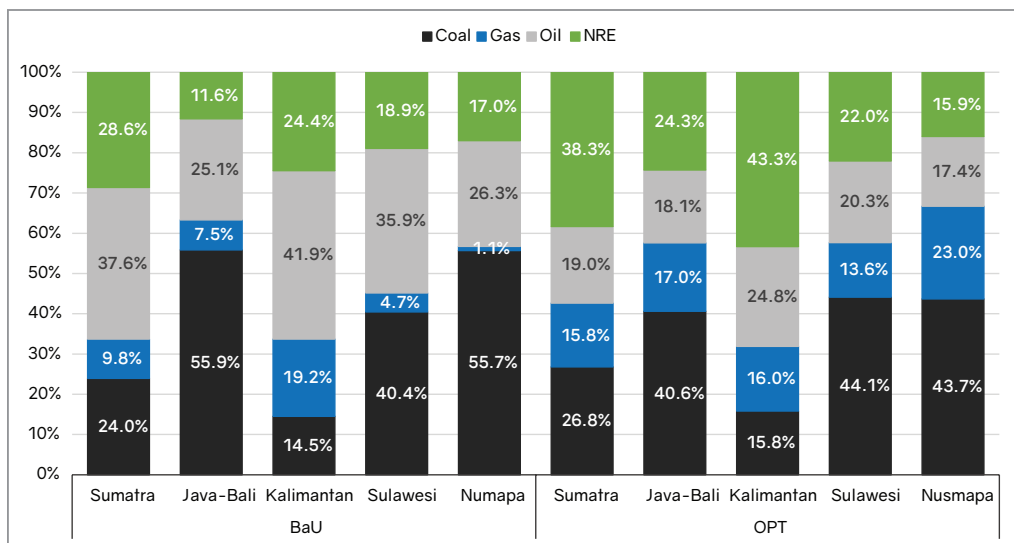


Picture 3.24 Primary Energy Mix in Both Scenarios in 2032

3.4.2 Primary Energy Supply per Region

In BaU scenario, the largest share of NRE primary energy in 2032 is projected to be in Sumatra region at around 28.6%, followed by Kalimantan region at around 24.4%. Meanwhile, the smallest share of NRE is in Java-Bali region at around 11.6% which is influenced by the presence of coal-fired power plants, which are mostly located in the island of Java.

In OPT scenario, the largest share of NRE primary energy is in Kalimantan region at 43.3%, followed by Sumatra region at around 38.3%. The potential for NRE power plants, especially hydro PP, which are quite large in both regions also have an effect on increasing the NRE mix. A complete overview of the share of primary energy supply per region in 2032 is shown in Picture 3.25.

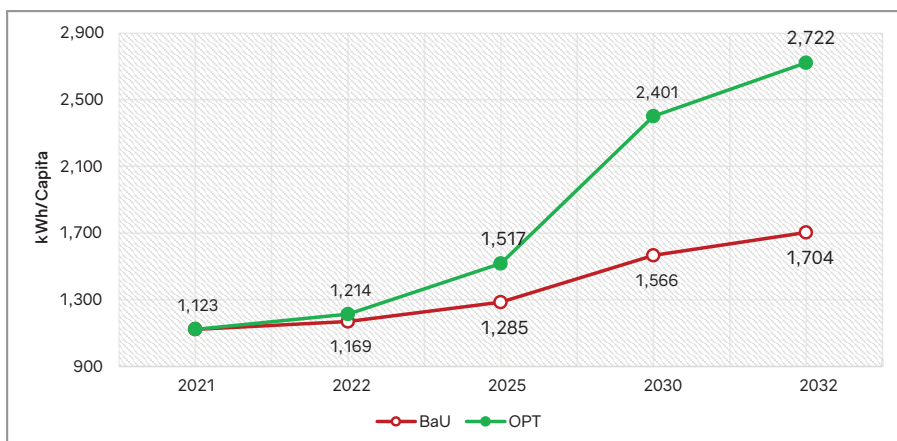


Picture 3.25 Primary Energy Supply per Region 2032

3.5 ENERGY INDICATOR

3.5.1 Electricity Consumption per Capita

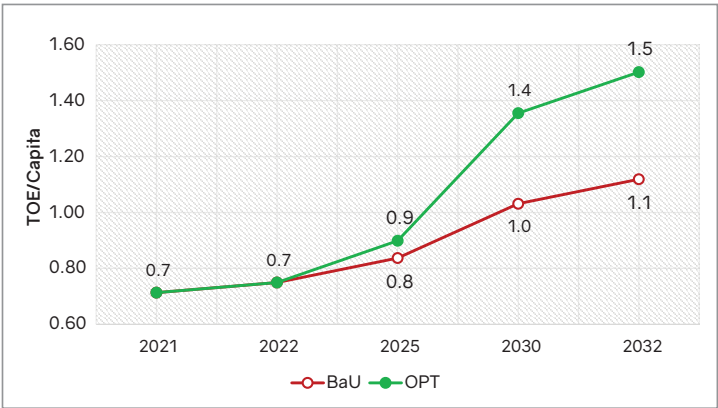
In 2025 the electricity demand per capita in both scenarios (BaU & OPT) are 1,285 kWh/Capita and 1,517 kWh/Capita respectively. It is still below the electricity target per capita in National Energy Policy of 2,500 kWh/Capita in 2025. Meanwhile in 2032 electricity consumption per capita will reach 2,722 kWh/Capita in OPT scenario and only 1,704 kWh/Capita in BaU scenario. The projection of electricity consumption per capita for each scenario is shown in Picture 3.26.



Picture 3.26 Electricity Consumption per Capita

3.5.2 Primary Energy per Capita

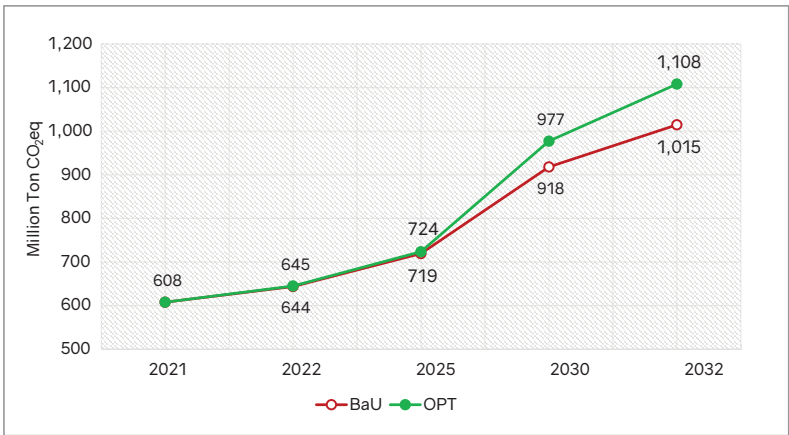
Similar to electricity consumption per capita, primary energy supply per capita increases accordingly in line with the trend of economic growth. Primary energy per capita in 2021 is 0.7 TOE/capita and is projected to increase in 2025 to 0.8 TOE/capita (BaU) and 0.9 TOE/capita (OPT). The projection of primary energy per capita in 2025 is still far from KEN target of 1.4 TOE/capita in 2025 as shown in Picture 3.27.



Picture 3.27 Primary Energy per Capita

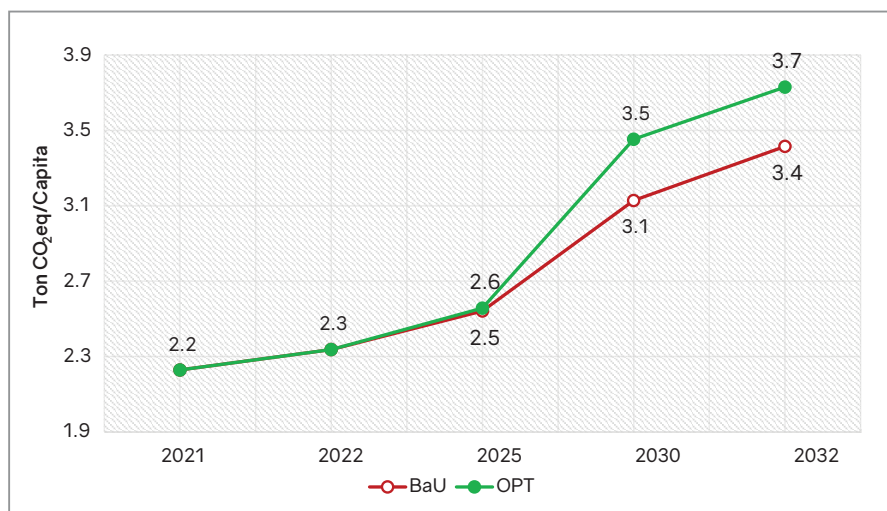
3.5.3 CO₂ Emission

Total emissions in 2032 are projected to increase to 1,015 million tons CO₂eq (BaU) and 1,108 million tons CO₂eq (OPT). The emission target of the two scenarios is lower than the emission target in the energy sector NDC, which is 1,355 million tons CO₂eq in 2030. The development of GHG emissions in the two scenarios per sector can be seen in Picture 3.28 below.



Picture 3.28 Carbon Emission in Both Scenarios

GHG emissions per capita are projected to increase to 3.4 million tonnes of CO₂eq (BaU) and 3.7 million tonnes of CO₂eq (OPT) as shown in Figure 3.29 below.



Picture 3.29 GHG Emission per Capita

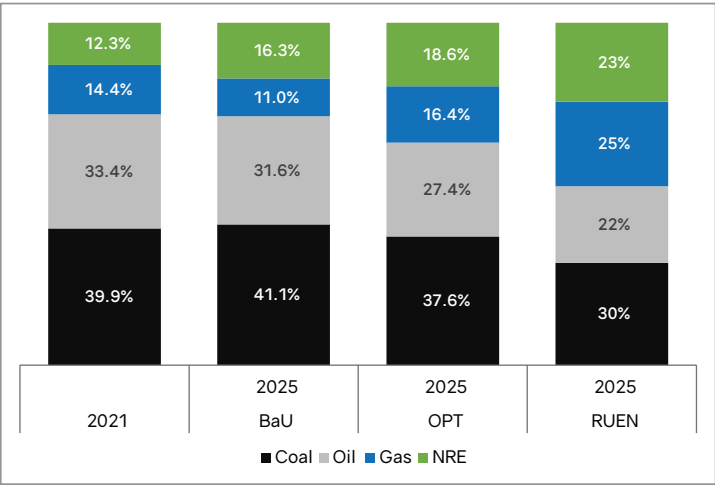
The resume of several energy indicators based on Presidential Regulation on National Energy Policy is shown in Table 3.1.

Table 3.1 Energy Indicator per Scenario

Energy Indicator	Unit	Current	BAU		OPT	
		2021	2025	2032	2025	2032
Primary energy supply	Million TOE	208	237	332	254	446
Primary Energy Mix	%					
Batubara		40%	41%	44%	38%	36%
Oil		33%	32%	30%	27%	19%
Gas		14%	11%	8%	16%	17%
NRE		12%	16%	17%	19%	28%
Electric consumption	kWh/Capita	1,123	1,285	1,704	1,517	2,722
Primary Energy Consumption	TOE/Capita	0.7	0.8	1.1	0.9	1.5
Emission	Million ton CO ₂ eq	608	719	1,015	724	1,108
Emission Per Capita	Ton CO ₂ eq/Capita	2.2	2.5	3.4	2.6	3.7

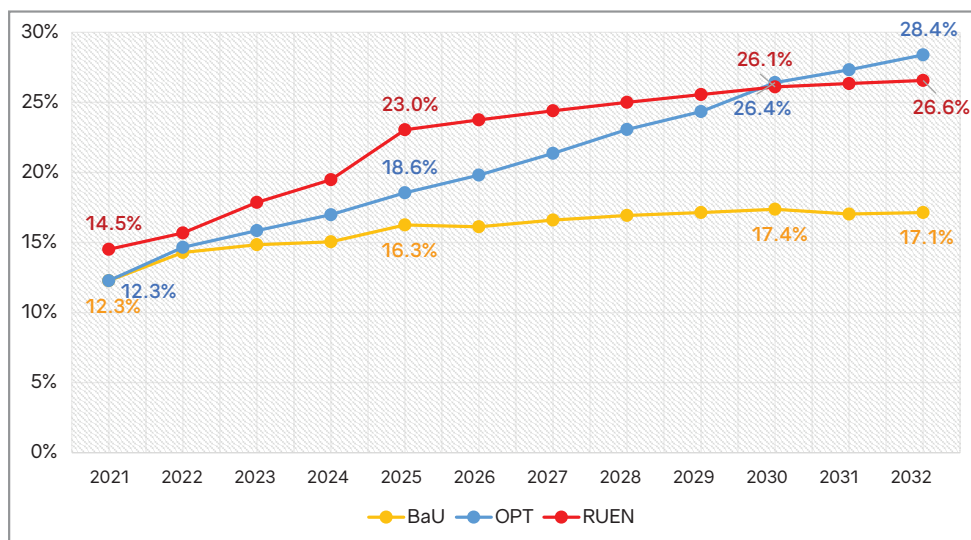
3.6 COMPARISON OF PRIMARY ENERGY MIX AS THE RESULT OF PROJECTION AND RUEN

In accordance with Government Regulation No. 79 of 2014 on National Energy Policy, by 2025 the primary energy mix is targeted to reach at least 23% of NRE, less than 25% of oil, at least 30% of coal, and at least 22% of gas. In more detail, RUEN projects the condition of the primary energy mix every year in order to achieve the National Energy Policy target. The comparison of the primary energy mix projection for BaU and OPT scenarios when compared to RUEN is shown in Picture 3.30 below.



Picture 3.30 Comparison of Primary Energy Mix Projection 2025

The graph shows that from the projection results, both in BaU and OPT scenarios, only coal is expected to reach the target of KEN-RUEN in 2025. Meanwhile, NRE is still quite far from the target. It is estimated that NRE mix in 2025 will only reach 16.3% in BaU scenario and 18.6% in OPT scenario considering that NRE target in 2021 has only reached 12.2%. However, this condition is quite good considering that NRE mix continues to increase every year. In addition, in OPT scenario, it is projected that NRE target can exceed RUEN projection in 2031 as shown in Picture 3.31. In line with Indonesia's commitment to reduce emissions, it is expected that the growth in NRE use can be further increased to meet Net Zero Emission target by 2060.



Picture 3.31 Comparison of NRE Mix Projection 2021–2032

3.7 REGIONAL ENERGY PROJECTION 2032

As an improvement of the previous publication, Indonesia Energy Outlook 2022 does not only project national energy, but also estimates the energy condition per region until 2032. The results of energy projection per region in 2032 are shown in Table 3.2 for BaU scenario and Table 3.3 for OPT scenario. Overall, until 2032 Indonesia's energy supply-demand is estimated to still be concentrated in Java-Bali region, followed by Sumatra region. This is in accordance with the distribution level of population, industry, and infrastructure. Likewise, in terms of NRE development, Java-Bali is projected to still be the region with the highest NRE use despite that Sumatra region has the largest percentage.

Table 3.2 Energy Projection per Region 2032 in BaU Scenario

Energy Indicator	Unit	Java-Bali	Sumatra	Kalimantan	Sulawesi	Nusmapa
Final energy consumption	Million TOE	95	42	27	22	21
Electricity production	TWh	344	84	22	21	18
Power plant capacity	GW	78	23	8	10	7
Primary energy supply	Million TOE	174	62	33	28	26
Primary energy mix						
➤ Coal	%	55.9%	24.0%	14.5%	40.4%	55.7%
➤ Gas	%	7.5%	9.8%	19.2%	4.7%	1.1%
➤ Oil	%	25.1%	37.6%	41.9%	35.9%	26.3%
➤ NRE	%	11.6%	28.6%	24.4%	18.9%	17.0%
CO ₂ Emission	Million TonCO ₂ eq	608	164	74	83	87

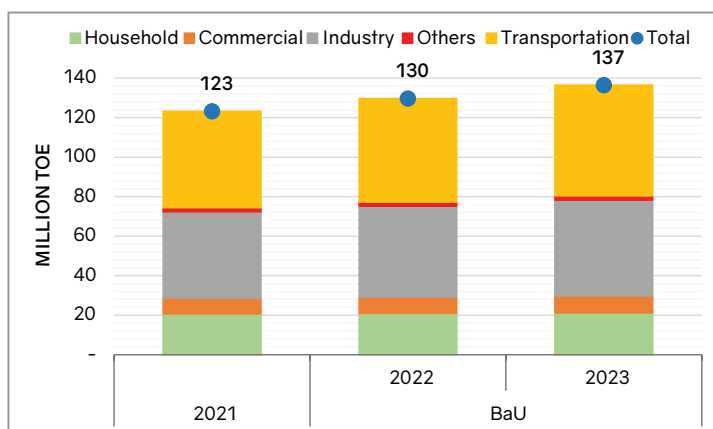
Table 3.3 Energy Projection per Region 2032 in OPT Scenario

Energy Indicator	Unit	Java-Bali	Sumatra	Kalimantan	Sulawesi	Nusmapa
Final energy consumption	Million TOE	120	47	33	28	21
Electricity production	TWh	509	154	51	44	33
Power plant capacity	GW	102	32	30	10	8
Primary energy supply	Million TOE	244	86	50	39	28
Primary energy mix						
➤ Coal	%	40.6%	26.8%	15.8%	44.1%	43.7%
➤ Gas	%	17.0%	15.8%	16.0%	13.6%	23.0%
➤ Oil	%	18.1%	19.0%	24.8%	20.3%	17.4%
➤ NRE	%	24.3%	38.3%	43.3%	22.0%	15.9%
CO ₂ Emission	Million Ton TonCO ₂ eq	657	186	87	102	77

3.8 ENERGY PROJECTION 2022-2023

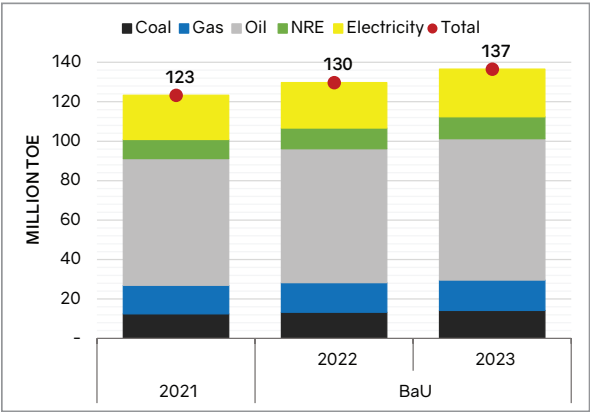
3.8.1 Final Energy

The level of energy consumption by the community continues to improve after the strike of Covid-19 pandemic in early 2020. As shown in Picture 3.32, in 2022 final energy consumption is projected to reach 130 million TOE and continues to reach 137 million TOE in 2023 or increases 10.8% from 2021. Thus, final energy consumption in 2023 has already exceeded final energy consumption in 2019 (before the Covid-19 pandemic). Final energy consumption in 2023 will still be dominated by transportation sector, up to 43% of the total energy demand. The main factor in increasing of the final energy consumption in transportation sector is the recovery of economic and industrial activities after Covid-19.



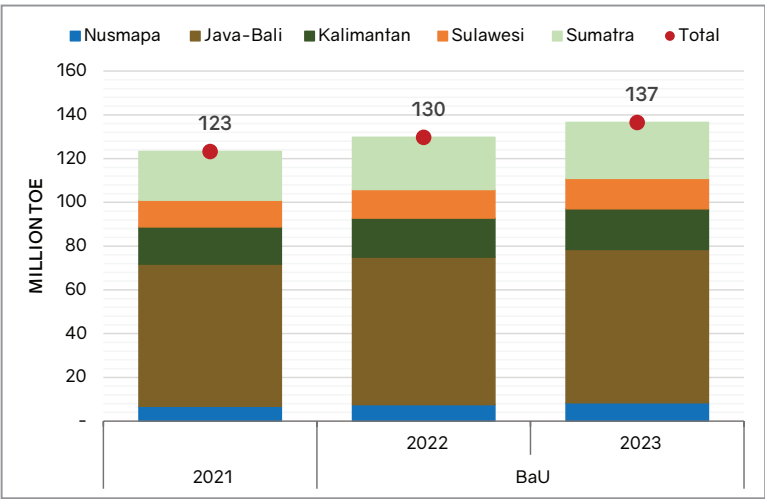
Picture 3.32 Final Energy Projection per Sector 2022-2023

From the type of energy, the highest growing energy is NRE. Until 2023, NRE is estimated to increase by 15% compared to the base year. This is supported by the increasing use of biodiesel especially in transportation sector. Meanwhile, the increase in coal consumption in the last 3 years is projected to reach 13.8% which is influenced by its use in industrial sector, especially in cement and smelter industries. The projected final energy consumption by energy type in details can be seen in Picture 3.33.



Picture 3.33 Projection of Final Consumption per Energy Type 2022–2023

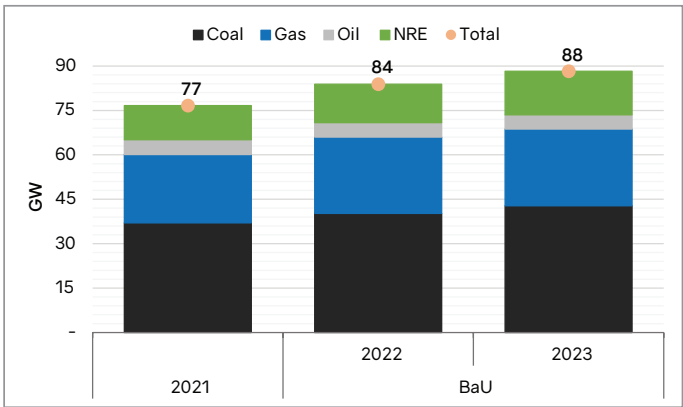
Meanwhile, Picture 3.34 shows the level of energy consumption by each region. Until 2023, the largest final energy will still be in Java-Bali and Sumatra reaching 51.3% and 18.7% of the national total. Meanwhile, consumption in the other 3 regions (Kalimantan, Sulawesi, and Nusmapa) are only around 13.6%, 10.2% and 6.1%, respectively.



Picture 3.34 Final Energy Projection per Region 2022–2023

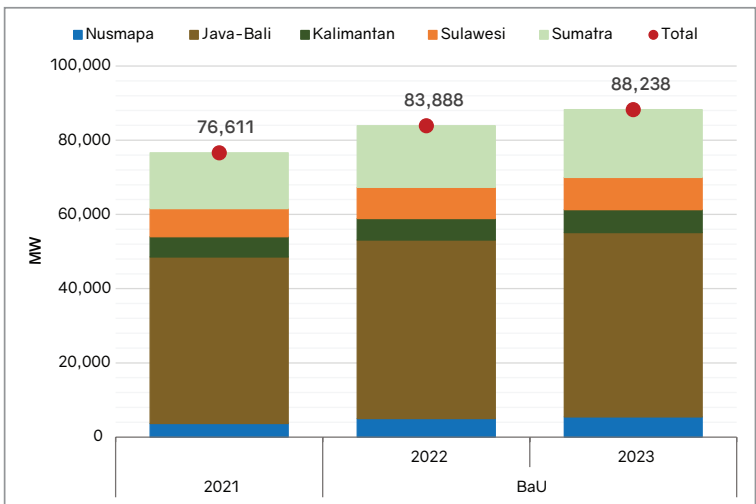
3.8.2 Power Plant

On transformation side, the development of power plant continues to be boosted. Until 2023, there will be additional 11 GW of power plants. As shown in Picture 3.35, power plants with a total of 88 GW in 2023 will still be dominated by coal-fired PP (48.5%) and gas PP (29.4%). Meanwhile, NRE power plants are estimated to contribute 16.7% or around 15 GW of power plant capacity in 2023, mainly supported by hydro PP (49.7%) and geothermal PP (17.2%).



Picture 3.35 Projection of Power Plant per Energy Type 2022–2023

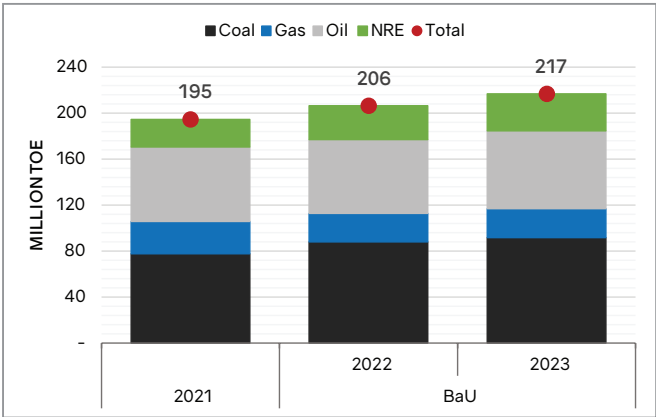
Based on location, the largest power plant capacity is in Java-Bali and Sumatra, which is estimated to reach 56.3% and 20.7% in 2023. Meanwhile, the smallest power plant capacity is in Nusmapa Region at 6.2%. The details of power plant projections per region can be seen in Picture 3.36 below.



Picture 3.36 Projection of Power Plant Capacity per Region 2022–2023

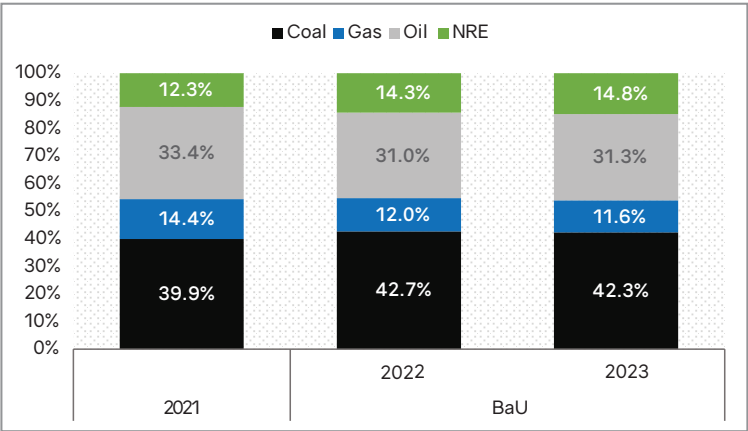
3.8.3 Primary Energy

To meet the energy demand, it is also necessary to ensure the availability of energy supply. As shown in Picture 3.37, the level of energy supply is projected to grow by around 11.4% from 2021 to 2023. Fossil energy is still the largest source of energy, mainly coal and oil, which are estimated to reach 42.3% and 31.3% respectively of the total supply. However, the highest growth in primary energy supply comes from NRE, which is around 34.8%, supported by the use of biofuels and the use of NRE in the power generation sector.



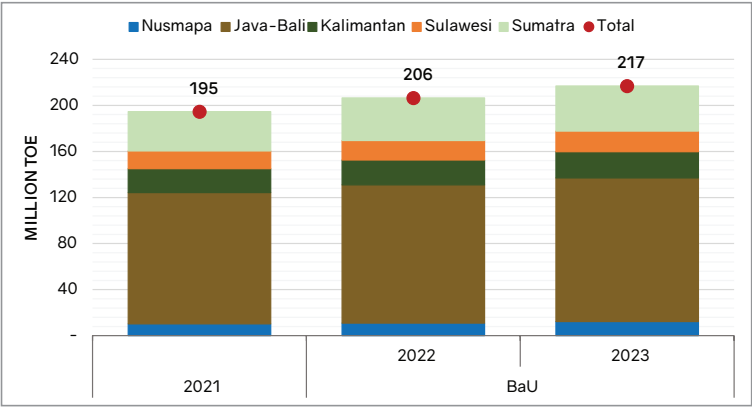
Picture 3.37 Projection of Primary Energy Supply 2022–2023

In percentage, in 2023 the energy mix is projected to consist of 14.8% of NRE, followed by 42.3% of coal, 11.6% of gas, and 31.3% of oil as shown in Picture 3.38. As a reference, in 2023, RUEN projects a mix of 18% of NRE, 33% of coal, 23% of gas, and 26% of oil. More efforts are needed to encourage the utilization of gas and NRE to be in line with the plan (RUEN).



Picture 3.38 Projection of Primary Energy Mix 2022–2023

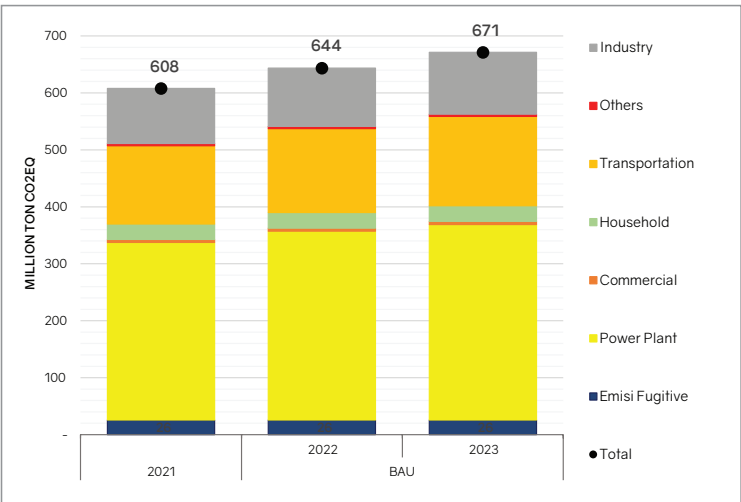
Based on projection per region, the largest supply of primary energy is in Java-Bali and Sumatra regions at 57.6% and 18% respectively. Meanwhile, Nusmapa's primary energy supply is only around 5.6%. The primary energy projection per region in more detail can be seen in Picture 3.39.



Picture 3.39 Projection of Primary Energy Supply per Region 2022–2023

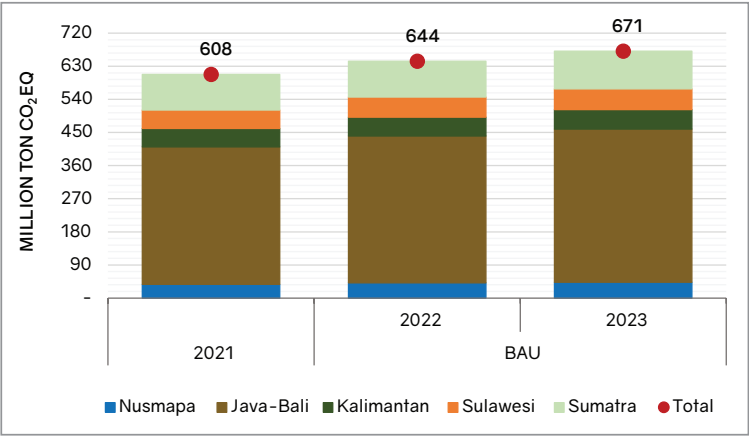
3.8.4 Emission

CO₂ emissions in energy sector in 2023 are projected to continue to increase to 671 million tons CO₂eq in line with the increasing use of fossil energy, especially in the industrial, transportation and power plant sectors. In 2023, emissions from electricity generation activities will reach 343 million tons CO₂eq or equivalent to 51.1% of total emissions. Meanwhile, the industrial sector contributes 16.2% of total emissions. In more detail, the projected emissions by sector can be seen in Picture 3.40.

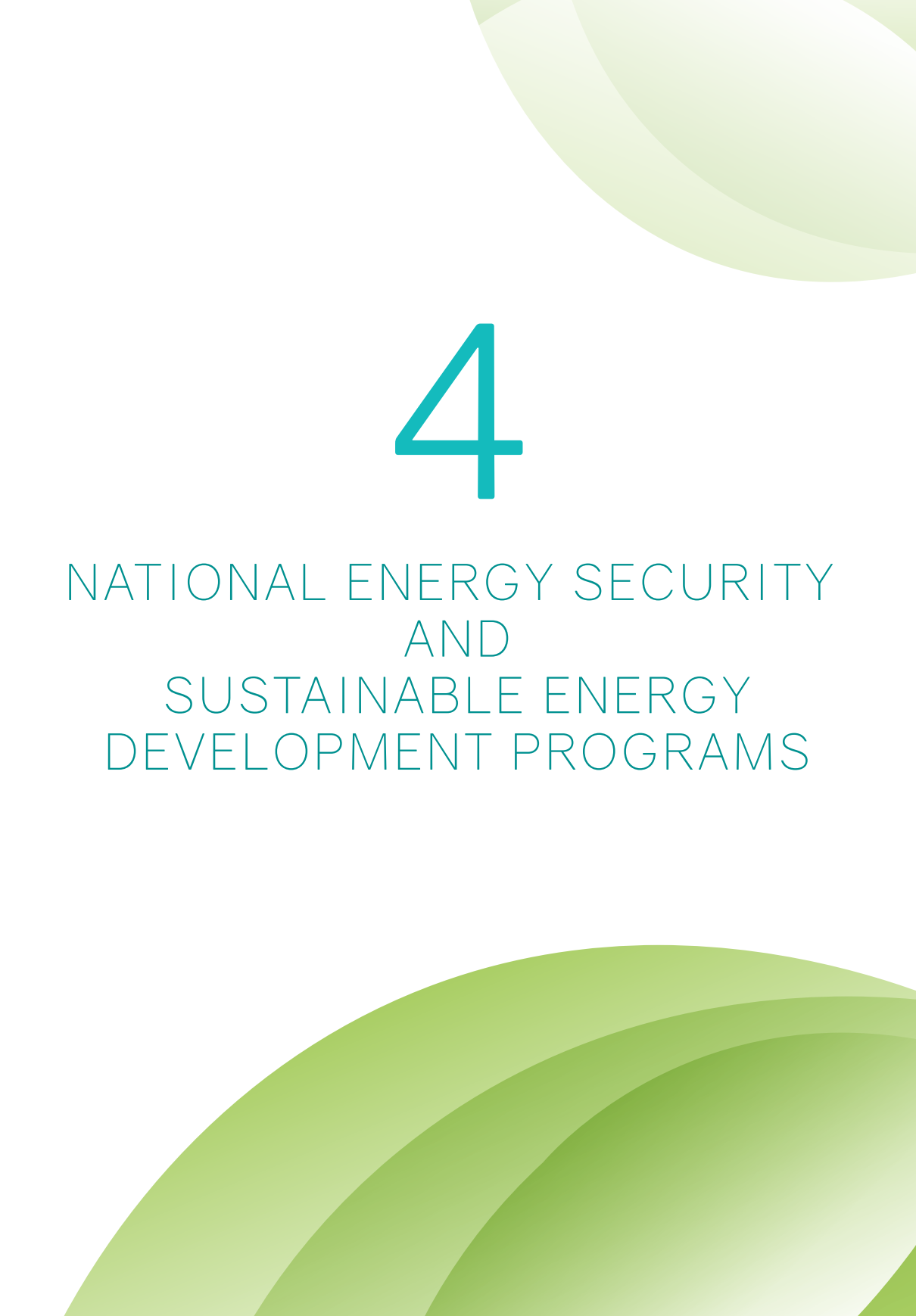


Picture 3.40 Projection of Energy Emission per Sector 2022–2023

Java-Bali region is the largest emission contributor, around 416 million tons of CO₂eq or equivalent to 62% of national emissions. This is in line with the large number of power plant and industrial developments. This condition is in the contrary to condition in Kalimantan, Sulawesi, and Nusmapa, each of which produces emissions of no more than 10% as can be seen in Picture 3.41.



Picture 3.41 Projection of Energy Emission per Region 2022–2023



4

NATIONAL ENERGY SECURITY AND SUSTAINABLE ENERGY DEVELOPMENT PROGRAMS

4 NATIONAL ENERGY SECURITY AND SUSTAINABLE ENERGY DEVELOPMENT PROGRAMS

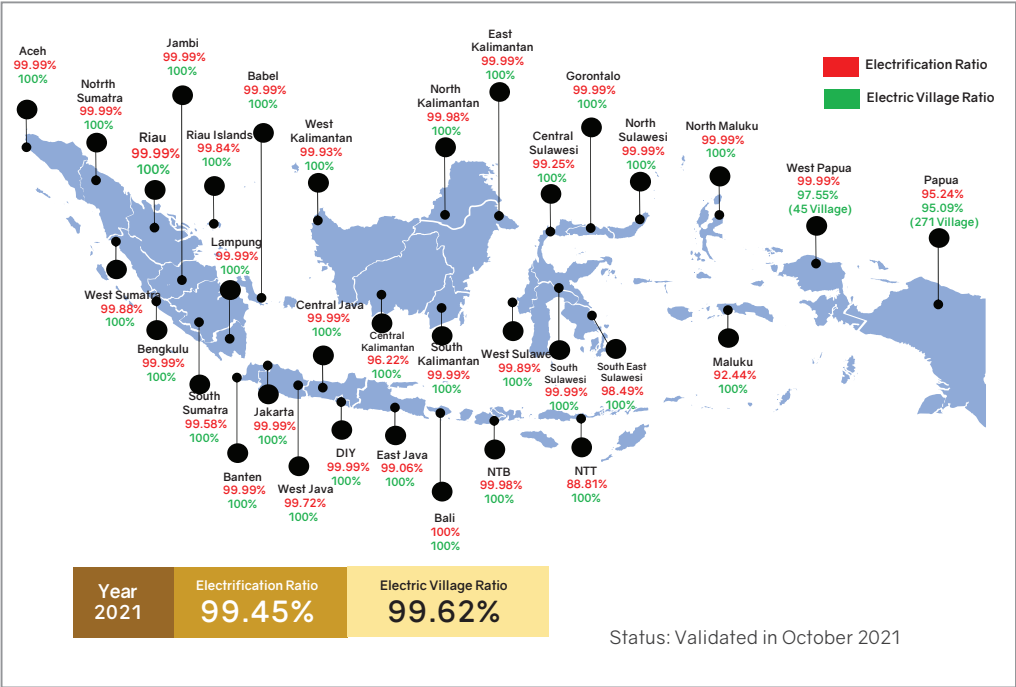
4.1 ELECTRICITY FOR RURAL AREAS AND ENERGY SUPPLY FOR OUTERMOST, REMOTE AND DISADVANTAGED AREAS

As an archipelago country with vast area and high population, access to energy for people especially in rural areas as well as in outermost, remote and disadvantaged areas is a challenge for the Government of Indonesia. Many ways and strategies have been taken to supply energy to rural areas as well as to outermost, remote and disadvantaged areas to give a just development. The problem of limited energy sources, especially in outermost, remote and disadvantaged areas, is caused by several factors, one of which is the distance from the village/area which is quite difficult to reach. The difficulty of access and mobility to outermost, remote and disadvantaged areas has an impact on increasing investment costs for energy infrastructure development, both network and power plant.

Based on MEMR data in 2021, the national electrification ratio reached 99.45%. It means around 0.55% people are not yet electrified. Meanwhile, electrified ratio in rural areas in 2020 reached 99.62%. It means that there still 316 unelectrified villages.

Electrified village is the number of villages which receive electricity from PLN, Non-PLN and Energy Saving Solar Lamp. The Non-PLN electrified village is village which receives electricity from regional government, private sector or self-subsistence that has been consolidated and verified by Office of Energy and Mineral Resources in provincial government. To meet the target of 100% electrification ratio, the government needs to allocate 12.02 trillion Rupiah to fund infrastructure. Several strategies by MEMR to increase the electrification ratio is to carry out grid extension massively to connect villages or households located near PLN grid. The government will also build mini grid and NRE based power plant for people

living in remote areas. In addition, there is also plan to build Electricity Recharge Station (SPEL) and Electrical Power Supply (APDAL) for people living scatteredly. Based on MEMR performance realization 2021 and work plan 2022, the government planned to distribute 11,347 APDAL packages or known as electricity box in 2022 funded by State Budget and PLN SPEL. The distribution of electrified village in 2021 is presented in Picture 4.1.



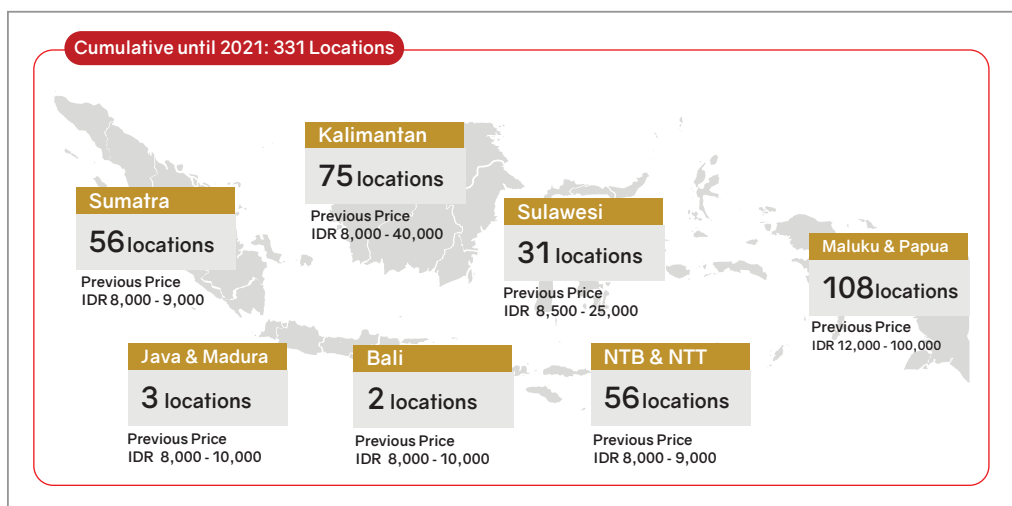
Source: MEMR performance realization 2021

Picture 4.1 Realization of Electrified Villages Distribution 2021

In addition to the electrification program, the Government has also established the One-Price Fuel program which has been running since 2017 in order to serve just energy throughout Indonesia. This policy was motivated by the high fuel price difference in several regions, especially in eastern Indonesia. The outermost, remote and disadvantaged areas are the focus of the Government in implementing the One-Price Fuel program so that affordable fuel prices can be accessed by people in rural areas and in the outermost, remote and disadvantaged areas.

In order to support the program, the Ministry of Energy and Mineral Resources has stipulated Ministerial Regulation No. 36 of 2016 on the Acceleration of One-Price for Certain Types of Fuel and Designated Fuel Nationally to support this policy. This regulation mandates Business Entity with fuel distribution license to establish distributor in certain locations

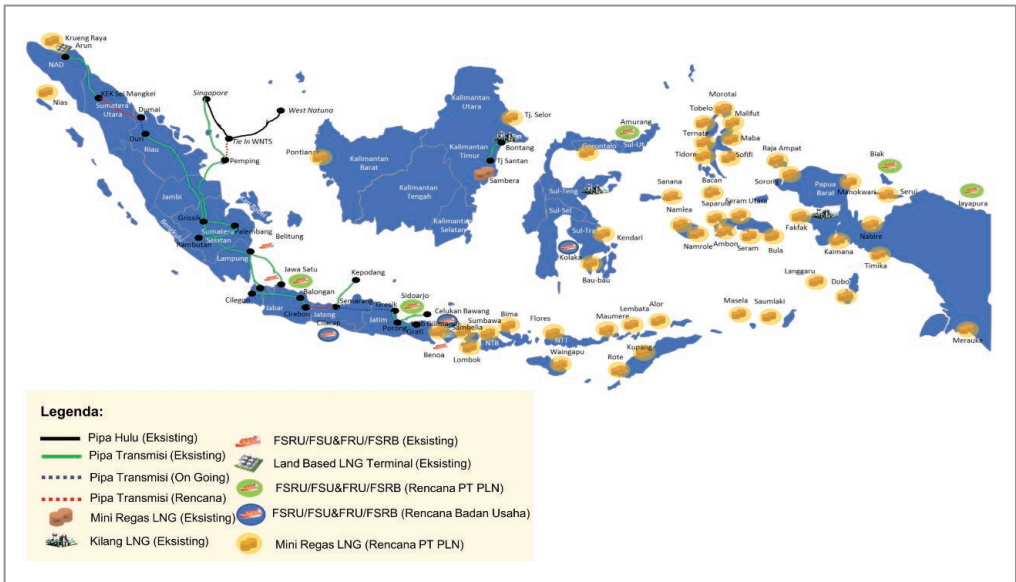
especially in areas where distributor of certain types of fuel and designated fuel has not existed yet to let people buy fuel with retail price as stipulated by the government. One-Price Fuel program is designed for Gasoline RON 88 (Premium) at the price of Rp 6,450 per liter and diesel at the price of Rp 5,150 per liter with the total target of 331 locations in 2021 (total additional 78 one-price fuel distribution locations in 2021). MEMR sets the total target of 423 One-Price Fuel locations until 2022. The distribution of one-price fuel locations until 2021 is presented in Picture 4.2 below.



Source: Performance realization 2021 and 2022 Program KESDM

Picture 4.2 Distribution of One-Price Fuel until 2020

In gas supply side, the government is promoting the development of Mini LNG Terminal for remote areas that are not covered by gas pipeline. Mini LNG Terminal in Sambera, East Kalimantan becomes the pilot project of LNG for power plant. This terminal has been used for Sambera Gas fired power plant, Kutai Kertanegara, East Kalimantan with the capacity of 2x20 MW. The development of this Mini LNG Terminal is based on the Ministry of Energy and Mineral Resources Decision No.13 of 2020 on Assignment on LNG Supply and Infrastructure Development and Fuel to LNG Conversion for Power Plant. This Ministerial Decision sets the target to develop 52 LNG power plant with total capacity of 1.7 GW and total gas supply of 167 BBTUD. This program is expected to save 2.6 million Kilo Liter fuel. The gas infrastructure development map can be seen in Picture 4.3.



Source: DG of Oil and Gas, 2021

Picture 4.3 Gas Infrastructure Development Map

4.2 CITY GAS NETWORK

To reduce imports of fuel oil and LPG which continue to increase, the Government is trying to divert the fulfillment of energy needs with gas. MEMR has built a gas network for households for the community for free. This natural gas distribution network development program for households is built in cities or areas close to gas sources.

In 2021, the household gas network or city gas has connected 127,000 households. The length of the gas pipelines that were successfully built in 2021 increased by 3,263.70 km or 76.08% from the previous year. The addition was obtained from network development in 25 regencies/cities carried out by PT Perusahaan Gas Negara (PGN) and PT Pertamina Niaga

Through this program, people are expected to obtain a cleaner and safer fuel. Furthermore, GSEN sets city gas, DME and electric stove program to stop LPG import by 2030.

The target of city gas network in RUEN in 2025 is to connect 4.7 million households. Based on RUEN, city gas network in RPJMN 2020-2024 is targeted to connect 4 million households in 2024, but the first development in 2009-2021 has connected only 799 thousand households. This number is far below the target due to budgeting issue. To give an overview, 1 household connection costs 8-10 million Rupiah.

City gas for household is still funded by the State Budget. The government takes over the project of city gas for households since it is considered as less interesting for business entity due to minimum profit from the project. Concerning the limited state budget, the Public Private Partnership (PPP) involving PT Pertamina and PT Perusahaan Gas Negara (PGN) is being formulated.

4.3 PROMOTING ELECTRIC STOVE AS FOSSIL FUEL SUBSTITUTION

One of the efforts to reduce LPG imports is to substituted with an electric stove. To reduce dependence on energy imports, the Government has launched several efforts to reduce LPG consumption. In GSEN document, several programs to reduce LPG imports (including a program to reduce LPG consumption) include: Development of City Gas Networks, Increase in Production Capacity of New Oil Refineries, Development of Dimethyl Ether (DME) and the use of electric stoves.

In contrast to LPG supply from imports, the electricity supply in Java-Bali, based on RUPTL 2021 data, shows that the installed capacity of power plants in Java-Bali is 41 GW, while the peak load is 29 GW. There is a reserve margin above 30% in Java-Bali region. One of the efforts to increase the growth of electricity consumption is by increasing the use of electric stoves in Java-Bali. The estimated energy consumption for the use of an electric stove with a power of 2,000 Watt based on a study by the Research and Development Center of the Ministry of Energy and Mineral Resources is around 82 kWh/month. Meanwhile, the number of household customers in Java and Bali with a power of 1,300 VA and above is around 10.3 million with details as shown in Table 4.1 below.

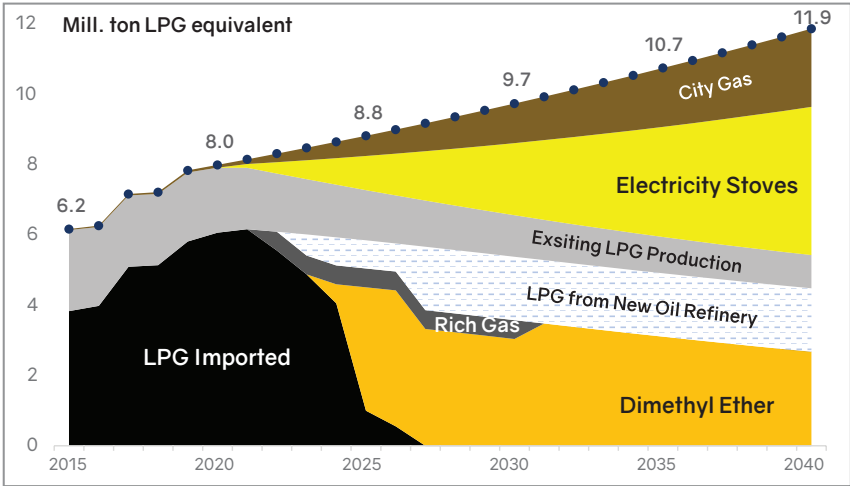
Table 4.1 Details of PLN Customer in Java- Bali

Customer Type	Banten	Jakarta	West Java	Yogyakarta and Central Java	East Java	Bali	Total
R1 / 1300 VA	703,393	1,776,808	2,113,980	913,834	1,076,458	356,599	6,941,072
R1 / 2200 VA	184,514	612,357	572,898	293,191	328,576	138,488	2,130,024
R2 / 3500-5500 VA	76,101	360,279	193,741	112,246	148,194	78,769	969,330
R3 / 6600 VA	12,776	114,590	27,869	16,605	27,441	20,721	220,002

Source: DG of Electricity, 2021

If it is assumed that there is an additional use of electric stoves for 1 million household customers per year, there will be an increase in electricity consumption of around 82 GWh or 414 MW. If all household customers with a power of 1,300 VA and above use electric stoves, there will be an increase in electricity consumption of 841 GWh or an increase in load of 4.2 GW.

Based on GSEN projections, the total electric stove users by the end of 2030 will be 19 million households. If the program in GSEN can be implemented, there will be no LPG imports in 2027. An overview of the GSEN Program related to reducing LPG imports can be seen in Picture 4.4 below.



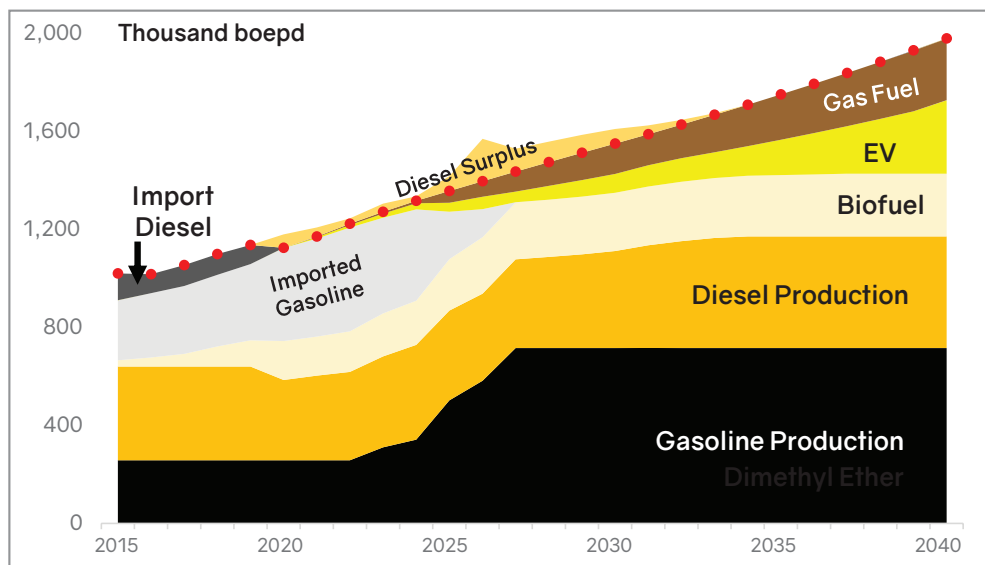
Picture 4.4 GSEN Program to Stop LPG Import

In 2022, the implementation of the electric stove program is targeted at around 300 thousand. Currently, a pilot project and 2,000 electric stoves program for each household in Solo and Bali have been carried out. The package consists of an electric stove unit, a set of cooking utensils, and an MCB to increase power. The estimated cost for the electric stove program is 1.8 million per package. Thus, the total funds needed to distribute packages to 300 thousand households are around Rp540 billion. It is hoped that the use of electric stoves can increasing electricity consumption in Java-Bali. Based on calculations, the use of electric stoves is estimated to absorb 4.2 GW from an excess supply of 12 GW.

4.4 THE DEVELOPMENT OF ELECTRIC VEHICLE

The government has regulated the use of electric vehicle in Presidential Regulation No. 55 of 2019 on Acceleration of Battery-based Electric Vehicle for Land Transportation. This Presidential Regulation sets the duties of related ministries in the area of infrastructure development, research and development, regulator, and 35% local content component. To promote this Presidential Regulation, the government through Ministry of Industry has formulated roadmap of electric vehicle development as set in Ministerial Regulation No. 27 of 2020. In this Minister of Industry Regulation, the target of electric vehicle in 2030 is 627 thousand units, while electric motorcycle is 9 million units. Currently, the Ministry of Industry is coordinating with Coordinating Ministry of Maritime and Investment to amend Minister of Industry Regulation No. 27 of 2020 on roadmap of battery-based electric vehicle.

Meanwhile in GSEN, there are programs to reduce fuel import through gas fueled vehicle, the use of biofuel, and the use of battery electric vehicle. In 2030, the target is 2 million units of electric car and 12 million units of electric motorcycle. The battery electric vehicle program is expected to stop fuel import before 2030 as shown in Picture 4.5 below.



Picture 4.5 GSEN Program to Stop Fuel Import

4.5 PREPARING ENERGY TRANSITION

Energy transition is done through transformation of fossil fuel using clean technology, acceleration of new and renewable energy use, and energy conservation. Energy transformation into Net Zero Emission is expected to create energy independence, energy resilience, sustainable development, climate resilience and low carbon condition.

Utilization of fossil fuels such as coal using clean technology is carried out by developing CCT (Clean Coal Technology). CCT is the use of high-efficiency technology in coal fired PP to reduce emissions from several pollutants and waste, as well as increase the energy from each ton of coal. Meanwhile, oil and gas as well as coal use CCUS (Carbon Capture, Utilization, and Storage) technology.

The Indonesian government has developed Ultra Super Critical (USC) technology in coal fired PP starting with the construction of Java-7 coal fired PP located in Serang (Banten) in 2017. In 2021, there are several USC coal fired PP under construction including Java 9 & 10 coal fired PP, Central Java (Batang) coal fired PP, and Java 4 (Tanjung Jati B) coal fired PP.

In 2020, the Government of Indonesia has conducted a feasibility study for the CCUS Pilot Project in Gundih Oil and Gas field, Central Java. The total CO₂ reduction potential in this pilot project is projected to be 2.92 Million Tons over 10 years. PT Pertamina also has a roadmap for the implementation of CCUS for 7 locations related to natural gas oil production, namely in Subang, Cilamaya, Jatibarang, Merbau, Jambaran Tiung Biru, Natuna and Matindok. The location has been studied and is planned to be implemented no later than 3 years from now. Indonesia itself has large geological storage resources that have the potential as CCUS locations in Java, Sumatra, and Kalimantan. Feasibility studies are also being carried out in these areas to develop CCUS technology.

The acceleration on new renewable energy use is done by substituting primary energy and final energy to biofuel, co-firing technology, fuel source from Refuse-Derived Fuel (RDF) and Solid Recovered Fuel (SFR), additional NRE power plant capacity with the focus on Rooftop solar PP, and non electricity NRE and non-biofuel utilization such as briquete, biogas and Compressed Biomethane Gas (CBG).

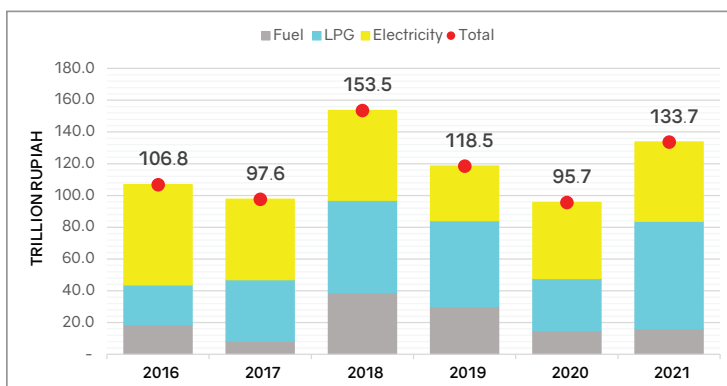
Through the use of battery-based electric vehicle, the battery industry and hydrogen technology are also expected to support the acceleration of new renewable energy use. In renewable energy sector, Smart Energy program optimizes renewable energy production system by maximizing efficiency, reducing cost, and enhancing security. In electricity, Smart Grid development is expected to reduce number of power plant needed to supply electricity and to integrate renewable energy sources. Meanwhile, the technology being developed is nuclear, coal gasification, liquified coal, tidal, hydrogen, and Carbon Capture Storage.

4.6 ENERGY SUBSIDY

The realization of fuel subsidy continues to decline despite of a significant increase in 2018 to 38.9 trillion Rupiah due to an increase in diesel oil subsidies. It decreased again in 2020 to 14.9 trillion Rupiah due to the Covid-19 pandemic. In 2021, it increased again by around 8.3% to 16.2 trillion Rupiah.

The realization of LPG subsidies in 2016 was 24.9 trillion Rupiah which continued to increase until 2018 to 58.1 Trillion Rupiah. This increased LPG burden is another form of commitment to shift the kerosene subsidy to clean energy. Meanwhile, in 2019 there was a decrease of around 6.9% caused by the decline in gas prices on the world market. It continued to decline in 2020 to 32.8 Trillion Rupiah due to the Covid-19 pandemic. However, in 2021 LPG subsidy increased again by around 106.1% from the previous year to 67.6 trillion Rupiah.

Similar to the realization of fuel subsidies, the realization of electricity subsidies also fluctuated from 63.1 trillion Rupiah in 2016 to 49.9 trillion Rupiah in 2021. The development of energy subsidy realization in 2016-2021 can be seen in Picture 4.6 below.



Picture 4.6 Development of Energy Subsidy 2016–2021

In early September 2022, the Government of Indonesia reduced fuel subsidy that it leads to the increasing price of several types of fuel. Based on the explanation from the Minister of Finance, this subsidy reduction was carried out because the burden of subsidies and compensation for the energy sector in 2022 was very high, reaching 502.4 trillion rupiah. The energy subsidy in 2022 is 208.9 trillion rupiah consisting of subsidy for fuel, LPG and electricity. Then there is energy compensation in 2022 of 234.6 trillion rupiah and underpayment of energy compensation in 2021 of 108.4 trillion rupiah.

4.7 THE ESTABLISHMENT OF NUCLEAR ENERGY PROGRAM IMPLEMENTING ORGANIZATION (NEPIO)

Nuclear Power Plants (Nuclear PP) have various advantages over other power plants. Nuclear power plants produce very large and reliable electricity. Nuclear power plants can operate for two years non-stop without changing fuel. Nuclear power plants have various types of technology, providing stable and large-scale electricity supplies ranging from 10 MWe to 1,600 MWe per unit. Nuclear power plants rely more on technological developments (technology bases) compared to energy sources (resource bases) so that they do not depend on the location of the nuclear fuel source. Nuclear power plants has the quality in terms of environmental externalities. Nuclear power plants have very little environmental impact with zero local emissions (SO_x, NO_x, and fly ash) and minimal global CO_x emissions. With a small environmental impact, nuclear power plants also have the lowest mortality percentage compared to other types of power plants. However, some countries are reluctant to build nuclear power plants due to security and safety risks.

The presence of nuclear minerals in Indonesia is spread over 26 locations in Sumatra, Kalimantan, Sulawesi, Maluku, and Papua. The estimate of uranium and thorium resources have been carried out in 9 deposit locations. In 2021, the total uranium resources reached 89,483 tons of U₃O₈ (the heat generation value for this amount of Uranium is equivalent to

the heat generation by 1.85 Trillion tons of coal), while the total thorium resources reached 143,234 tons. Details of radioactive mineral resources in Indonesia can be seen in Table 4.2.

Table 4.2 Indonesia Radioactive Mineral Resources 2021

No.	Potential Sector	Resource								
		Measured		Indi- cated	Concessed		Hypothetic		Speculative	
		U ₃ O ₈	Th	U ₃ O ₈	U ₃ O ₈	Th	U ₃ O ₈	Th	U ₃ O ₈	Th
1	Aloban-Sibolga				490					
2	Singkep-Riau	*2,840					* 1,298	* 433		
3	Bangka Belitung		*4,729				* 1,224	*10,361	*25,715	*111,298
4	Ketapang						* 736	*4,767		
5	Kalan-Melawai	*2,394		* 5,903	*2,914		*5,058			
6	Mentawa and Darab				* 623		* 9,669			
7	Katinga						* 572	*2,261		
8	Kawat - Mahakam Hulu						* 17,861			
9	Mamuju				* 769	*3,424	*3,023	*3,138	* 8,393	*2,823
Total		* 5,234	**4,729	*5,903	*4,796	**3,424	* 39,441	**20,960	* 34,108	**114,121
		Measu- red U	Measu- red Th	Indicated U	Indicated U	Measu- red Th	Hipothe- tic U	Hipothe- tic Th	Specu- lative U	Specu- lative Th
Total U ₃ O ₈		*89,483								
Total TH		**143,234								

Source: ORTN BRIN, 2021

Note: (*) Unit Ton U₃O₈

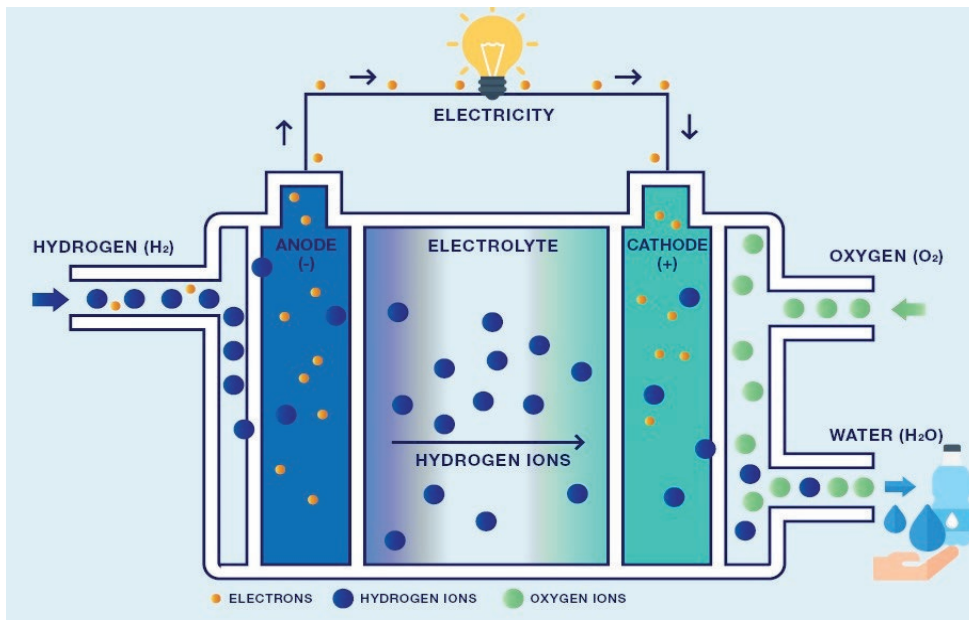
(**) Unit Ton Th

For countries that do not have nuclear power plants and intend to develop nuclear power plants, the International Atomic Energy Agency (IAEA) highly recommends them to form a Nuclear Energy Program Implementing Organization (NEPIO) as part of the government's commitment to nuclear energy development.

Therefore, in 2021 the Minister of Energy and Mineral Resources Decision No. 250.K/HK.02/MEM/2021 on the Preparation Team for the Establishment of a Nuclear Energy Program Implementing Organization has been issued.

4.8 HYDROGEN DEVELOPMENT PLAN

Hydrogen fuel is a zero emissions fuel that is used as fuel for electricity generation along with oxygen by using a unit called a hydrogen fuel cell. Similar to a battery, a fuel cell has an anode and cathode poles where hydrogen (H₂) and oxygen (O₂) are supplied to the two different poles as shown in Picture 4.7 below.



Source: MEMR, 2021

Picture 4.7 Hydrogen Fuel Cell

Hydrogen can be produced through the following processes, including:

a. Gas Steam Reforming

In this process, the methane contained in natural gas is extracted and reacted with steam to produce hydrogen. Around 95% of the hydrogen on the market is produced by this process. This process can produce gray hydrogen and blue hydrogen if the carbon produced is absorbed through Carbon Capture and Storage technology.

b. NRE Power Plant Electrolysis

The electrolysis process uses a device called an electrolyzer. The main concept in this process is the separation of hydrogen molecules and oxygen from water by a reaction caused by an electric current. If this process is carried out at an NRE power plant, green hydrogen will be obtained.

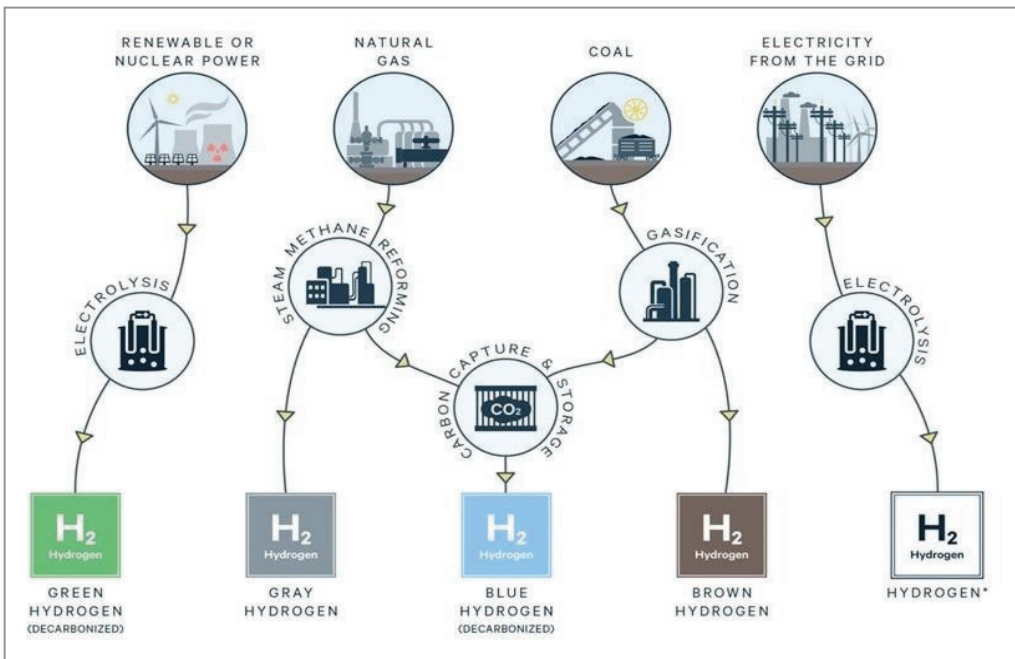
c. Coal Gasification

In the gasification process, coal or biomass material is reacted with oxygen and steam to produce synthesis gas. Then, the hydrogen molecules are separated from the synthesis gas using a separation system. The coal gasification process will produce brown hydrogen or blue hydrogen if the carbon produced is absorbed through Carbon Capture and Storage technology.

d. Biological Process

Microbes such as bacteria and microalgae can produce hydrogen by biological reactions using sunlight or organic matter. This technology is still at the research and development stage.

The details of hydrogen production technology can be seen in Picture 4.8 below.



Source: BBPT, 2021

Picture 4.8 Hydrogen Production Technology

In Indonesia, hydrogen development is still in the research and pilot project stages and there are no commercial projects yet. Research on hydrogen as a fuel has been carried out by the Agency for the Assessment and Application of Technology (BPPT), the University of Indonesia (UI), the Bandung Institute of Technology (ITB) and Lemigas since 2007. Furthermore, in 2012 ITB made the concept of a fuel cell car and the Korea International Cooperation Agency (KOICA) provided a grant to the Indonesian government for the construction of a fuel cell pilot project with a capacity of 300 kilowatts (kW) in Ancol, Jakarta. Then in 2014, the Indonesian Association for Fuel Cell and Hydrogen Energy (INAFHE) was established to accelerate the development of hydrogen in Indonesia. In 2017, PT Telkomsel developed a fuel cell at the Base Transceiver Station (BTS) as backup power. After that, in 2018, BPPT and Toshiba ESS signed a cooperation agreement regarding the Development of the H₂One Autonomous Hydrogen Energy System for Off Grid Systems. In 2019, the Ministry of Energy and Mineral Resources, PT Kereta Api Indonesia (KAI) and Allstorm began collaborating in the development of hydrogen-fueled trains. PT HDF Energi took the initiative to develop green hydrogen from hybrid Solar PP and Wind PP in Sumba Island with a capacity of 7-8 MW during the day and 1-2 MW at night from hydrogen storage. In addition, there is also cooperation between PT Pertamina and GIZ in the development of a green hydrogen pilot project from geothermal energy.

To achieve the Net Zero Emission (NZE) target by 2060, the Government of Indonesia plans to start developing hydrogen technology in 2030.

4.9 ENERGY CONSERVATION AND EFFICIENCY

The application of energy conservation and efficiency can reduce energy imports, maximize energy use for domestic needs and reduce exports, increase Reliability, control energy demand growth and encourage energy diversification by maximizing the use of NRE in order to increase national energy security. On the other hand, energy conservation and efficiency can increase economic growth and competitiveness because it can reduce energy intensity, increase industrial competitiveness, reduce energy production costs to lead to more affordable energy costs. Energy conservation and efficiency is also seen as a solution to reduce the greenhouse gas emissions and address the climate change issue which is very much needed to fulfill international obligations under the United Nations Framework Convention on Climate Change (UNFCCC).

An effective and efficient energy use management is aimed to produce maximum products through structured and well-documented technical actions that it gives an impact on the optimal use of raw materials and supporting materials. The systematic implementation of Energy Management can be done through the implementation of ISO 50001-Energy Management System. There are four main stages in ISO 50001-Energy Management System, including:

- a. Plan stage, namely to review energy and establish baselines, energy performance indicators, goals, targets and action plans needed to provide results that will improve energy performance in accordance with the organization's energy policy
- b. Do stage, namely to implement the energy management action plan
- c. Check stage, namely to monitor and measure the processes and key characteristics of operations that determine energy performance toward energy policies and objectives and report the results
- d. Act stage, namely to take action to continuously improve energy performance and Energy Management System (ENMs)

In 2021, there will be five ISO 50001-certified buildings, namely The Agency for the Assessment and Application of Technology (BPPT); Directorate General of New, Renewable Energy and Energy Conservation, Ministry of Energy and Mineral Resources; Secretariat General of the Ministry of Energy and Mineral Resources; Ngurah Rai Airport (PT. Angkasa Pura I); and Terminal 3 Soekarno Hatta Airport (PT. Angkasa Pura II). In addition, there are 49 energy sector companies that have been certified to ISO 50001 consisting of 31 power plants, 14 oil and gas fields, and 4 mining companies. On the other hand, there are also 78 industrial sector companies that have been certified to ISO 50001 consisting of 21 food and beverage companies, 17 manufacturing companies, 6 textile companies, 14 agro and paper companies, and 20 chemical companies.

In addition, there are energy efficiency activities in the use of LED Public Street Lighting compared to HPS Public Street Lighting. LED Public Street Lighting only needs the power of 90 W, while HPS Public Street Lighting needs the power of 250 W. In 2021, it is estimated that the energy savings obtained from the use of LED Public Street Lighting is 96 GWh or equivalent to a savings in electricity costs of 141 billion Rupiah.

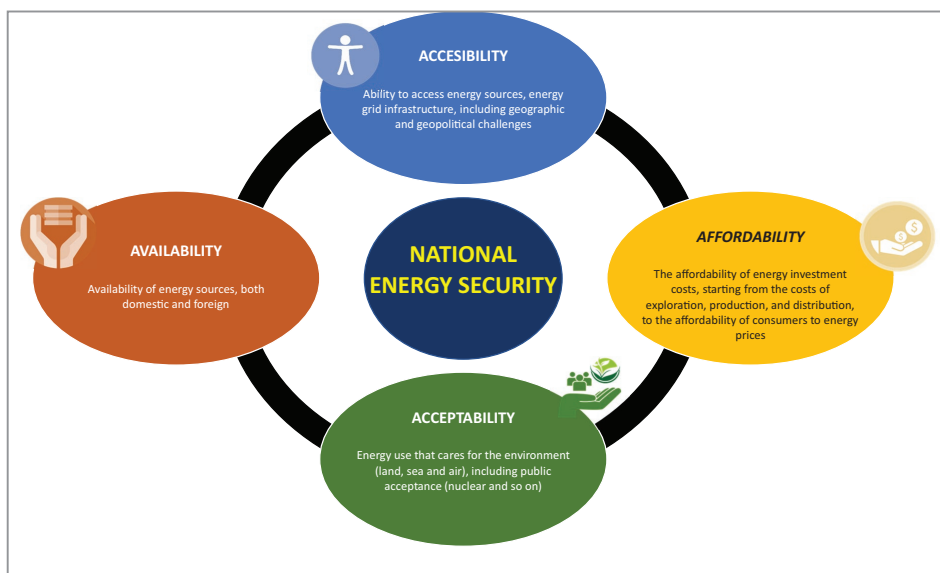
4.10 CONDITION OF NATIONAL ENERGY SECURITY

Energy Security according to Article 1 point 10 of Government Regulation No. 79 of 2014 on National Energy Policy (KEN) is defined as a condition of ensuring the availability of energy and public access to energy at affordable prices in the long term while taking into account the protection of the environment.

The condition of energy security is dynamic which is influenced by various strategic environmental factors, both internal and external, including the development of energy demand and supply, investment in energy infrastructure development, and the impact of policies and regulations on the energy sector and related sectors. The dynamics of energy security conditions need to be monitored regularly to identify the change in energy sector development in order to ensure the sustainability of energy supply and use and support the increase in value added of energy use for the national economy and people's welfare towards the targets in the National Energy Policy.

The assessment of Indonesia's energy security has been carried out with an energy security model using 4A aspects (Picture 4.9), namely Availability (availability of energy sources both from domestic and abroad), Affordability (affordability of energy investment costs, ranging from exploration, production and distribution costs to consumer affordability to energy prices), Accessibility (ability to access energy sources, energy network infrastructure, including geographic and geopolitical challenges), Acceptability (energy use that puts concern on land, sea and air environments including public acceptance of nuclear, and so on). It also uses weighting method using AHP (Hierarchical Process Analysis) and assessment by experts. Since the assessment from 2004 to 2019, the scale of Indonesia's energy security value has always increased from year to year.

Energy security assessment is conducted to see Indonesia's ability to provide energy in the country through selected aspects and indicators during the previous one-year period.



Source: Indonesia Energy Security, 2019

Picture 4.9 Energy Security Model

In order to enhance energy security in the future, it is necessary to make improvements in every aspect and indicator, one of which is by realizing the existing programs in GSEN, such as by: Accelerating the utilization of 38 GW of NRE power plants by 2035 (Solar PP and other NRE); Increasing crude oil production up to 1 million bopd and acquisition of overseas oil fields for refinery needs; Increasing the capacity of existing refineries and building new refineries; Providing gas-based energy for industrial and transportation areas (such as CNG); Increasing the use of battery-based electric vehicles; Optimizing biofuel production (biodiesel or biohydrocarbons); Increasing the development of the city gas network; Increasing domestic LPG production; Encouraging the use of electric stoves; Developing DME production; Building gas transmission, LNG receiving terminal, and energy buffer reserve infrastructure; Developing methanol, fertilizer & syngas production, as well as synergy between coal mining and smelters; Building electricity transmission & distribution, smart grid, off grid and nuclear power plants as needed and the establishment of the Nuclear Energy Program Implementing Organization (NEPIO); Encouraging efficiency, energy conservation and innovation in energy such as Hydrogen, NH_3 , CCS and CCUS.



5

CONCLUSION AND RECOMMENDATION

5 CONCLUSION AND RECOMMENDATION

5.1. CONCLUSION

1. In 2032, final energy consumption of BaU scenario will increase with an average growth of 4.8% per year to 207 million TOE, with transportation and industrial sectors are the largest share of final energy consumption with 42.6% and 37.9% respectively. While in the OPT scenario, final energy consumption in the next 10 years will increase by 6.6% per year to 248 million TOE with the largest share coming from the industrial sector by 49.2% and 36.4% from transportation, this growth were in line with targets towards developed countries in 2045.
2. In 2021, household sector is the largest electricity consumption, but the situation would change in 2032 OPT scenario, where both industrial and transportation sector are the largest electricity consumers with 46.6% and 19.7% respectively, as a result of the high energy demand growth in outside Java industry sector and the growing use of electric vehicles since 2022.
3. Java-Bali region will still being the biggest consumer of final energy in 2032, around 45.7% (BaU) and 48.3% (OPT) shares from total final energy consumption. However, the largest primary energy mix from NRE in 2032 in the BaU scenario is the Sumatra region (28.6%), and in the OPT scenario is the Kalimantan region (43.3%).
4. The NRE share on primary energy mix will still below the RUEN target in 2025, around 16.3% in the BaU scenario and 18.6% in the OPT scenario. However, based on the OPT scenario, in 2030 the NRE achievements will exceed the RUEN target, reaching 26.4%.

5.2. RECOMMENDATION

1. To achieve the NZE 2060 target, Indonesia must have high economic growth so that it can become a developed country through growth in the industrial sector, especially outside Java Island.
2. Massive use of NRE in the power generation sector need to be achieved, especially coal co-firing PP and Solar PP as well as increasing the blend of biodiesel and bioethanol utilization so that the achievement of the EBT mix in 2030 can exceed the RUEN target and be in line with the NZE target in 2060.
3. Massive use of electric vehicles and electric stoves, especially in the Java-Bali region, needs to be implemented immediately to absorb excess power/electricity.



ATTACHMENT

ATTACHMENT 1

OUTLOOK SUMMARY

No	Energy Indicator	Unit	National Projection in BaU Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	130	137	143	150	157	165	173	181	189	198	207
2	Electricity Demand	TWh	269	281	293	306	321	337	354	372	391	412	433
3	Industrial Sector Final Energy Demand	Million TOE	46	48	51	54	56	59	62	66	70	74	79
4	Transportation Sector Final Energy Demand	Million TOE	53	56	60	63	67	70	74	77	81	85	88
5	Household Sector Final Energy Demand	Million TOE	21	21	21	21	22	22	22	22	22	22	22
6	Commercial Sector Final Energy Demand	Million TOE	8	9	9	10	10	11	12	12	13	14	15
7	Others Sector Final Energy Demand	Million TOE	2	2	2	2	2	2	3	3	3	3	3
8	Primary Energy Supply	Million TOE	206	217	227	237	250	263	275	289	303	317	332
9	Green House Gas Emission	Million Ton CO ₂	644	671	700	719	765	802	838	877	918	968	1,015
10	Power Plan Capacity	GW	84	88	91	96	102	107	111	115	119	122	125
11	Electricity Production	TWh	310	322	335	348	365	383	402	422	443	465	489

No	Energy Indicator	Unit	National Projection in OPT Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	130	138	147	156	166	178	190	203	218	232	248
2	Electricity Demand	TWh	280	305	333	365	400	440	483	532	587	642	702
3	Industrial Sector Final Energy Demand	Million TOE	48	52	57	63	69	76	83	92	102	111	122
4	Transportation Sector Final Energy Demand	Million TOE	52	55	58	61	65	68	72	76	81	86	91
5	Household Sector Final Energy Demand	Million TOE	21	21	21	21	22	22	22	22	22	22	22
6	Commercial Sector Final Energy Demand	Million TOE	8	8	8	9	9	9	9	10	10	10	10
7	Others Sector Final Energy Demand	Million TOE	2	2	2	2	2	3	3	3	3	3	3
8	Primary Energy Supply	Million TOE	209	223	238	254	276	300	325	352	383	414	446
9	Green House Gas Emission	Million Ton CO ₂	645	671	699	724	774	824	872	927	977	1,042	1,108
10	Power Plan Capacity	GW	84	91	97	104	112	123	133	144	157	169	182
11	Electricity Production	TWh	322	350	380	414	454	499	548	603	664	725	791

No	Energy Indicator	Unit	Projection of Sumatra Region in BaU Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	24	26	27	29	31	32	34	36	38	40	42
2	Electricity Demand	TWh	44	46	48	51	53	56	59	62	66	70	74
3	Industrial Sector Final Energy Demand	Million TOE	6	6	7	7	7	8	8	8	9	9	10
4	Transportation Sector Final Energy Demand	Million TOE	12	13	15	16	17	18	19	21	22	23	25
5	Household Sector Final Energy Demand	Million TOE	4	4	4	4	4	4	4	4	4	4	4
6	Commercial Sector Final Energy Demand	Million TOE	1	1	2	2	2	2	2	2	2	3	3
7	Others Sector Final Energy Demand	Million TOE	0.4	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5
8	Primary Energy Supply	Million TOE	37	39	41	43	46	48	51	54	57	60	63
9	Green House Gas Emission	Million Ton CO ₂	97	103	108	104	114	125	130	137	142	153	164
10	Power Plan Capacity	GW	17	18	19	21	21	21	21	22	23	23	23
11	Electricity Production	TWh	51	53	55	58	61	64	67	71	75	79	84

No	Energy Indicator	Unit	Projection of Sumatra Region in OPT Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	24	25	27	29	31	33	35	38	40	43	47
2	Electricity Demand	TWh	47	52	57	64	71	79	88	99	110	123	137
3	Industrial Sector Final Energy Demand	Million TOE	7	7	8	9	10	11	12	14	15	17	19
4	Transportation Sector Final Energy Demand	Million TOE	12	12	13	14	15	15	16	17	19	20	21
5	Household Sector Final Energy Demand	Million TOE	4	4	4	4	4	4	4	4	4	4	4
6	Commercial Sector Final Energy Demand	Million TOE	1	1	1	1	1	2	2	2	2	2	2
7	Others Sector Final Energy Demand	Million TOE	0.4	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.6
8	Primary Energy Supply	Million TOE	37	40	43	47	51	55	60	65	72	78	86
9	Green House Gas Emission	Million Ton CO ₂	99	106	114	115	129	142	151	161	165	174	186
10	Power Plan Capacity	GW	17	18	19	21	21	22	23	24	27	29	32
11	Electricity Production	TWh	54	59	65	72	80	90	100	112	125	139	154

No	Energy Indicator	Unit	Projection of Java-Bali Region in BaU Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	67	70	73	75	78	81	83	86	89	92	95
2	Electricity Demand	TWh	188	197	205	214	225	237	249	262	276	290	305
3	Industrial Sector Final Energy Demand	Million TOE	19	20	21	21	22	23	24	25	26	27	28
4	Transportation Sector Final Energy Demand	Million TOE	30	32	33	35	36	37	39	40	41	43	44
5	Household Sector Final Energy Demand	Million TOE	13	13	14	14	14	14	14	14	14	14	14
6	Commercial Sector Final Energy Demand	Million TOE	4	4	5	5	5	6	6	6	7	7	8
7	Others Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
8	Primary Energy Supply	Million TOE	120	125	130	135	142	149	155	161	168	175	182
9	Green House Gas Emission	Million Ton CO ₂	400	416	433	449	476	494	516	540	563	588	608
10	Power Plant Capacity	GW	48	50	51	53	58	63	66	68	71	74	78
11	Electricity Production	TWh	217	225	234	244	256	269	283	297	312	328	344

No	Energy Indicator	Unit	Projection of Java-Bali Region in OPT Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	68	72	75	80	84	89	94	100	106	113	120
2	Electricity Demand	TWh	193	207	223	241	263	288	315	345	379	414	452
3	Industrial Sector Final Energy Demand	Million TOE	20	21	23	25	27	30	33	36	40	44	48
4	Transportation Sector Final Energy Demand	Million TOE	30	32	34	36	38	40	42	44	47	49	52
5	Household Sector Final Energy Demand	Million TOE	13	13	14	14	14	14	14	14	14	14	14
6	Commercial Sector Final Energy Demand	Million TOE	4	4	4	4	4	5	5	5	5	5	5
7	Others Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
8	Primary Energy Supply	Million TOE	122	128	135	142	154	166	180	195	210	226	244
9	Green House Gas Emission	Million Ton CO ₂	399	410	422	435	463	491	521	553	587	622	657
10	Power Plan Capacity	GW	48	50	52	54	61	68	74	81	88	95	102
11	Electricity Production	TWh	222	237	254	273	299	326	357	391	428	467	509

No	Energy Indicator	Unit	Projection of Kalimantan Region in BaU Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	18	19	19	20	21	22	23	24	25	26	27
2	Electricity Demand	TWh	12	13	13	14	14	15	16	17	17	18	19
3	Industrial Sector Final Energy Demand	Million TOE	11	11	11	12	12	12	13	13	14	14	15
4	Transportation Sector Final Energy Demand	Million TOE	4	4	4	5	5	5	6	6	6	7	7
5	Household Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
6	Commercial Sector Final Energy Demand	Million TOE	1	2	2	2	2	2	2	2	2	3	3
7	Others Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
8	Primary Energy Supply	Million TOE	22	23	24	25	26	27	28	29	30	32	33
9	Green House Gas Emission	Million Ton CO ₂	51	53	55	56	59	60	63	64	67	70	74
10	Power Plant Capacity	GW	6	6	7	7	7	7	7	8	8	8	8
11	Electricity Production	TWh	14	14	15	16	16	17	18	19	20	21	22

No	Energy Indicator	Unit	Projection of Kalimantan Region in OPT Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	18	19	20	21	22	23	25	27	29	30	33
2	Electricity Demand	TWh	14	17	20	24	26	29	32	36	40	42	45
3	Industrial Sector Final Energy Demand	Million TOE	11	11	12	13	14	15	16	17	19	20	22
4	Transportation Sector Final Energy Demand	Million TOE	4	4	4	4	5	5	5	5	6	6	6
5	Household Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
6	Commercial Sector Final Energy Demand	Million TOE	1	1	2	2	2	2	2	2	2	2	2
7	Others Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
8	Primary Energy Supply	Million TOE	22	24	27	29	32	34	37	41	44	47	50
9	Green House Gas Emission	Million Ton CO ₂	53	56	58	63	64	65	67	69	73	80	87
10	Power Plan Capacity	GW	6	10	12	14	16	18	20	22	25	27	30
11	Electricity Production	TWh	16	20	23	27	30	33	37	40	45	48	51

No	Energy Indicator	Unit	Projection of Sulawesi Region in BaU Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	13	14	15	15	16	17	18	19	20	21	22
2	Electricity Demand	TWh	12	13	13	14	15	15	16	17	17	18	19
3	Industrial Sector Final Energy Demand	Million TOE	7	8	8	8	9	9	10	10	11	11	12
4	Transportation Sector Final Energy Demand	Million TOE	4	4	5	5	5	6	6	7	7	7	8
5	Household Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
6	Commercial Sector Final Energy Demand	Million TOE	0.4	0.5	0.5	0.5	0.5	0.6	0.6	0.6	0.7	0.7	0.8
7	Others Sector Final Energy Demand	Million TOE	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2
8	Primary Energy Supply	Million TOE	17	18	19	20	21	22	23	24	25	27	28
9	Green House Gas Emission	Million Ton CO ₂	54	57	59	62	64	67	69	72	75	79	83
10	Power Plan Capacity	GW	8	9	9	9	9	9	9	9	10	10	10
11	Electricity Production	TWh	14	15	15	16	17	17	18	19	20	21	21

No	Energy Indicator	Unit	Projection of Sulawesi Region in OPT Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	13	14	15	17	18	19	21	23	24	26	28
2	Electricity Demand	TWh	13	15	17	19	21	23	26	29	33	36	39
3	Industrial Sector Final Energy Demand	Million TOE	7	8	9	10	11	12	13	15	16	17	19
4	Transportation Sector Final Energy Demand	Million TOE	4	4	4	5	5	5	5	6	6	7	7
5	Household Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
6	Commercial Sector Final Energy Demand	Million TOE	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.6	0.6
7	Others Sector Final Energy Demand	Million TOE	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2
8	Primary Energy Supply	Million TOE	17	19	20	22	24	26	28	31	34	36	39
9	Green House Gas Emission	Million Ton CO ₂	55	59	62	66	70	74	79	84	89	96	102
10	Power Plan Capacity	GW	8	8	8	8	9	9	9	9	10	10	10
11	Electricity Production	TWh	15	17	19	21	24	27	30	33	37	41	44

No	Energy Indicator	Unit	Projection of Nusmapa Region in BaU Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	8	8	9	10	11	13	14	16	17	19	21
2	Electricity Demand	TWh	12	13	13	13	14	14	14	15	15	15	16
3	Industrial Sector Final Energy Demand	Million TOE	3	4	4	5	6	7	8	9	11	12	14
4	Transportation Sector Final Energy Demand	Million TOE	3	3	3	3	3	4	4	4	5	5	5
5	Household Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
6	Commercial Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
7	Others Sector Final Energy Demand	Million TOE	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2
8	Primary Energy Supply	Million TOE	11	12	13	14	16	17	19	20	22	24	27
9	Green House Gas Emission	Million Ton CO ₂	41	43	46	49	52	56	60	65	71	78	87
10	Power Plant Capacity	GW	5	5	6	6	6	7	7	7	7	7	7
11	Electricity Production	TWh	14	15	15	15	15	16	16	17	17	17	18

No	Energy Indicator	Unit	Projection of Nusmapa Region in OPT Scenario										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	Final Energy Demand	Million TOE	8	8	9	11	12	13	14	16	18	19	21
2	Electricity Demand	TWh	13	14	16	17	19	20	22	24	26	27	29
3	Industrial Sector Final Energy Demand	Million TOE	3	4	5	6	7	8	9	10	12	13	15
4	Transportation Sector Final Energy Demand	Million TOE	2	3	3	3	3	3	3	3	4	4	4
5	Household Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
6	Commercial Sector Final Energy Demand	Million TOE	1	1	1	1	1	1	1	1	1	1	1
7	Others Sector Final Energy Demand	Million TOE	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2
8	Primary Energy Supply	Million TOE	11	12	13	15	16	18	19	22	24	26	28
9	Green House Gas Emission	Million Ton CO ₂	39	41	43	45	48	51	55	59	64	70	77
10	Power Plan Capacity	GW	5	5	5	6	6	6	7	7	8	8	8
11	Electricity Production	TWh	15	16	18	20	21	23	25	27	29	31	33



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